Joint Communication-Control Strategy Optimization with Partially Nested Information Structures: The Linear-Quadratic Case

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***Abstract*— In this paper, we formalize a joint communication-control strategy optimization (JCCO) problem in multi-agent linear systems with quadratic costs, under the common-information-based (CIB) framework from decentralized stochastic control. Towards computational tractability, we focus on such JCCO problems with partially nested (PN) information structures (ISs). In particular, with a baseline communication protocol that leads to a PN IS, we establish a series of conditions under which the partial nestedness is preserved under the (additional) communication strategies to be optimized, while missing them may cause nonlinearity of the optimal strategies in general, with open-loop communication strategies. We then develop a dynamic-programming-based approach to compute the optimal control strategies of JCCO with open-loop communication strategies, which yields a set of closed-form Riccati Equations. Such an approach also offers a way to solve decentralized linear-quadratic control with PN ISs and *output* feedback, under the CIB framework. Finally, we extend such an approach to JCCO with closed-loop communication strategies, yielding a more tractable dynamic program than an infinite-dimensional CIB-belief-based one.**

I. Introduction

The design of communication protocols for control has been extensively studied in both the control theory and multi-agent learning literature, from different perspectives and under different models. For example, [1], [2] investigated the design of communication channels with capacity constraints in networked control systems; [3], [4] studied communication architecture and control strategy co-design via norm minimization; and [5], [6] developed heuristic learning-to-communicate algorithms to jointly learn the control and communication strategies for reward maximization.

In this paper, we formalize a joint communication-control strategy optimization (JCCO) problem in multi-agent linear systems with quadratic costs. Motivated by the empirical studies [5], [6], we model communication as some *actions/strategies* to be optimized jointly with the control strategies, under an optimal control formulation to minimize some accumulated costs. The agents may follow some fixed baseline communication protocols for information sharing, and then share their private information through *additional sharing*, following their communication strategies. We formalize the problem under the common-information-based (CIB) framework from decentralized stochastic control [7], [8], which has also been adopted in several recent studies on control-communication strategy co-optimization [9], [10], [11]. In comparison, [9], [10] focused specifically on the sharing of *instantaneous observations* only, and the systems with *decoupled* state dynamics. The formalism in [11] is more general and allows the sharing of the private-information history, with computational and sample complexity analyses. The present paper studies an instantiation of the formalism in [11] to the linear-quadratic (LQ) setting.

To facilitate tractable computation of the optimal strategies, we focus on deriving *dynamic-programming* (DP)-based approaches to solve JCCO. To this end, we concentrate on such co-optimization problems with partially nested (PN) information structures (ISs) [12]. Indeed, with a *fixed* communication strategy (and thus a fixed IS), partial nestedness is known to be a favorable IS that not only ensures the *linearity* of an optimal control strategy, but also yields a *convex reformulation* of the decentralized LQ control problem [12]. However, as pointed out in [13], the reformulated convex program can be too large to be computationally efficient, which precisely motivated the DP-based solution therein. Moreover, the DP-based approach can provide more insights into the structure of the optimal controller [13], by identifying the proper *sufficient statistics* for decision-making [13], [7], [14], [15].

Yet, the results in [13], [15] do not apply to our setting, as they focused on the special cases with *factorized* states, and with either *state-feedback* [13] or *multi-tree* coupling and communication graphs [15]. Under the CIB framework [7], [14] considered the output-feedback setting, and showed that when control strategies are restricted to be linear, and the private-information component of the strategy is *fixed*, the resulting problem is linear-quadratic-Gaussian (LQG) control, which can be solved via Riccati Equations. However, it remains unclear how to solve the overall decentralized LQ control problem by further optimizing over the private-information component. Since given a *fixed* communication strategy, our JCCO problem reduces to a decentralized LQ control one, we need to advance these results in order to fully address JCCO. We thus make the following contributions:

Contributions. (i) We formalize *joint communication-control strategy optimization* as an optimal control problem with linear systems and quadratic costs, under the common-information-based framework [7]. (ii) Targeting computational tractability, we focus on JCCO with *partially nested* baseline ISs [12], and identify structural conditions under which partial nestedness is preserved under additional information sharing, while missing them may cause nonlinearity or non-existence of the optimal control strategies under

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open-loop communication strategies. (iii) We then develop a closed-form dynamic program, i.e., a set of Riccati Equations, to solve for the optimal control strategy with respect to fixed open-loop communication strategies in JCCO, which may be viewed as a new approach, of independent interest, for solving decentralized LQ control with PN IS and *output* feedback under the CIB framework. The key is to leverage a connection between the (strictly) PN IS [12] and the strategy-independent CIB-belief (SI-CIB) condition [8]. (iv) We then extend such an approach to JCCO with *closed-loop* communication strategies, yielding reduced-dimensional sufficient statistics and a more tractable dynamic program, than an (infinite-dimensional) belief-based one, when applying the CIB framework to general LQ settings directly.

II. Preliminaries

Notation. Random variables are denoted by bold upper case letters, and their realizations are denoted by the corresponding non-bold upper case letters. For any matrix X, we use $X^\dagger$ to denote the pseudoinverse of X if $X \succeq 0$. For any two integers $0 \le a < b$, we denote $[a:b] := \{a, a+1, \cdots, b\}$, and denote $[a] = [1:a]$. For any vector space $\mathcal{X}$, we denote $\mathcal{P}(\mathcal{X})$ the space of all probability distributions over $\mathcal{X}$.

A. Problem formulation

For $n > 1$ agents, a *joint communication-control strategy optimization* problem with *output feedback* can be described as $\mathcal{D} = \langle H, \mathcal{X}, \{\mathcal{Y}_i\}_{i\in[n]}, \{\mathcal{U}_i\}_{i\in[n]}, \{\mathcal{M}_{i,h}\}_{i\in[n],h\in[H]}, \{A_h\}_{h\in[H]}, \{B_{i,h}\}_{i\in[n],h\in[H]}, \{E_{i,h}\}_{i\in[n],h\in[H]}, \{Q_h^1\}_{h\in[H+1]}, \{Q_h^2\}_{h\in[H]}, \{\mathcal{K}_h\}_{h\in[H]}\rangle$, where H is the time horizon and $\boldsymbol{X}_h \in \mathcal{X} = \mathbb{R}^{d_x}$ is the state. At each timestep $h \in [H]$, each agent $i \in [n]$ receives a noisy observation $\boldsymbol{Y}_{i,h} \in \mathcal{Y}_i = \mathbb{R}^{d_y^i}$ of the state $\boldsymbol{X}_h$, and chooses a control action $\boldsymbol{U}_{i,h} \in \mathcal{U}_i = \mathbb{R}^{d_u^i}$. At timestep $h \in [H]$, we denote by $\boldsymbol{U}_h = \begin{bmatrix}\boldsymbol{U}_{1,h}^\top & \boldsymbol{U}_{2,h}^\top & \cdots & \boldsymbol{U}_{n,h}^\top\end{bmatrix}^\top$ the joint control action of all the n agents, and by $\mathcal{U} = \mathbb{R}^{\sum_{i=1}^n d_u^i}$ the joint control action space; we denote by $\boldsymbol{Y}_h = \begin{bmatrix}\boldsymbol{Y}_{1,h}^\top & \boldsymbol{Y}_{2,h}^\top & \cdots & \boldsymbol{Y}_{n,h}^\top\end{bmatrix}^\top$ the joint observation, and by $\mathcal{Y} = \mathbb{R}^{\sum_{i=1}^n d_y^i}$ the joint observation space. The system evolves as follows: for each timestep $h \in [H]$

$$\begin{aligned}\boldsymbol{X}_{h+1} &= A_h \boldsymbol{X}_h + \sum_{i=1}^n B_{i,h}\boldsymbol{U}_{i,h} + \boldsymbol{W}_{0,h},\\ \boldsymbol{Y}_{i,h} &= E_{i,h}\boldsymbol{X}_h + \boldsymbol{W}_{i,h}, \forall i \in [n],\end{aligned} \tag{II.1}$$

where for each $i \in [0:n], h \in [H]$, $\boldsymbol{W}_{i,h}$ is a Gaussian random variable with distribution $\mathcal{N}(\boldsymbol{0}, \Sigma_{i,h})$ for some covariance matrix $\Sigma_{i,h} \succeq 0$, and $A_h, E_{i,h}$ are matrices of proper dimensions, and we define $B_h := \begin{bmatrix}B_{1,h} & B_{2,h} & \cdots & B_{n,h}\end{bmatrix}$ and $E_h := \begin{bmatrix}E_{1,h}^\top & E_{2,h}^\top & \cdots & E_{n,h}^\top\end{bmatrix}^\top$. Here, we assume that the initial state $\boldsymbol{X}_1$ follows a Gaussian distribution $\mathcal{N}(\boldsymbol{0}, \Sigma_1)$ with covariance matrix $\Sigma_1 \succeq 0$, and that $\boldsymbol{X}_1$ and $\{\boldsymbol{W}_{i,h}\}_{i\in[0:n],h\in[H]}$ are mutually independent.

At timestep $h \in [H]$, each agent will share part of her information to other agents. The shared information $\boldsymbol{Z}_h := \boldsymbol{Z}_h^b \cup \boldsymbol{Z}_h^a$ consists of two parts, the baseline-sharing part $\boldsymbol{Z}_h^b$, which comes from some existing sharing protocol, and the additional-sharing part $\boldsymbol{Z}_{i,h}^a$, which is *decided/learned* by agents, with joint additional-sharing information $\boldsymbol{Z}_h^a := \cup_{i=1}^n \boldsymbol{Z}_{i,h}^a$. At timestep h, the common information among all the agents is defined as the union of all the shared information so far: $\boldsymbol{C}_{h^-} = \cup_{t=1}^{h-1}\boldsymbol{Z}_t \cup \boldsymbol{Z}_h^b, \boldsymbol{C}_{h^+} = \cup_{t=1}^h \boldsymbol{Z}_t$, where $\boldsymbol{C}_{h^-}$ and $\boldsymbol{C}_{h^+}$ denote the common information *before* and *after* additional sharing, respectively. The private information of agent i at timestep h before and after additional sharing is denoted as $\boldsymbol{P}_{i,h^-}, \boldsymbol{P}_{i,h^+}$, respectively, where $\boldsymbol{P}_{i,h^-} \subseteq \{\boldsymbol{Y}_{1:h}, \boldsymbol{U}_{1:h-1}\}\backslash\boldsymbol{C}_{h^-}, \boldsymbol{P}_{i,h^+} \subseteq \{\boldsymbol{Y}_{1:h}, \boldsymbol{U}_{1:h-1}\}\backslash\boldsymbol{C}_{h^+}$. Then, we define $\boldsymbol{I}_{i,h^-} := \boldsymbol{P}_{i,h^-} \cup \boldsymbol{C}_{h^-}, \boldsymbol{I}_{i,h^+} := \boldsymbol{P}_{i,h^+} \cup \boldsymbol{C}_{h^+}$ as the information available to agent i before and after additional sharing, respectively. We denote by $\boldsymbol{P}_{h^-} := [\boldsymbol{P}_{1,h^-}^\top \ \cdots \ \boldsymbol{P}_{n,h^-}^\top]^\top$ the joint private information at timestep h before additional sharing, and similarly by $\boldsymbol{P}_{h^+}$ the joint private information after additional sharing. We denote by $\mathcal{C}_{h^-}, \mathcal{C}_{h^+}, \mathcal{P}_{i,h^-}, \mathcal{P}_{i,h^+}, \mathcal{P}_{h^-}, \mathcal{P}_{h^+}, \mathcal{I}_{i,h^-}, \mathcal{I}_{i,h^+}$ the sets of all possible values of the random variables $\boldsymbol{C}_{h^-}, \boldsymbol{C}_{h^+}, \boldsymbol{P}_{i,h^-}, \boldsymbol{P}_{i,h^+}, \boldsymbol{P}_{h^-}, \boldsymbol{P}_{h^+}, \boldsymbol{I}_{i,h^-}, \boldsymbol{I}_{i,h^+}$.

At each timestep h, in addition to the control action, each agent i needs to choose a communication action $\boldsymbol{M}_{i,h} \in \mathcal{M}_{i,h}$ to determine what information $\boldsymbol{Z}_{i,h}^a$ she will share, through the way to be specified later. We use $\boldsymbol{M}_h := (\boldsymbol{M}_{1,h}, \cdots, \boldsymbol{M}_{n,h}) \in \mathcal{M}_h$ to denote the joint communication action at timestep h.

We assume that the evolution of common and private information in $\mathcal{D}$ follows the rules below, where we adopt the convention that any quantity at $h = 0$ is empty/null.

Assumption II.1 (Information Evolution):

(a) (Baseline sharing) The baseline sharing evolves as $\boldsymbol{Z}_h^b = \chi_h(\boldsymbol{P}_{(h-1)^+}, \boldsymbol{U}_{h-1}, \boldsymbol{Y}_h)$ for some fixed projection function χ_h.

(b) (Additional sharing). For each $i \in [n]$, the additional sharing evolves as $\boldsymbol{Z}_{i,h}^a = \phi_{i,h}(\boldsymbol{M}_{i,h}, \boldsymbol{P}_{i,h^-})$ for some function $\phi_{i,h}$, where for each realization $M_{i,h} \in \mathcal{M}_{i,h}$, $\phi_{i,h}(M_{i,h}, \cdot)$ is a fixed projection function. The joint additional sharing $\boldsymbol{Z}_h^a$ is thus generated by $\boldsymbol{Z}_h^a = \phi_h(\boldsymbol{M}_h, \boldsymbol{P}_{h^-})$ for some function ϕ_h.

(c) For each $i \in [n], \boldsymbol{P}_{i,h^-} = \zeta_{i,h}(\boldsymbol{P}_{i,(h-1)^+}, \boldsymbol{U}_{i,h-1}, \boldsymbol{Y}_{i,h})$ for some fixed projection function $\zeta_{i,h}$, and the joint private information thus evolves as $\boldsymbol{P}_{h^-} = \zeta_h(\boldsymbol{P}_{(h-1)^+}, \boldsymbol{U}_{h-1}, \boldsymbol{Y}_h)$ for some projection function ζ_h.

(d) For each $i \in [n], \boldsymbol{P}_{i,h^+} = \boldsymbol{P}_{i,h^-}\backslash\boldsymbol{Z}_{i,h}^a$.

(e) For each $i \in [n], \boldsymbol{I}_{i,h^-} \subseteq \boldsymbol{I}_{i,h^+} \subseteq \boldsymbol{I}_{i,(h+1)^-}, \boldsymbol{Y}_{i,h} \in \boldsymbol{I}_{i,h^-}$.

Note that (a) and (c) on baseline sharing follow from those in [8], [16], [11]. (b) and (d) on additional sharing dictate how the communication action affects the additional sharing. For example, a common choice of $(\mathcal{M}_{i,h}, \phi_{i,h})$ is that $\mathcal{M}_{i,h} = \{0,1\}^{\max_{P_{i,h^-}\in\mathcal{P}_{i,h^-}} |P_{i,h^-}|}$ where $|P_{i,h^-}|$ denotes the number of elements in P_{i,h^-}, and for any $P_{i,h^-} \in \mathcal{P}_{i,h^-}, M_{i,h} \in \mathcal{M}_{i,h}$, $\phi_{i,h}(\boldsymbol{P}_{i,h^-} = P_{i,h^-}, \boldsymbol{M}_{i,h} = M_{i,h})$ consists of the k-th element ($k \in [|P_{i,h^-}|]$) of P_{i,h^-} if and only if the k-th element of $M_{i,h}$ is 1. Condition (e) means that each agent has full memory of the available information, and has a *closed-loop* (baseline) information structure with $\boldsymbol{Y}_{i,h} \in \boldsymbol{I}_{i,h^-}$ [11], [17].

B. Objective and strategies

At each timestep $h \in [H]$, agents will incur two types of cost, communication cost κ_h and control cost c_h. The control cost inherits the cost of a decentralized LQG problem without communication, see e.g., [14], [13], and is defined as $c_h = \boldsymbol{X}_h^\top Q_h^1 \boldsymbol{X}_h + \boldsymbol{U}_h^\top Q_h^2 \boldsymbol{U}_h$, with a terminal control cost of $c_{H+1} = \boldsymbol{X}_{H+1}^\top Q_{H+1}^1 \boldsymbol{X}_{H+1}$. Here, $\forall h \in [H], Q_h^1, Q_h^2 \succeq 0, Q_{H+1}^1 \succeq 0$ are symmetric matrices with proper dimensions. The communication cost is determined by the additional sharing and is defined as $\kappa_h = \mathcal{K}_h(\boldsymbol{M}_h)$ for some function $\mathcal{K}_h : \mathcal{M}_h \to \mathbb{R}_+$, depending on the communication action $\boldsymbol{M}_h$.

At each timestep h, each agent i needs to choose a communication action $\boldsymbol{M}_{i,h}$ and a control action $\boldsymbol{U}_{i,h}$, based on her communication strategy $g_{i,h}^m$ and control strategy $g_{i,h}^a$, respectively. We may consider two different classes of communication strategies: 1) the *open-loop* ones where $\mathcal{G}_{i,h}^m := \{g_{i,h}^m : \emptyset \to \mathcal{M}_{i,h}\}$ and $\boldsymbol{M}_{i,h}$ is thus pre-selected. In this case, we consider the control strategies $g_{i,h}^a \in \mathcal{G}_{i,h}^a := \{g_{i,h}^a : \mathcal{I}_{i,h^+} \to \mathcal{U}_i\}$; 2) the *closed-loop* ones $\mathcal{G}_{i,h}^m := \{g_{i,h}^m : \mathcal{I}_{i,h^-} \times \mathcal{M}_{1:h-1} \to \mathcal{M}_{i,h}\}$ where $\boldsymbol{M}_{i,h}$ is thus chosen based on $\boldsymbol{I}_{i,h^-}$ and $\boldsymbol{M}_{1:h-1}$. Note that as $\boldsymbol{M}_{i,h}$ (dictating what to share) will be known given $\boldsymbol{Z}_{i,h}^a$ (what has been shared), $\boldsymbol{M}_{i,h}$ is thus also assumed to be known and used by the agents in this closed-loop setting, following [9], [10], [11]. In this case, we consider the control strategies $g_{i,h}^a \in \mathcal{G}_{i,h}^a := \{g_{i,h}^a : \mathcal{I}_{i,h^+} \times \mathcal{M}_{1:h} \to \mathcal{U}_i\}$ for generality. We will focus on the open-loop setting throughout, and extend to the closed-loop setting in §V. We use $g_h^m := (g_{1,h}^m, \cdots, g_{n,h}^m)$ and $g_h^a := (g_{1,h}^a, \cdots, g_{n,h}^a)$ to denote the joint communication and control strategies, respectively. We denote by $\mathcal{G}_h^a, \mathcal{G}_h^m$ the spaces of these joint strategies.

The total cost (objective) of all the agents is defined as

$$J_{\mathcal{D}}(g_{1:H}^m, g_{1:H}^a) := \mathbb{E}\left[\left(\sum_{h=1}^{H} c_h + \kappa_h\right) + c_{H+1} \,\middle|\, g_{1:H}^m, g_{1:H}^a\right],$$

which is the expected accumulated sum of the control and communication costs over H timesteps. With this objective, we define the solution concept of *team-optimum* below.

Definition II.2: Given a JCCO problem $\mathcal{D}$, a strategy $(g_{1:H}^{a,*}, g_{1:H}^{m,*})$ is *team-optimal* if $\forall g_{1:H}^m \in \mathcal{G}_{1:H}^m, g_{1:H}^a \in \mathcal{G}_{1:H}^a$,

$$J_{\mathcal{D}}(g_{1:H}^m, g_{1:H}^a) \geq J_{\mathcal{D}}(g_{1:H}^{m,*}, g_{1:H}^{a,*}),$$

with $g_{1:H}^{m,*}, g_{1:H}^{a,*}$ being referred to as the *optimal communication* and *control* strategy, respectively.

Note that the team-optimal strategies depend on the (communication) strategy space being used (i.e., open- v.s closed-loop), which will be clear later from the context.

C. Information structures

Information structure is a well-studied notion in decentralized stochastic control that captures *who knows what and when* [18], [19]. For any JCCO problem $\mathcal{D}$, as the additional sharing via communication will also affect the IS and is not determined beforehand, when we discuss the IS of $\mathcal{D}$, we will refer to that of the problem *without additional sharing*, which is essentially a decentralized LQG control problem (with potential baseline information sharing), denoted by $\check{\mathcal{D}}$. We refer to $\check{\mathcal{D}}$ as the *decentralized LQG control problem induced by* $\mathcal{D}$, and define the IS of $\mathcal{D}$ as that of $\check{\mathcal{D}}$.

Partially nested [12] ISs are an important subclass of ISs well studied in decentralized stochastic control. We extend such a categorization to JCCO. Formally, we call a JCCO problem $\mathcal{D}$ PN if the induced decentralized LQG control problem $\check{\mathcal{D}}$ is PN. Namely, for any $i_1, i_2 \in [n], h_1 < h_2$, if there is no additional sharing and $\boldsymbol{U}_{i_1,h_1}$ influences $\boldsymbol{I}_{i_2,h_2^-}$, then agent (i_2, h_2) can access $\boldsymbol{I}_{i_1,h_1^-}$, i.e., $\boldsymbol{I}_{i_1,h_1^-} \subseteq \boldsymbol{I}_{i_2,h_2^-}$.

III. Linearity and Structural Conditions for JCCO with Open-Loop Communication Strategies

We now focus on the setting with *open-loop* communication strategies. Given any such fixed $g_{1:H}^m$ (and thus pre-selected $M_{1:H}$), the problem becomes a decentralized LQG control one. However, it is known that the optimal control strategy of a decentralized LQG control problem may be *nonlinear* in general [20], making it computationally challenging. Fortunately, with partial-nested ISs, the optimal control strategies can be linear [12]. Hence, it is natural to ask: when it comes to JCCO (with open-loop $g_{1:H}^m$), when would the optimal control strategy $g_{i,h}^a$ still be *linear*, i.e., a linear function of the available information $\boldsymbol{I}_{i,h^+}$, for all $i \in [n], h \in [H]$?

To answer this question, firstly, if the IS of JCCO (i.e., the IS of baseline sharing) is not even PN, then in general it is hard for the IS of the decentralized LQG to be PN *after* the optimal additional sharing, especially with communication costs. This may thus further break the linearity of the optimal control strategies. To formalize this intuition, we introduce the following result, whose proof, together with other omitted proofs for this section, can be found in §B.

Lemma III.1: There exists a JCCO problem $\mathcal{D}$ with open-loop communication strategies that is not PN but satisfies Assumptions II.1, III.2, and III.4, such that the optimal control strategy is nonlinear.

In the proof, we can construct a $\mathcal{D}$ with high enough communication cost, but satisfies Assumptions II.1, III.2, and III.4. Then, the optimal communication strategy leads to no additional sharing, and $\mathcal{D}$ reduces to the Witsenhausen's counterexample [20], whose optimal strategy is nonlinear.

Due to Lemma III.1, we focus on PN JCCO problems, in order to obtain linearity of the control strategies and thus computational tractability. However, even when the baseline sharing is PN, the additional sharing may *break* the PN IS, and thus cause computational hardness again. Specifically, suppose agent i_1's control action $\boldsymbol{U}_{i_1,h_1}$ does not influence another agent i_2's information $\boldsymbol{I}_{i_2,h_2}$ in baseline sharing, but her control action $\boldsymbol{U}_{i_1,h_1}$ is shared with others via additional sharing. Then, $\boldsymbol{U}_{i_1,h_1}$ starts to influence $\boldsymbol{I}_{i_2,h_2}$ and the PN IS may thus break. Here, the *non-influence* of $\boldsymbol{U}_{i_1,h_1}$ in baseline sharing may come from two cases: 1) $\boldsymbol{U}_{i_1,h_1}$ does not influence the underlying *state* $\boldsymbol{X}_{h_1+1}$; or 2) $\boldsymbol{U}_{i_1,h_1}$ does influence the underlying state $\boldsymbol{X}_{h_1+1}$, but the effect is *not*

observed by other agents. To avoid the impact of such non-influential actions, we make the following two assumptions, followed by justifications of their necessity regarding the *existence/linearity* of the optimal control strategies.

For case 1) above, we make the following assumption that such useless actions will *not* be used or shared later. In fact, for the linear systems considered in (II.1), a useless control action will not appear as a random variable in the system evolution, and it is natural to disregard it in the available information of later agents. Moreover, such an assumption has been implicitly made in the common examples in decentralized stochastic control [13], [16].

Assumption III.2: For any $i \in [n], h \in [H]$, if $B_{i,h} = \mathbf{0}$, then $\boldsymbol{U}_{i,h} \notin \boldsymbol{I}_{j,(h')^-}, \boldsymbol{U}_{i,h} \notin \boldsymbol{I}_{j,(h')^+}, \forall h' > h, j \in [n]$.

The significance of Assumption III.2 in JCCO is reflected in two aspects: First, it helps to preserve the PN IS (as to be shown formally in Theorem III.6); second, without it, the optimal control strategy may either not exist or be nonlinear, as shown in the following lemma.

Lemma III.3: There exist PN JCCO problems $\mathcal{D}_1, \mathcal{D}_2$ with open-loop communication strategies that satisfy Assumptions II.1, III.4 but not Assumption III.2, such that the optimal control strategy of $\mathcal{D}_1$ is nonlinear, and the team-optimal strategy of $\mathcal{D}_2$ does not exist.

In the proof, we can construct JCCO problems where there is no communication cost for sharing some specific *useless action*. Then, the optimal communication strategy will share such an action, which will be designed to encode (and thus implicitly share) some observation, as a *compression* of the observation. The nonlinearity result then stems from the fact that there is no *linear* bijection mapping between a high-dimensional vector space and a compressed lower-dimensional one. The non-existence result originates from the fact that there is no optimal compression function that minimizes the second moment of the compressed information (i.e., the useless action).

For case 2) above, the PN IS may break due to the *non-informativeness* of agents' observations. We thus make the following assumption.

Assumption III.4: For any $h \in [H-1], i \in [n]$, $\operatorname{rank}(E_{-i,h+1}B_{i,h}) = \operatorname{rank}(B_{i,h})$, where $E_{-i,h+1} := \begin{bmatrix} E_{1,h+1}^\top & \cdots & E_{i-1,h+1}^\top & E_{i+1,h+1}^\top & \cdots & E_{n,h+1}^\top \end{bmatrix}^\top$.

Assumption III.4 means that for any $i \in [n], h \in [H-1]$, any effect on the state $\boldsymbol{X}_{h+1}$ caused by the control action $\boldsymbol{U}_{i,h}$ will be observed by other agents. Note that a similar assumption was also made in the discrete setting [11], but this Assumption III.4 is weaker in that it does not require other agents to observe the *whole state* $\boldsymbol{X}_h$, i.e., $E_{-i,h+1}$ being full-rank, but only requires that other agents can jointly observe the *part* of state $\boldsymbol{X}_h$ *affected by* $\boldsymbol{U}_{i,h}$. For instance, if $B_{i,h} = \mathbf{0}$, then there is no requirement for the matrix $E_{-i,h+1}$. A property under Assumption III.4 is that, for any $h \in [H-1], i \in [n]$, if $\boldsymbol{U}_{i,h}$ influences $\boldsymbol{X}_{h+1}$, i.e., $B_{i,h} \neq \mathbf{0}$, then there exists $j \neq i$ such that $\boldsymbol{U}_{i,h}$ influences $\boldsymbol{Y}_{j,h+1}$. As shown below, Assumption III.4 is necessary to ensure the linearity of the optimal control strategy of $\mathcal{D}$.

Lemma III.5: There exists a PN JCCO problem $\mathcal{D}$ with open-loop communication strategies that satisfies Assumptions II.1, III.2 but not Assumption III.4, such that the optimal control strategy of $\mathcal{D}$ is nonlinear.

In the proof, we construct a JCCO $\mathcal{D}$ based on the Witsenhausen's counterexample, but change the observation of the second agent to be null, which can only happen without Assumption III.4, then $\mathcal{D}$ has PN IS. However, we allow such an agent to obtain such an observation via additional sharing, making $\mathcal{D}$ reduce to the Witsenhausen's counterexample, where the optimal control strategy is nonlinear.

We have thus justified the importance of the above assumptions and the partial nestedness of baseline sharing for JCCO problems, by showing that missing any one of them (while keeping others) may cause the nonlinearity (or even non-existence) of the optimal control strategy.

On the other hand, for any fixed open-loop communication strategy $g^m_{1:H} \in \mathcal{G}^m_{1:H}$, we denote by $\mathcal{D}(g^m_{1:H})$ the subproblem of finding the optimal control strategy with respect to $g^m_{1:H}$, where we add $\bar{\ }$ to the notation in $\mathcal{D}(g^m_{1:H})$ to distinguish them from that in the original JCCO problem. We remove $+$ in the timestep index for all the notation in $\mathcal{D}(g^m_{1:H})$ for convenience, as the IS is now *fixed* and only the control problem will be analyzed. Then, we can show that if $\mathcal{D}$ satisfies all the assumptions above, the optimal control strategy of $\mathcal{D}$ is linear, by showing that under any fixed open-loop $g^m_{1:H} \in \mathcal{G}^m_{1:H}$, the subproblem $\mathcal{D}(g^m_{1:H})$ is a *decentralized LQG* problem with PN IS (see §A for a formal definition).

Theorem III.6: Consider a PN JCCO problem $\mathcal{D}$ that satisfies Assumptions II.1, III.2, and III.4. For any open-loop communication strategy $g^m_{1:H} \in \mathcal{G}^m_{1:H}$, the subproblem $\mathcal{D}(g^m_{1:H})$ is a decentralized LQG control problem with a PN IS, and its information evolution rules follow:

$$\bar{\boldsymbol{C}}_h = \bar{\boldsymbol{C}}_{h-1} \cup \bar{\boldsymbol{Z}}_h, \bar{\boldsymbol{Z}}_h = \bar{\chi}_h(\bar{\boldsymbol{P}}_{h-1}, \bar{\boldsymbol{U}}_{h-1}, \bar{\boldsymbol{Y}}_h),\ \bar{\boldsymbol{Y}}_{i,h} \in \bar{\boldsymbol{I}}_{i,h},$$
$$\bar{\boldsymbol{P}}_{i,h} = \bar{\zeta}_{i,h}(\bar{\boldsymbol{P}}_{i,h-1}, \bar{\boldsymbol{U}}_{i,h-1}, \bar{\boldsymbol{Y}}_{i,h}), \quad \bar{\boldsymbol{I}}_{i,h-1} \subseteq \bar{\boldsymbol{I}}_{i,h}, \forall i \in [n],$$

for some projection functions $\bar{\chi}_h, \{\bar{\zeta}_{i,h}\}_{i\in[n]}$. Furthermore, there exists an optimal strategy $(g^{m,*}_{1:H}, g^{a,*}_{1:H})$ of $\mathcal{D}$ with open-loop communication strategies such that $g^{a,*}_{1:H}$ is linear.

Theorem III.6 provides a way to solve $\mathcal{D}$ by finding the optimal control strategy $g^{a,*}_{1:H}$ with respect to any fixed open-loop communication strategy $g^m_{1:H}$ by solving $\mathcal{D}(g^m_{1:H})$, and then optimizing over the open-loop communication strategies $g^m_{1:H}$. We will thus investigate how to solve $\mathcal{D}(g^m_{1:H})$ next.

IV. Dynamic Programming for Decentralized LQG with Partially Nested IS

Since $\mathcal{D}(g^m_{1:H})$ is PN under the assumptions in Theorem III.6, it now suffices to develop an algorithm that can solve the PN decentralized LQG control problem. For notational convenience, in this section, we will omit the superscript a for the control strategies $\bar{g}^a_{1:H}$ in this $\mathcal{D}(g^m_{1:H})$. The algorithm to be introduced can be viewed as a dynamic-programming-based algorithm for solving a class of PN decentralized LQG with *output* feedback, which may be of independent interest. Due to space constraints, detailed proofs of results in this section are deferred to §C.

Specifically, under PN IS, the known dynamic-programming-based approaches and sufficient statistics in [13], [15] do not apply to our setting, as they focused on the special settings with *factorized* states, and with either *state-feedback* [13] or *multi-tree* coupling and communication graphs [15].

On the other hand, under the common-information-based framework [7], [14] showed that if agents are restricted to use *linear* control strategies, then the decentralized control problem can be reformulated as a *centralized LQG* one when *fixing* the private-information component of the strategies. However, it is not clear how to further *optimize* the private-information component, which induces a non-convex optimization problem in general [14]. Furthermore, [8] showed that, if additionally the CIB-beliefs from [7] are *strategy independent* [8, Section 2.4], then the CIB-beliefs can be further simplified as the conditional mean of the state and private information of *finite dimensions* in this LQ setting[1].

Inspired by [8], and to address the non-convexity issue in [14], we exploit the PN IS to obtain a finite-dimensional, conditional-mean-based sufficient statistic, by establishing a new connection between the PN IS and the SI-CIB condition.

To this end, we adapt our technique in [11] to this linear-quadratic setting. First, we expand the PN IS into a *strictly* PN one, where the *actions* of agents who affect some other agent will also be known by the affected agent (in addition to their *information*). More formally, we construct a new problem $\widetilde{\mathcal{D}}(g^m_{1:H})$, whose elements are almost identical to those in $\mathcal{D}(g^m_{1:H})$, except that the agents' information is now expanded as: for any $h \in [H]$

$$\begin{aligned}\widetilde{\boldsymbol{C}}_h &= \bar{\boldsymbol{C}}_h \cup \{\bar{\boldsymbol{U}}_{i,t} \,|\, i \in [n], t < h, \bar{\boldsymbol{I}}_{i,t} \subseteq \bar{\boldsymbol{C}}_h, \bar{B}_{i,t} \neq \boldsymbol{0}\} \\ \forall i \in [n], \widetilde{\boldsymbol{P}}_{i,h} &= \bar{\boldsymbol{P}}_{i,h} \backslash \{\bar{\boldsymbol{U}}_{i,t} \,|\, i \in [n], t < h, \bar{\boldsymbol{I}}_{i,t} \subseteq \bar{\boldsymbol{C}}_h\},\end{aligned} \tag{IV.1}$$

where we add $\widetilde{\ }$ to the notation in $\widetilde{\mathcal{D}}(g^m_{1:H})$, and the incremental common information is correspondingly defined as: $\widetilde{\boldsymbol{Z}}_h := \widetilde{\boldsymbol{C}}_h \backslash \widetilde{\boldsymbol{C}}_{h-1}, \forall h \in [H]$. Through such an expansion, we can leverage the new problem $\widetilde{\mathcal{D}}(g^m_{1:H})$ to solve the original $\mathcal{D}(g^m_{1:H})$, as formalized in the following lemma.

Lemma IV.1: Let $\mathcal{D}$ be PN and satisfy Assumptions II.1, III.2, and III.4, and for any fixed open-loop $g^m_{1:H} \in \mathcal{G}^m_{1:H}$, let $\mathcal{D}(g^m_{1:H})$ and $\widetilde{\mathcal{D}}(g^m_{1:H})$ be the decentralized LQG control problems constructed above. Then, there exists a function φ such that for any linear optimal control strategy $\widetilde{g}^*_{1:H}$ of problem $\widetilde{\mathcal{D}}(g^m_{1:H})$, $\bar{g}^*_{1:H} = \varphi(\widetilde{g}^*_{1:H}, \mathcal{D}(g^m_{1:H}))$ is a linear optimal control strategy of $\mathcal{D}(g^m_{1:H})$, and $J_{\mathcal{D}(g^m_{1:H})}(\bar{g}^*_{1:H}) = J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}^*_{1:H})$, where $J_{\mathcal{D}(g^m_{1:H})}(\bar{g}^*_{1:H})$ is the expected accumulated cost of $\mathcal{D}(g^m_{1:H})$ under the strategy $\bar{g}^*_{1:H}$, and similarly for $J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}^*_{1:H})$ (cf. §A for formal definitions).

Now, we introduce some additional notation in $\widetilde{\mathcal{D}}(g^m_{1:H})$. For any $h \in [H+1]$, we define $\widetilde{\boldsymbol{S}}_h := \begin{bmatrix}\widetilde{\boldsymbol{X}}_h^\top & \widetilde{\boldsymbol{P}}_{1,h}^\top & \cdots & \widetilde{\boldsymbol{P}}_{n,h}^\top\end{bmatrix}^\top$ and use $\widetilde{\mathcal{S}}_h := \widetilde{\mathcal{X}} \times \widetilde{\mathcal{P}}_{1,h} \times \cdots \times \widetilde{\mathcal{P}}_{n,h}$ to denote the space of $\widetilde{\boldsymbol{S}}_h$. Then, we use $\widetilde{\boldsymbol{B}}_h \in \mathcal{P}(\widetilde{\mathcal{S}}_h)$ with $\widetilde{\boldsymbol{B}}_h(\mathrm{d}\widetilde{\boldsymbol{S}}_h) := \mathbb{P}^{\widetilde{\mathcal{D}}(g^m_{1:H})}(\mathrm{d}\widetilde{\boldsymbol{S}}_h \,|\, \widetilde{\boldsymbol{C}}_h, \widetilde{g}_{1:h-1})$ to denote the conditional probability measure of $\widetilde{\boldsymbol{S}}_h$ given the common information $\widetilde{\boldsymbol{C}}_h$ and the past strategies $\widetilde{g}_{1:h-1}$. Hence, we have $\widetilde{\boldsymbol{B}}_h = \Pi_h(\widetilde{\boldsymbol{C}}_h, \widetilde{g}_{1:h-1})$ for some functional Π_h. The benefit of solving $\widetilde{\mathcal{D}}(g^m_{1:H})$ instead of $\mathcal{D}(g^m_{1:H})$ is that $\widetilde{\mathcal{D}}(g^m_{1:H})$ can satisfy the SI-CIB condition under given assumptions, as shown below.

Theorem IV.2: If $\mathcal{D}$ is PN and satisfies Assumptions II.1, III.2, and III.4, then for any fixed open-loop $g^m_{1:H} \in \mathcal{G}^m_{1:H}$, $\widetilde{\mathcal{D}}(g^m_{1:H})$ constructed as above is (strictly) PN and satisfies the *SI-CIB condition*: for any $h \in [H]$, any realization $\widetilde{C}_h \in \widetilde{\mathcal{C}}_h$, and any two control strategies $\widetilde{g}_{1:h-1}, \widetilde{g}'_{1:h-1}$ that can reach $\widetilde{C}_h$, it holds that $\Pi_h(\widetilde{C}_h, \widetilde{g}_{1:h-1}) = \Pi_h(\widetilde{C}_h, \widetilde{g}'_{1:h-1})$. Meanwhile, $\widetilde{\mathcal{D}}(g^m_{1:H})$ has the information evolution rules as follows: for any $i \in [n]$,

$$\begin{aligned}&\widetilde{\boldsymbol{C}}_h = \widetilde{\boldsymbol{C}}_{h-1} \cup \widetilde{\boldsymbol{Z}}_h, \widetilde{\boldsymbol{Z}}_h = \widetilde{\chi}_h(\widetilde{\boldsymbol{P}}_{h-1}, \widetilde{\boldsymbol{U}}_{h-1}, \widetilde{\boldsymbol{Y}}_h), \\ &\widetilde{\boldsymbol{P}}_{i,h} = \widetilde{\zeta}_{i,h}(\widetilde{\boldsymbol{P}}_{i,h-1}, \widetilde{\boldsymbol{U}}_{i,h-1}, \widetilde{\boldsymbol{Y}}_{i,h}), \widetilde{\boldsymbol{Y}}_{i,h} \in \widetilde{\boldsymbol{I}}_{i,h}, \widetilde{\boldsymbol{I}}_{i,h-1} \subseteq \widetilde{\boldsymbol{I}}_{i,h},\end{aligned} \tag{IV.2}$$

for some projection functions $\widetilde{\chi}_h, \{\widetilde{\zeta}_{i,h}\}_{i \in [n]}$.

Furthermore, we define the *prescription* of each agent i at timestep h as $\widetilde{\gamma}_{i,h} : \widetilde{\mathcal{P}}_{i,h} \to \widetilde{\mathcal{U}}_i$. We denote by $\widetilde{\gamma}_{i,h} = (\widetilde{\gamma}_{1,h}, \cdots, \widetilde{\gamma}_{n,h})$ the joint prescriptions of all the agents, and by $\widetilde{\Gamma}_{i,h}$ and $\widetilde{\Gamma}_h$ the spaces of agent i's and the joint prescriptions at timestep h, respectively. For convenience, we write $\widetilde{\gamma}_h(\widetilde{\boldsymbol{P}}_h) = \begin{bmatrix}\widetilde{\gamma}_{1,h}(\widetilde{\boldsymbol{P}}_{1,h})^\top & \cdots & \widetilde{\gamma}_{n,h}(\widetilde{\boldsymbol{P}}_{n,h})^\top\end{bmatrix}^\top$.

Under the SI-CIB condition, [8] showed that the belief $\widetilde{\boldsymbol{B}}_h$ in $\widetilde{\mathcal{D}}(g^m_{1:H})$ is a Gaussian distribution with mean $\widetilde{\boldsymbol{\Theta}}_h = \begin{bmatrix}\mathbb{E}[\widetilde{\boldsymbol{X}}_h \,|\, \widetilde{\boldsymbol{C}}_h]^\top & \mathbb{E}[\widetilde{\boldsymbol{P}}_{1,h} \,|\, \widetilde{\boldsymbol{C}}_h]^\top & \cdots & \mathbb{E}[\widetilde{\boldsymbol{P}}_{n,h} \,|\, \widetilde{\boldsymbol{C}}_h]^\top\end{bmatrix}^\top$ and covariance $\widetilde{\Sigma}_h$ that evolve over time, as formalized below.

Lemma IV.3 ([8] Lemmas 2.2, 2.3): Consider a decentralized LQG problem $\widetilde{\mathcal{D}}(g^m_{1:H})$ that has information evolution rules in Eq. (IV.2) and satisfies the SI-CIB condition, then the CIB belief $\widetilde{\boldsymbol{B}}_h$ admits a Gaussian distribution $\mathcal{N}(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\Sigma}_h)$, and the conditional mean $\widetilde{\boldsymbol{\Theta}}_h$ and the conditional covariance $\widetilde{\Sigma}_h$ of $\widetilde{\boldsymbol{B}}_h$ evolve as: for any $h \in [2:H]$

$$\widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Psi}^1_h(\widetilde{\boldsymbol{\Theta}}_{h-1}, \widetilde{\boldsymbol{Z}}_h), \qquad \widetilde{\Sigma}_h = \widetilde{\Psi}^2_h(\widetilde{\Sigma}_{h-1}),$$

where $\widetilde{\Psi}^1_h$ is a fixed linear transformation and $\widetilde{\Psi}^2_h$ is a fixed transformation, which do not depend on the strategies $\widetilde{g}_{1:h-1}$.

According to Lemma IV.3, we know that $\widetilde{\boldsymbol{\Theta}}_h$ only depends on the $\{\widetilde{\boldsymbol{Z}}_t\}_{t=1}^h$. Hence, there exists a fixed linear transformation $\widetilde{\Psi}^3_h$ such that $\widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Psi}^3_h(\widetilde{\boldsymbol{C}}_h)$. Then, following [8], we can construct a new problem $\widetilde{\mathcal{D}}^\ddagger(g^m_{1:H})$: At timestep $h \in [H]$, the state of $\widetilde{\mathcal{D}}^\ddagger(g^m_{1:H})$ is defined as $\widetilde{\boldsymbol{\Theta}}_h \in \widetilde{\mathcal{S}}_h$ and the action is defined as $\widetilde{\gamma}_h \in \widetilde{\Gamma}_h$. The cost at timestep $h \in [H]$ is a function of the state $\widetilde{\boldsymbol{\Theta}}_h$ and the action $\widetilde{\gamma}_h$, given by:

$$c^\ddagger_h(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\gamma}_h) := \mathbb{E}_{\substack{[\widetilde{\boldsymbol{X}}_h^\top \ \widetilde{\boldsymbol{P}}_h^\top]^\top \\ \sim \mathcal{N}(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\Sigma}_h)}}\left[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\gamma}_h(\widetilde{\boldsymbol{P}}_h)^\top \widetilde{Q}^2_h \widetilde{\gamma}_h(\widetilde{\boldsymbol{P}}_h)^\top\right],$$

and the cost at timestep $H+1$ is $c^\ddagger_{H+1}(\widetilde{\boldsymbol{\Theta}}_{H+1}) := \mathbb{E}_{[\widetilde{\boldsymbol{X}}_{H+1}^\top, \widetilde{\boldsymbol{P}}_{H+1}^\top]^\top \sim \mathcal{N}(\widetilde{\boldsymbol{\Theta}}_{H+1}, \widetilde{\Sigma}_{H+1})}\left[\widetilde{\boldsymbol{X}}_{H+1}^\top \widetilde{Q}_{H+1} \widetilde{\boldsymbol{X}}_{H+1}\right]$. The admissible strategy at timestep h in $\widetilde{\mathcal{D}}^\ddagger(g^m_{1:H})$ is denoted as $g^\ddagger_h : \widetilde{\mathcal{C}}_h \to \widetilde{\Gamma}_h$, and we use $\mathcal{G}^\ddagger_h$ to denote its space. The objective of $\widetilde{\mathcal{D}}^\ddagger(g^m_{1:H})$ is then defined as:

[1]Note that [8] focused on a *game* setting with strategic agents, instead of the *team* setting in [14], [13] and the present paper.

$$J_{\widetilde{\mathcal{D}}^{\ddagger}(g^m_{1:H})}(g^{\ddagger}_{1:H}) := \mathbb{E}[c^{\ddagger}_{H+1}(\widetilde{\boldsymbol{\Theta}}_{H+1}) + \sum_{h=1}^{H} c^{\ddagger}_h(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\gamma}_h)].$$

As shown in the following theorem, the constructed problem $\widetilde{\mathcal{D}}^{\ddagger}(g^m_{1:H})$ is a Markov decision problem, and the optimal strategy of $\widetilde{\mathcal{D}}^{\ddagger}(g^m_{1:H})$ can be used to compute the team-optimal strategy of $\widetilde{\mathcal{D}}(g^m_{1:H})$, and vice versa.

Theorem IV.4: Let $\widetilde{\mathcal{D}}^{\ddagger}(g^m_{1:H})$ be the problem constructed from $\widetilde{\mathcal{D}}(g^m_{1:H})$ satisfying the SI-CIB condition. Then, $\widetilde{\mathcal{D}}^{\ddagger}(g^m_{1:H})$ is a Markov decision problem with horizon H, state $\{\boldsymbol{\Theta}_h\}_{h\in[H+1]}$, control action $\{\widetilde{\gamma}_h\}_{h\in[H]}$, and cost function $\{c^{\ddagger}_h\}_{h\in[H+1]}$. Moreover, there exist bijection mappings $\varsigma_h : \widetilde{\mathcal{G}}_h \to \mathcal{G}^{\ddagger}_h$ for all $h \in [H]$ defined as: for any strategy $\widetilde{g}_h \in \widetilde{\mathcal{G}}_h$ and realization of common information $\widetilde{C}_h \in \widetilde{\mathcal{C}}_h, \varsigma_h(\widetilde{g}_h)(\widetilde{C}_h)(\cdot) = \widetilde{g}_h(\widetilde{C}_h, \cdot)$. Also, such bijection mappings $\{\varsigma_h\}_{h\in[H]}$ satisfy that for any strategy $\widetilde{g}_{1:H} \in \widetilde{\mathcal{G}}_{1:H}, J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}_{1:H}) = J_{\widetilde{\mathcal{D}}^{\ddagger}(g^m_{1:H})}(g^{\ddagger}_{1:H})$, where $g^{\ddagger}_h = \varsigma_h(\widetilde{g}_h), \forall h \in [H]$.

By Theorem IV.4, if $\widetilde{\mathcal{D}}^{\ddagger}(g^m_{1:H})$ admits an optimal strategy, then there must exist a *Markovian* optimal strategy $g^{\ddagger,*}_{1:H}$ such that $g^{\ddagger,*}_h$ only depends on $\widetilde{\boldsymbol{\Theta}}_h$ for each $h \in [H]$. In other words, for any $h \in [H]$, we can write $g^{\ddagger,*}_h$ as $g^{\ddagger,*}_h(\widetilde{\boldsymbol{C}}_h) = \mathscr{G}^*_h(\widetilde{\Psi}^3_h(\widetilde{\boldsymbol{C}}_h))$ for some function $\mathscr{G}^*_h$. Furthermore, if we define the strategy $\widetilde{g}^*_h = \varsigma^{-1}_h(g^{\ddagger,*}_h), \forall h \in [H]$, then $\widetilde{g}^*_{1:H}$ is an optimal strategy of $\widetilde{\mathcal{D}}(g^m_{1:H})$. Therefore, there is no loss of optimality in restricting attention to $\widetilde{g}_{i,h}$ that chooses action $\widetilde{\boldsymbol{U}}_h$ only depending on $\widetilde{\boldsymbol{\Theta}}_h$ and $\widetilde{\boldsymbol{P}}_{i,h}$ for any $h \in [H], i \in [n]$ in solving the problem $\widetilde{\mathcal{D}}(g^m_{1:H})$. We categorize this type of strategy as *common-information-based Markovian (CIB-Markovian) strategy*. Moreover, we can define the associated value functions as: $\forall h \in [H]$

$$\widetilde{V}_h(\widetilde{\boldsymbol{\Theta}}_h) := \min_{\widetilde{g}_{h:H}} \mathbb{E}\left[\sum_{t=h}^{H} (\widetilde{\boldsymbol{X}}^\top_t \widetilde{Q}^1_t \widetilde{\boldsymbol{X}}_t + \widetilde{\boldsymbol{U}}^\top_t \widetilde{Q}^2_t \widetilde{\boldsymbol{U}}_t) \,\middle|\, \widetilde{\boldsymbol{\Theta}}_h, \widetilde{g}_{h:H}\right],$$

and $\widetilde{V}_{H+1}(\widetilde{\boldsymbol{\Theta}}_{H+1}) := \mathbb{E}\big[\widetilde{\boldsymbol{X}}^\top_{H+1}\widetilde{Q}_{H+1}\widetilde{\boldsymbol{X}}_{H+1}\big]$, which satisfy the following Bellman Equations: $\forall h \in [H]$, $\widetilde{V}_h(\widetilde{\boldsymbol{\Theta}}_h) = \min_{\widetilde{\gamma}_h\in\widetilde{\Gamma}_h} \mathbb{E}[\widetilde{\boldsymbol{X}}^\top_h \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}^\top_h \widetilde{Q}^2_h \widetilde{\boldsymbol{U}}_h + \widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) \,|\, \widetilde{\boldsymbol{\Theta}}_h, \widetilde{\gamma}_h]$.

Then, we can derive an approach to obtain the optimal strategy $\widetilde{g}^*_{1:H}$ by dynamic programming: for all $h \in [H], \widetilde{C}_h \in \widetilde{\mathcal{C}}_h$, $\widetilde{g}^*_h(\widetilde{C}_h, \cdot) \in \operatorname{argmin}_{\widetilde{\gamma}_h\in\widetilde{\Gamma}_h} \mathbb{E}[\widetilde{\boldsymbol{X}}^\top_h \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}^\top_h \widetilde{Q}^2_h \widetilde{\boldsymbol{U}}_h + \widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) \,|\, \widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Psi}^3_h(\widetilde{C}_h), \widetilde{\gamma}_h]$. However, such an approach requires enumerating all possible $\widetilde{C}_h$ (or $\widetilde{\Theta}_h$), which is uncountably many, making it computationally intractable, if no closed-form solution can be established.

For computational tractability, we leverage the linearity property of the optimal strategy. Although it is known that there exists a *linear* optimal strategy under PN IS [12], and if it exists, the optimal strategy can be *CIB-Markovian* under the SI-CIB condition (cf. Theorem IV.4), the existence of an optimal strategy with both properties remains unclear. We show the existence of such an optimal strategy below.

Lemma IV.5: Let $\widetilde{\mathcal{D}}(g^m_{1:H})$ be a decentralized LQG control problem with PN IS that satisfies the information evolution rules in Eq. (IV.2) and the SI-CIB condition, then there exists an optimal linear strategy $\widetilde{g}^*_{1:H}$, and matrices $\{\widetilde{\mathcal{E}}^*_{i,h}\}_{i\in[n],h\in[H]}, \{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]}$ such that for each $\widetilde{\Theta}_h \in \widetilde{\mathcal{S}}_h$, $\widetilde{\gamma}^*_{i,h}(\cdot) = \widetilde{\mathcal{E}}^*_{i,h}\widetilde{\Theta}_h + \widetilde{\mathcal{F}}^*_{i,h}\cdot$, $\widetilde{\gamma}^*_h \in \operatorname{argmin}_{\widetilde{\gamma}_h\in\widetilde{\Gamma}_h} \mathbb{E}[\widetilde{\boldsymbol{X}}^\top_h \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}^\top_h \widetilde{Q}^2_h \widetilde{\boldsymbol{U}}_h + \widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) \,|\, \widetilde{\gamma}_h, \widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Theta}_h]$, and $\widetilde{g}^*_{i,h}(\widetilde{C}_h, \widetilde{P}_{i,h}) = \widetilde{\mathcal{E}}^*_{i,h}\widetilde{\Psi}^3_h(\widetilde{C}_h) + \widetilde{\mathcal{F}}^*_{i,h}\widetilde{P}_{i,h}$ for any $i \in [n], h \in [H], \widetilde{C}_h \in \widetilde{\mathcal{C}}_h, \widetilde{P}_{i,h} \in \widetilde{\mathcal{P}}_{i,h}$.

Lemma IV.5 shows that it suffices to compute the optimal strategy that is linear in $\widetilde{\boldsymbol{\Theta}}_h$ and $\widetilde{\boldsymbol{P}}_{i,h}$. Then, for any $i \in [n], h \in [H]$, we can parameterize the strategy $\widetilde{g}_{i,h}$ by two matrices $\widetilde{\mathcal{E}}_{i,h}, \widetilde{\mathcal{F}}_{i,h}$, where $\widetilde{g}_{i,h}(\widetilde{\boldsymbol{C}}_h, \widetilde{\boldsymbol{P}}_{i,h}) = \widetilde{\mathcal{E}}_{i,h}\widetilde{\Psi}^3_h(\widetilde{\boldsymbol{C}}_h) + \widetilde{\mathcal{F}}_{i,h}\widetilde{\boldsymbol{P}}_{i,h}$. Define $\widetilde{\mathcal{E}}^*_h := \begin{bmatrix}(\widetilde{\mathcal{E}}^*_{1,h})^\top & \cdots & (\widetilde{\mathcal{E}}^*_{n,h})^\top\end{bmatrix}^\top$ and $\widetilde{\mathcal{F}}^*_h := \operatorname{diag}(\widetilde{\mathcal{F}}^*_{1,h}, \cdots, \widetilde{\mathcal{F}}^*_{n,h})$, then we can find the optimal control strategy $\widetilde{g}^*_h$ by solving for the optimal matrices:

$$\begin{aligned}(\widetilde{\mathcal{E}}^*_h, \widetilde{\mathcal{F}}^*_h) \in \operatorname*{argmin}_{\widetilde{\mathcal{E}}_h, \widetilde{\mathcal{F}}_h} \mathbb{E}[&\widetilde{\boldsymbol{X}}^\top_h \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}^\top_h \widetilde{Q}^2_h \widetilde{\boldsymbol{U}}_h \\ &+ \widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) \,|\, \widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Theta}_h, \widetilde{\mathcal{E}}_h, \widetilde{\mathcal{F}}_h],\end{aligned} \tag{IV.3}$$

for any $\widetilde{\Theta}_h \in \widetilde{\mathcal{S}}_h$. As shown in the following theorem, such optimal matrices can be computed through a set of Riccati Equations with better computational efficiency.

Theorem IV.6: Suppose $\widetilde{D}(g^m_{1:H})$ is a decentralized LQG control problem with PN IS that satisfies the information evolution rules in Eq. (IV.2) and the SI-CIB condition. Then, for any $h \in [H], \widetilde{\Theta}_h \in \widetilde{\mathcal{S}}_h$, the conditional mean evolves as $\widetilde{\boldsymbol{\Theta}}_{h+1} = K^1_{h+1}\widetilde{\boldsymbol{\Theta}}_h + K^2_{h+1}\widetilde{\boldsymbol{P}}_h + K^3_{h+1}\widetilde{\boldsymbol{U}}_h + K^4_{h+1}\widetilde{\boldsymbol{Y}}_{h+1}$ for some matrices $\{K^k_{h+1}\}_{k\in[4]}$, and the value function $\widetilde{V}_h(\widetilde{\boldsymbol{\Theta}}_h)$ has a quadratic form as $\widetilde{V}_h(\widetilde{\boldsymbol{\Theta}}_h) = \widetilde{\boldsymbol{\Theta}}^\top_h \widetilde{R}_h \widetilde{\boldsymbol{\Theta}}_h + \widetilde{c}_h$ for some matrix $\widetilde{R}_h$ and constant $\widetilde{c}_h$ that do not depend on $\widetilde{\boldsymbol{\Theta}}_h$, which can be computed through the following Riccati Equations:

$$\begin{aligned}
&\widetilde{R}_h = \widetilde{Q}^1_h + (\widehat{\mathcal{E}}^*_h)^\top \widetilde{Q}^2_h \widehat{\mathcal{E}}^*_h + (\widetilde{L}^1_h \widehat{\mathcal{E}}^*_h + \widetilde{L}^2_h)^\top \widetilde{R}_{h+1}(\widetilde{L}^1_h \widehat{\mathcal{E}}^*_h + \widetilde{L}^2_h),\\
&\widehat{\mathcal{E}}^*_h = -(\widetilde{Q}^2_h + (\widetilde{L}^1_h)^\top \widetilde{R}_{h+1}\widetilde{L}^1_h)^\dagger (\widetilde{L}^1_h)^\top \widetilde{R}_{h+1}\widetilde{L}^2_h,\\
&\widetilde{L}^1_h = K^3_{h+1} + K^4_{h+1}\widetilde{E}_{h+1}\widetilde{B}_h,\\
&\widetilde{L}^2_h = K^1_{h+1} + K^2_{h+1}\mathbb{I}_{p,h} + K^4_{h+1}\widetilde{E}_{h+1}\widetilde{A}_h\mathbb{I}_{x,h},
\end{aligned} \tag{IV.4}$$

where the $\mathbb{I}_{x,h}, \mathbb{I}_{p,h}$ are the identity projection matrices that project vectors in $\widetilde{\mathcal{S}}_h = \widetilde{\mathcal{X}} \times \widetilde{\mathcal{P}}_h$ to the corresponding vectors in $\widetilde{\mathcal{X}}$ and $\widetilde{\mathcal{P}}_h$, respectively. Also, $\{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n]}, \widetilde{c}_h$ can be computed correspondingly as:

$$\begin{aligned}
&\widetilde{L}^3_h = K^3_{h+1} + K^4_{h+1}\widetilde{E}_{h+1}\widetilde{B}_h,\\
&\widetilde{L}^4_h = K^2_{h+1}\mathbb{I}_{p,h} + K^4_{h+1}\widetilde{E}_{h+1}\widetilde{A}_h\mathbb{I}_{x,h},\\
&\widetilde{\mathcal{F}}^*_h \in \operatorname*{argmin}_{\widetilde{\mathcal{F}}_h = \operatorname{diag}(\widetilde{\mathcal{F}}_{1,h}, \cdots, \widetilde{\mathcal{F}}_{n,h})} \operatorname{Tr}(\widetilde{\mathcal{F}}_h\mathbb{I}_{p,h}\widetilde{\Sigma}_h\mathbb{I}^\top_{p,h}\widetilde{\mathcal{F}}^\top_h\widetilde{Q}^2_h)\\
&\qquad + \operatorname{Tr}((\widetilde{L}^3_h\widetilde{\mathcal{F}}_h\mathbb{I}_{p,h} + \widetilde{L}^4_h)\widetilde{\Sigma}_h(\widetilde{L}^3_h\widetilde{\mathcal{F}}_h\mathbb{I}_{p,h} + \widetilde{L}^4_h)^\top\widetilde{R}_{h+1}),\\
&\widetilde{\mathcal{F}}^*_h =: \operatorname{diag}(\widetilde{\mathcal{F}}^*_{1,h}, \cdots, \widetilde{\mathcal{F}}^*_{n,h}),\\
&\widetilde{c}_h = \operatorname{Tr}(\widetilde{\mathcal{F}}^*_h\mathbb{I}_{p,h}\widetilde{\Sigma}_h\mathbb{I}^\top_{p,h}(\widetilde{\mathcal{F}}^*_h)^\top\widetilde{Q}^2_h) + \operatorname{Tr}(\mathbb{I}_{x,h}\widetilde{\Sigma}_h\mathbb{I}^\top_{x,h}\widetilde{Q}^1_h)\\
&\quad + \operatorname{Tr}((\widetilde{L}^3_h\widetilde{\mathcal{F}}^*_h\mathbb{I}_{p,h} + \widetilde{L}^4_h)\widetilde{\Sigma}_h(\widetilde{L}^3_h\widetilde{\mathcal{F}}^*_h\mathbb{I}_{p,h} + \widetilde{L}^4_h)^\top\widetilde{R}_{h+1}) +\\
&\quad \operatorname{Tr}(K^4_{h+1}\operatorname{diag}(\widetilde{\Sigma}_{1,h+1}, \cdots, \widetilde{\Sigma}_{n,h+1})(K^4_{h+1})^\top\widetilde{R}_{h+1}) +\\
&\quad \operatorname{Tr}(K^4_{h+1}\widetilde{E}_{h+1}\Sigma_{0,h}\widetilde{E}^\top_{h+1}(K^4_{h+1})^\top\widetilde{R}_{h+1}) + \widetilde{c}_{h+1}.
\end{aligned} \tag{IV.5}$$

Finally, $\widetilde{\mathcal{E}}_h^* := \left[(\widetilde{\mathcal{E}}_{1,h}^*)^\top \quad \cdots \quad (\widetilde{\mathcal{E}}_{n,h}^*)^\top\right]^\top$ is computed as

$$\widetilde{\mathcal{E}}_h^* = \widehat{\mathcal{E}}_h^* - \mathrm{diag}(\widetilde{\mathcal{F}}_{1,h}^*, \widetilde{\mathcal{F}}_{2,h}^*, \cdots, \widetilde{\mathcal{F}}_{n,h}^*)\mathbb{I}_{p,h}. \tag{IV.6}$$

Solving JCCO with open-loop $g_{1:H}^m$: Now we are ready to solve the problem $\mathcal{D}(g_{1:H}^m)$. First, from Theorem IV.6, we can solve for $\widetilde{g}_{1:H}^*$ as $\widetilde{g}_{i,h}^*(\widetilde{\boldsymbol{C}}_h, \widetilde{\boldsymbol{P}}_{i,h}) = \widetilde{\mathcal{E}}_{i,h}^* \widetilde{\Psi}_h^3(\widetilde{\boldsymbol{C}}_h) + \widetilde{\mathcal{F}}_{i,h}^* \widetilde{\boldsymbol{P}}_{i,h}$ for any $i \in [n], h \in [H]$, and obtain the optimal value of $\min_{\widetilde{g}_{1:H}} J_{\widetilde{\mathcal{D}}(g_{1:H}^m)}(\widetilde{g}_{1:H}) = \mathbb{E}[\widetilde{V}_1(\widetilde{\boldsymbol{\Theta}}_1)] := \widetilde{c}_0$. Note that $\widetilde{\boldsymbol{\Theta}}_1 = \widetilde{\Psi}_1^3(\widetilde{\boldsymbol{C}}_1) = K_1^1 \widetilde{\boldsymbol{C}}_1, \widetilde{\boldsymbol{C}}_1 = \widetilde{\boldsymbol{Z}}_1 = \widetilde{\chi}_1(\widetilde{\boldsymbol{Y}}_1) = K_1^2 \widetilde{\boldsymbol{Y}}_1, \widetilde{\boldsymbol{Y}}_1 = \widetilde{E}_1 \widetilde{\boldsymbol{X}}_1$, where $\boldsymbol{X}_1 \sim \mathcal{N}(\boldsymbol{0}, \Sigma_1)$ for some matrices K_1^1, K_1^2. Therefore, we can write $\widetilde{c}_0 = \mathrm{Tr}(\widetilde{E}_1^\top (K_1^2)^\top (K_1^1)^\top R_1 K_1^1 K_1^2 \widetilde{E}_1 \Sigma_1) + \widetilde{c}_1$. Then, by Lemma IV.1, we can obtain an optimal control strategy of $\mathcal{D}(g_{1:H}^m)$ as $\bar{g}_{1:H}^* = \varphi(\widetilde{g}_{1:H}^*, \mathcal{D}(g_{1:H}^m))$ such that $J_{\mathcal{D}(g_{1:H}^m)}(\bar{g}_{1:H}^*) = J_{\widetilde{\mathcal{D}}(g_{1:H}^m)}(\widetilde{g}_{1:H}^*)$. Further, if we assign $g_{1:H}^* = \bar{g}_{1:H}^*$, then $g_{1:H}^*$ is an optimal control strategy with respect to the communication strategy $g_{1:H}^m$ in $\mathcal{D}$. Let $M_{1:H}$ be the communication actions chosen by $g_{1:H}^m$ such that $M_h = g_h^m$, then $J_{\mathcal{D}}(g_{1:H}^m, g_{1:H}^*) = J_{\widetilde{\mathcal{D}}(g_{1:H}^m)}(\widetilde{g}_{1:H}^*) + \sum_{h=1}^H \mathcal{K}_h(g_h^m)$. Therefore, we can finally solve the JCCO problem by solving $g_{1:H}^{m,*} \in \mathrm{argmin}_{g_{1:H}^m} J_{\widetilde{\mathcal{D}}(g_{1:H}^m)}(\widetilde{g}_{1:H}^*) + \sum_{h=1}^H \mathcal{K}_h(g_h^m)$.

V. Extension to JCCO with Closed-loop Communication Strategies

In this section, we extend to studying the JCCO problem with *closed-loop* communication strategies, i.e., for any $h \in [H], i \in [n], g_{i,h}^m : \mathcal{I}_{i,h^-} \times \mathcal{M}_{1:h-1} \to \mathcal{M}_{i,h}$, with associated (closed-loop) control strategies $g_{i,h}^a : \mathcal{I}_{i,h^+} \times \mathcal{M}_{1:h} \to \mathcal{U}_i$. Since each agent may influence others through the additional sharing determined by communication strategy $g_{i,h}^m$, while others may not access the input of $g_{i,h}^m$, PN IS may break. Therefore, the SI-CIB condition does not hold in general and the common-information-based beliefs are no longer Gaussian distributions that can be characterized by the finite-dimensional conditional mean and covariance.

Fortunately, we can still apply the approach designed in [11], which expanded the problem into a new one that satisfies the SI-CIB condition, and then solved it via dynamic programming. To guarantee the SI-CIB condition, [11] introduced the following assumption.

Assumption V.1: The communication strategies take common information and past communication actions as input, i.e., $\forall i \in [n], h \in [H], g_{i,h}^m : \mathcal{C}_{h^-} \times \mathcal{M}_{1:h-1} \to \mathcal{M}_{i,h}$.

Assumption V.1 was shown necessary in [11] for discrete-space, (closed-loop) communication-control strategy optimization, in that missing it (i.e., when $g_{i,h}^m$ may use the *private information* $\boldsymbol{P}_{i,h^-}$) can cause computational intractability in general. More importantly, it was also shown in [11] that this assumption can help *preserve* the PN IS after additional sharing, which will then lead to the SI-CIB condition, as we will show for the LQ case next.

To this end, we first introduce some additional notation. We denote by $\mathcal{P}_{h^-}(M_{1:h-1}) \subseteq \mathcal{P}_{h^-}$ the set of all possible P_{h^-} given $M_{1:h-1}$, which is a finite-dimensional vector space. Similarly, we denote by $\mathcal{P}_{h^+}(M_{1:h}) \subseteq \mathcal{P}_{h^+}$ the set of all possible P_{h^+} given $M_{1:h}$. Then, given any $M_{1:h-1}$, we can define $\boldsymbol{B}_{h^-} \in \mathcal{P}(\mathcal{X} \times \mathcal{P}_{h^-}(M_{1:h-1}))$ as the conditional probability measure of state $\boldsymbol{X}_h$ and private information $\boldsymbol{P}_{h^-}$, given the past strategies $g_{1:h-1}^m, g_{1:h-1}^a$, communication actions $M_{1:h-1}$, and common information $\boldsymbol{C}_{h^-}$ before additional sharing. Similarly, we define $\boldsymbol{B}_{h^+} \in \mathcal{P}(\mathcal{X} \times \mathcal{P}_{h^+}(M_{1:h}))$ as the conditional probability measure of $\boldsymbol{X}_h$ and $\boldsymbol{P}_{h^+}$ given $g_{1:h}^m, g_{1:h-1}^a, \boldsymbol{M}_{1:h}$, and $\boldsymbol{C}_{h^+}$. Hence, we can write $\boldsymbol{B}_{h^-} = \Pi_{h^-}(\boldsymbol{C}_{h^-}, \boldsymbol{M}_{1:h-1}, g_{1:h-1}^m, g_{1:h-1}^a)$ and $\boldsymbol{B}_{h^+} = \Pi_{h^+}(\boldsymbol{C}_{h^+}, \boldsymbol{M}_{1:h}, g_{1:h}^m, g_{1:h-1}^a)$ for some functionals Π_{h^-} and Π_{h^+}, respectively. Then, we can extend the SI-CIB condition as follows.

Definition V.2 (SI-CIB Condition of $\mathcal{D}$): We say that a JCCO problem $\mathcal{D}$ satisfies the *strategy-independent common-information-based belief* condition if

- For any $h \in [H]$, any realization $C_{h^-} \in \mathcal{C}_{h^-}, M_{1:h-1} \in \mathcal{M}_{1:h-1}$ and any two strategies $(g_{1:h-1}^m, g_{1:h-1}^a)$ and $(g_{1:h-1}^{m,'}, g_{1:h-1}^{a,'})$ that can reach C_{h^-} and $M_{1:h-1}$, it holds that $\Pi_{h^-}(C_{h^-}, M_{1:h-1}, g_{1:h-1}^m, g_{1:h-1}^a) = \Pi_{h^-}(C_{h^-}, M_{1:h-1}, g_{1:h-1}^{m,'}, g_{1:h-1}^{a,'})$.
- For any $h \in [H]$, any realization $C_{h^+} \in \mathcal{C}_{h^+}, M_{1:h} \in \mathcal{M}_{1:h}$ and any two strategies $(g_{1:h}^m, g_{1:h-1}^a)$ and $(g_{1:h}^{m,'}, g_{1:h-1}^{a,'})$ that can reach C_{h^+} and $M_{1:h}$, it holds that $\Pi_{h^+}(C_{h^+}, M_{1:h}, g_{1:h}^m, g_{1:h-1}^a) = \Pi_{h^+}(C_{h^+}, M_{1:h}, g_{1:h}^{m,'}, g_{1:h-1}^{a,'})$.

Under Assumption V.1, we can expand the problem $\mathcal{D}$ into a new one by adding some actions that affect other agents into the common information for both timesteps h^- and h^+, similarly as Eq. (IV.1). After the strict expansion, the new problem satisfies the SI-CIB condition, and the team-optimal strategy of the expended JCCO problem can be used to obtain that of the original one. More importantly, satisfying the SI-CIB condition ensures that $\boldsymbol{B}_{h^-}$ and $\boldsymbol{B}_{h^+}$ are Gaussian distributions, as shown below.

Theorem V.3: Let $\mathcal{D}$ be the problem after the above strict expansion from a JCCO problem with PN IS that satisfies Assumptions II.1, III.2, III.4, and V.1, then $\mathcal{D}$ is a JCCO problem satisfying the SI-CIB condition. Moreover, for any $h \in [H]$, $\boldsymbol{B}_{h^-}, \boldsymbol{B}_{h^+}$ are Gaussian distributions. Formally, $\boldsymbol{B}_{h^-} = \mathcal{N}(\boldsymbol{\Theta}_{h^-}, \Sigma_{h^-}), \boldsymbol{B}_{h^+} = \mathcal{N}(\boldsymbol{\Theta}_{h^+}, \Sigma_{h^+})$, and

$$\begin{aligned} \boldsymbol{\Theta}_{h^-} &= \Psi_{h^-}^1(\boldsymbol{\Theta}_{(h-1)^+}, \boldsymbol{M}_{1:h-1}, \boldsymbol{Z}_h^b), \\ \boldsymbol{\Theta}_{h^+} &= \Psi_{h^+}^1(\boldsymbol{\Theta}_{h^-}, \boldsymbol{M}_{1:h}, \boldsymbol{Z}_h^a), \\ \Sigma_{h^-} &= \Psi_{h^-}^2(\boldsymbol{M}_{1:h-1}), \quad \Sigma_{h^+} = \Psi_{h^+}^2(\boldsymbol{M}_{1:h}), \end{aligned} \tag{V.1}$$

for some functions $\Psi_{h^-}^1, \Psi_{h^+}^1, \Psi_{h^-}^2, \Psi_{h^+}^2$ that do not depend on the strategies $g_{1:h}^m, g_{1:h}^a$.

Theorem V.3 provides a way to solve JCCO with closed-loop communication strategies via dynamic programming. Specifically, we can conduct dynamic programming over the spaces of $(M_{1:h}, \Theta_{h^+}) \in \mathcal{M}_{1:h} \times (\mathcal{X} \times \mathcal{P}_{h^+}(M_{1:h}))$ and the spaces of $(M_{1:h-1}, \Theta_{h^-}) \in \mathcal{M}_{1:h-1} \times (\mathcal{X} \times \mathcal{P}_{h^-}(M_{1:h-1}))$, to solve for the optimal strategies $g_h^{a,*}$ and $g_h^{m,*}$, respectively. The overall algorithm is tabulated as Algorithm 1 in §D.

VI. Numerical Example

This section illustrates our method in a numerical example of a linear-quadratic JCCO problem $\mathcal{D}$ with $n = 2$ and $H =$

2 and open-loop communication strategies. The baseline sharing of $\mathcal{D}$ is *one-step measurement delay*, i.e., for any $i \in [2], h \in [2], \boldsymbol{C}_{h^-} = \{\boldsymbol{Y}_{1:h-1}\}, \boldsymbol{P}_{i,h^-} = \{\boldsymbol{Y}_{i,h}\}$, and $\boldsymbol{C}_{3^-} = \{\boldsymbol{Y}_{1:2}\}, \boldsymbol{P}_{3^-} = \emptyset$. The communication action spaces are defined as $\mathcal{M}_{i,h} = \{0,1\}, \forall i \in [2], h \in [2]$, where $\boldsymbol{M}_{i,h} = 1$ represents $\boldsymbol{Z}^a_{i,h} = \{\boldsymbol{Y}_{i,h}\}$ and otherwise $\boldsymbol{Z}^a_{i,h} = \emptyset$, and the communication cost is defined as $\mathcal{K}_h(\boldsymbol{M}_h) = 0.2(\boldsymbol{M}_{1,h} + \boldsymbol{M}_{2,h})$. The system is defined as $\mathcal{X} = \mathcal{Y}_{i,h} = \mathcal{U}_i = \mathbb{R}, \forall i \in [2], h \in [2]$ with the following dynamics data:

$$A_1 = 1.25, A_2 = 1.15, B_{1,1} = 1, B_{2,1} = 0.9, B_{1,2} = 0.85,$$
$$B_{2,2} = 1.1, E_{1,1} = 1.1, E_{2,1} = 0.8, E_{1,2} = 0.9, E_{2,2} = 1.2$$
$$\Sigma_{0,1} = 1.5, \Sigma_{0,2} = 2, \Sigma_{1,1} = 1, \Sigma_{2,1} = 1.5, \Sigma_{1,2} = 2,$$
$$\Sigma_{2,2} = 0.8, \Sigma_1 = 2, Q^1_1 = 5, Q^1_2 = 6, Q^1_3 = 8, Q^2_1 = Q^2_2 = \mathbb{I}_2.$$

$\mathcal{D}$ has PN IS and satisfies Assumptions II.1,III.2, and III.4, and we can use our method to solve this problem as follows.

Step 1: There are 16 different communication strategies with $g^m_{i,h} \in \{0,1\}, \forall h \in [2], i \in [2]$. For each $g^m_{1:2}$, we can construct the associated decentralized LQG problem $\mathcal{D}(g^m_{1:2})$.

Step 2: We expand $\mathcal{D}(g^m_{1:2})$ to $\widetilde{\mathcal{D}}(g^m_{1:2})$, and compute the corresponding matrices $\{K^j_{h+1}\}_{h\in[H],j\in[4]}$ and $\{\widetilde{\Sigma}_h\}_{h\in[3]}$ based on the information structure of $\widetilde{\mathcal{D}}(g^m_{1:2})$. We illustrate this with the following example: $g^m_{1,1} = g^m_{2,2} = 1, g^m_{2,1} = g^m_{1,2} = 0$, making $\widetilde{\boldsymbol{C}}_1 = \{\widetilde{\boldsymbol{Y}}_{1,1}\}, \widetilde{\boldsymbol{C}}_2 = \{\widetilde{\boldsymbol{Y}}_1, \widetilde{\boldsymbol{U}}_1, \widetilde{\boldsymbol{Y}}_{2,2}\}, \widetilde{\boldsymbol{C}}_3 = \{\widetilde{\boldsymbol{Y}}_{1:2}, \widetilde{\boldsymbol{U}}_{1:2}\}, \widetilde{\boldsymbol{P}}_{2,1} = \{\widetilde{\boldsymbol{Y}}_{2,1}\}, \widetilde{\boldsymbol{P}}_{1,2} = \{\widetilde{\boldsymbol{Y}}_{1,2}\}, \widetilde{\boldsymbol{P}}_{1,1} = \widetilde{\boldsymbol{P}}_{2,2} = \emptyset$. For timestep $h = 2$, $\{K^j_2\}_{j\in[4]}, \widetilde{\Sigma}_2$ can be computed as:

$$K^1_2 = E^{\text{ext}}_{1,2}\left(1 - \frac{\mathring{P}_2 E^2_{2,2}}{E^2_{2,2}\mathring{P}_2 + \Sigma_{2,2}}\right) A_1 \mathring{P}_1 \frac{E^2_{1,1}\Sigma_1 + \Sigma_{1,1}}{\Sigma_1\Sigma_{1,1}} \begin{bmatrix}1 & 0\end{bmatrix},$$
$$K^2_2 = E^{\text{ext}}_{1,2}\left(1 - \frac{\mathring{P}_2 E^2_{2,2}}{E^2_{2,2}\mathring{P}_2 + \Sigma_{2,2}}\right) A_1 \mathring{P}_1 \frac{E_{2,1}}{\Sigma_{2,1}},$$
$$K^3_2 = E^{\text{ext}}_{1,2}\left(1 - \frac{\mathring{P}_2 E^2_{2,2}}{E^2_{2,2}\mathring{P}_2 + \Sigma_{2,2}}\right) B_1, K^4_2 = E^{\text{ext}}_{1,2}\frac{\mathring{P}_2 E_{2,2}\begin{bmatrix}0 & 1\end{bmatrix}}{E^2_{2,2}\mathring{P}_2 + \Sigma_{2,2}},$$
$$\widetilde{\Sigma}_2 = \begin{bmatrix} \frac{\mathring{P}_2\Sigma_{2,2}}{E^2_{2,2}\mathring{P}_2+\Sigma_{2,2}} & \frac{E_{1,2}\mathring{P}_2\Sigma_{2,2}}{E^2_{2,2}\mathring{P}_2+\Sigma_{2,2}} \\ \frac{E_{1,2}\mathring{P}_2\Sigma_{2,2}}{E^2_{2,2}\mathring{P}_2+\Sigma_{2,2}} & \Sigma_{1,2} + \frac{E^2_{1,2}\mathring{P}_2\Sigma_{2,2}}{E^2_{2,2}\mathring{P}_2+\Sigma_{2,2}} \end{bmatrix}, E^{\text{ext}}_{1,2} = \begin{bmatrix}1 & E_{1,2}\end{bmatrix}^\top,$$

where $\mathring{P}_1 = (\frac{1}{\Sigma_1} + \frac{E^2_{1,1}}{\Sigma_{1,1}} + \frac{E^2_{2,1}}{\Sigma_{2,1}})^{-1}, \mathring{P}_2 = A^2_1\mathring{P}_1 + \Sigma_{0,1}$. For timesteps $h = 1, 3$, $\{K^j_h\}_{j\in[4]}, \widetilde{\Sigma}_h$ are computed similarly.

Step 3: Apply the Riccati Equations in Theorem IV.6 to compute $\widetilde{c}_0$ and $\widetilde{g}^*_{1:2}$ with respect to such $g^m_{1:2}$, and define $f(g^m_{1:2}) := \min_{g^a_{1:2}} J_{\mathcal{D}}(g^a_{1:2}, g^m_{1:2}) = \widetilde{c}_0 + \mathcal{K}_1(g^m_1) + \mathcal{K}_2(g^m_2)$.

Step 4: Compute $g^{m,*}_{1:2} \in \operatorname{argmin}_{g^m_{1:2}\in\mathcal{G}^m_{1:2}} f(g^m_{1:2})$ as $g^{m,*}_{1,1} = g^{m,*}_{2,2} = 1, g^{m,*}_{2,1} = g^{m,*}_{1,2} = 0$, and the optimal strategy $\widetilde{g}^*_{1:2}$ of the problem $\widetilde{\mathcal{D}}(g^{m,*}_{1:2})$ can be computed as:[2]

$$\widetilde{g}^*_{1,1}(\widetilde{\boldsymbol{I}}_{1,1}) = -0.410\widetilde{\boldsymbol{Y}}_{1,1}, \widetilde{g}^*_{2,1} = -0.219\widetilde{\boldsymbol{Y}}_{1,1} - 0.292\widetilde{\boldsymbol{Y}}_{2,1},$$
$$\widetilde{g}^*_{1,2}(\widetilde{\boldsymbol{I}}_{1,2}) = 0.038\widetilde{\boldsymbol{Y}}_{1,1} - 0.019\widetilde{\boldsymbol{Y}}_{2,1} - 0.060\widetilde{\boldsymbol{U}}_{1,1} - 0.054\widetilde{\boldsymbol{U}}_{2,1}$$
$$- 0.196\widetilde{\boldsymbol{Y}}_{1,2} - 0.200\widetilde{\boldsymbol{Y}}_{2,2}, \qquad \widetilde{g}^*_{2,2}(\widetilde{\boldsymbol{I}}_{2,2}) = -0.079\widetilde{\boldsymbol{Y}}_{1,1}$$
$$- 0.038\widetilde{\boldsymbol{Y}}_{2,1} - 0.123\widetilde{\boldsymbol{U}}_{1,1} - 0.110\widetilde{\boldsymbol{U}}_{2,1} - 0.410\widetilde{\boldsymbol{Y}}_{2,2}.$$

Further, the optimal objective of $\widetilde{\mathcal{D}}(g^{m,*}_{1:2})$ is $\widetilde{c}_0 = 45.947$, and we can compute $g^{a,*}_{1:2} = \varphi(\widetilde{g}^*_{1:2}, \mathcal{D}(g^{m,*}_{1:2}))$ as: $g^{a,*}_1 = \widetilde{g}^*_1$,

$$g^{a,*}_{1,2}(\boldsymbol{I}_{1,2^+}) = -0.002\boldsymbol{Y}_{1,1} - 0.003\boldsymbol{Y}_{2,1} - 0.196\boldsymbol{Y}_{1,2} - 0.200\boldsymbol{Y}_{2,2},$$
$$g^{a,*}_{2,2}(\boldsymbol{I}_{2,2^+}) = -0.004\boldsymbol{Y}_{1,1} - 0.006\boldsymbol{Y}_{2,1} - 0.410\boldsymbol{Y}_{2,2},$$

and the optimal objective of $\mathcal{D}$ is $J_{\mathcal{D}}(g^{m,*}_{1:2}, g^{a,*}_{1:2}) = 46.347$.

[2] All numerical values are rounded to three decimal places.

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APPENDIX

A. Decentralized Linear–Quadratic–Gaussian Control Problem

For $n > 1$ agents, a (cooperative) decentralized linear-quadratic-Gaussian (decentralized LQG) control problem $\check{\mathcal{D}}$ is described by a tuple $\langle H, \mathcal{X}, \{\mathcal{Y}_{i,h}\}_{i\in[n],h\in[H]}, \{\mathcal{U}_i\}_{i\in[n]}, \{A_h\}_{h\in[H]}, \{B_{i,h}\}_{i\in[n],h\in[H]}, \{E_{i,h}\}_{i\in[n],h\in[H]}, \{Q_h^1\}_{h\in[H+1]}, \{Q_h^2\}_{h\in[H]}\rangle$, where H is the time horizon and $\boldsymbol{X}_h \in \mathcal{X} = \mathbb{R}^{d_x}$ is the state. At each timestep $h \in [H]$, each agent $i \in [n]$ receives a noisy observation $\boldsymbol{Y}_{i,h} \in \mathcal{Y}_i = \mathbb{R}^{d_y^i}$ of the state $\boldsymbol{X}_h$, and chooses a control action $\boldsymbol{U}_{i,h} \in \mathcal{U}_i = \mathbb{R}^{d_u^i}$. At timestep $h \in [H]$, we denote by $\boldsymbol{U}_h = \begin{bmatrix}\boldsymbol{U}_{1,h}^\top & \boldsymbol{U}_{2,h}^\top & \cdots & \boldsymbol{U}_{n,h}^\top\end{bmatrix}^\top$ the joint control action of all the n agents, and by $\mathcal{U} = \mathbb{R}^{\sum_{i=1}^n d_u^i}$ the joint control action space; we denote by $\boldsymbol{Y}_h = \begin{bmatrix}\boldsymbol{Y}_{1,h}^\top & \boldsymbol{Y}_{2,h}^\top & \cdots & \boldsymbol{Y}_{n,h}^\top\end{bmatrix}^\top$ the joint observation, and by $\mathcal{Y} = \mathbb{R}^{\sum_{i=1}^n d_y^i}$ the joint observation space. The system evolves as follows: for each timestep $h \in [H]$

$$\begin{aligned}\boldsymbol{X}_{h+1} &= A_h\boldsymbol{X}_h + B_h\boldsymbol{U}_h + \boldsymbol{W}_{0,h} = A_h\boldsymbol{X}_h + \sum_{i=1}^n B_{i,h}\boldsymbol{U}_{i,h} + \boldsymbol{W}_{0,h},\\ \boldsymbol{Y}_{i,h} &= E_{i,h}\boldsymbol{X}_h + \boldsymbol{W}_{i,h}, \forall i \in [n],\end{aligned} \tag{.1}$$

where $\boldsymbol{W}_{i,h}$ is a random variable taking values in some real-vector space $\mathcal{W}_{i,h}$ for all $h \in [H], i = 0, 1, \cdots, n$, and $B_h = \begin{bmatrix}B_{1,h} & \cdots & B_{n,h}\end{bmatrix}$. $A_h, B_h, E_{i,h}$ are matrices of proper dimensions, with $B_h = \begin{bmatrix}B_{1,h} & B_{2,h} & \cdots & B_{n,h}\end{bmatrix}$. Here, $\boldsymbol{W}_{i,h} \sim \mathcal{N}(\boldsymbol{0}, \Sigma_{i,h})$ for some covariance matrices $\Sigma_{i,h} \succeq 0$ for all $i \in [0:n], h \in [H]$ and $\boldsymbol{X}_1 \sim \mathcal{N}(\boldsymbol{0}, \Sigma_1)$ for some covariance matrix $\Sigma_1 \succeq 0$. The random variables $\boldsymbol{X}_1$ and $\{\boldsymbol{W}_{i,h}\}_{i\in[0:n],h\in[H]}$ are mutually independent.

At timestep $h \in [H]$, each agent can access some information $\boldsymbol{I}_{i,h} \subseteq \{\boldsymbol{Y}_{1:h}, \boldsymbol{U}_{1:h-1}\}$, and the collection of all possible such available information is denoted by $\mathcal{I}_{i,h}$. We use $\boldsymbol{I}_h$ to denote the joint available information at timestep h. Meanwhile, agents may share part of their information with each other, and the shared information is denoted as $\boldsymbol{Z}_h$. At timestep h, the common information among all agents is defined as the union of all the shared information so far: $\boldsymbol{C}_h = \cup_{t=1}^h \boldsymbol{Z}_t$. The private information of agent i at timestep h is thus defined as $\boldsymbol{P}_{i,h} = \boldsymbol{I}_{i,h} \backslash \boldsymbol{C}_h$, and the joint private information is denoted as $\boldsymbol{P}_h := [\boldsymbol{P}_{1,h}^\top \ \cdots \ \boldsymbol{P}_{n,h}^\top]^\top$. We denote by $\mathcal{C}_h, \mathcal{P}_h, \mathcal{P}_{i,h}, \mathcal{I}_{i,h}, \mathcal{I}_h$ the sets of values of the random variables $\boldsymbol{C}_h, \boldsymbol{P}_h, \boldsymbol{P}_{i,h}, \boldsymbol{I}_{i,h}, \boldsymbol{I}_h$ for each $h \in [H], i \in [n]$.

Each agent i at timestep h chooses the control action $\boldsymbol{U}_{i,h}$ based on some strategy $g_{i,h} : \mathcal{I}_{i,h} \to \mathcal{U}_i$. We denote by $g_h := (g_{1,h}, g_{2,h}, \cdots, g_{n,h})$ the joint control strategy of all the agents, and by $g_{1:h} := (g_1, g_2, \cdots, g_h), \forall h \in [H]$ the sequence of joint strategies from timesteps 1 to h. We use $\mathcal{G}_{i,h}$ to denote the strategy space of $g_{i,h}$, and use $\mathcal{G}_h, \mathcal{G}_{1:h}$ to denote the joint strategy spaces, correspondingly. At each timestep $h \in [H]$, the cost is defined as $c_h = \boldsymbol{X}_h^\top Q_h^1 \boldsymbol{X}_h + \boldsymbol{U}_h^\top Q_h^2 \boldsymbol{U}_h$, and the cost of final timestep is $c_{H+1} = \boldsymbol{X}_{H+1}^\top Q_{H+1}^1 \boldsymbol{X}_{H+1}$. Here, $\forall h \in [H], Q_h^1, Q_h^2 \succeq 0, Q_{H+1}^1 \succeq 0$ are symmetric matrices with proper dimensions. The objective among all the agents is thus defined as

$$J_{\check{\mathcal{D}}}(g_{1:H}) := \mathbb{E}\left[\sum_{h=1}^H c_h + c_{H+1} \,\middle|\, g_{1:H}\right].$$

With the objective, we define the solution concept of *(team)-optimum* of the problem $\check{\mathcal{D}}$ as follows.

Definition .1 (Team optimum): We call a joint strategy $g_{1:H}^* \in \mathcal{G}_{1:H}$ a *team-optimal strategy* of the decentralized LQG control problem $\check{\mathcal{D}}$ if

$$\forall g_{1:H} \in \mathcal{G}_{1:H}, \qquad J_{\check{\mathcal{D}}}(g_{1:H}) \geq J_{\check{\mathcal{D}}}(g_{1:H}^*). \tag{.2}$$

B. Deferred Details of §III

We start with the following auxiliary lemma.

Lemma .2: For any $n_1, n_2 \in \mathbb{N}$, any $Q_1, Q_2 \succeq 0$ being $n_1 \times n_1, n_2 \times n_2$ symmetric matrices, and any matrices A, B with proper dimensions, it holds that $\ker(Q_1 + B^\top Q_2 B) \subseteq \ker(A^\top Q_2 B)$.

Proof: For any $v \in \ker(Q_1 + B^\top Q_2 B)$, we have $v^\top Q_1 v + v^\top B^\top Q_2 B v = 0$. Since $Q_1 \succeq 0, Q_2 \succeq 0$, we know $v^\top Q_1 v = v^\top B^\top Q_2 B v = 0$. Then, from $v^\top B^\top Q_2 B v = 0$, we have $Q_2^{\frac{1}{2}} B v = 0$ and thus $Q_2 B v = 0$, which further implies that $A^\top Q_2 B v = 0$ and thus $v \in \ker(A^\top Q_2 B)$. This completes the proof. ■

1) Proof of Lemma III.1: *Proof:* To prove this lemma, we leverage the following proposition.

Proposition .3 (Adapted from Theorems 1 and 2 of [20]): There exists a decentralized LQG control problem with $n = 2, H = 2$ and parameters k and σ_0 such that the optimal control strategy is nonlinear:

$$\begin{aligned}&\text{All random variables are scalar, } \boldsymbol{X}_1 \sim \mathcal{N}(0, \sigma_0^2), \boldsymbol{X}_2 = \boldsymbol{X}_1 + \boldsymbol{U}_{1,1}, \boldsymbol{X}_3 = \boldsymbol{X}_2 - \boldsymbol{U}_{2,2},\\ &\boldsymbol{Y}_{1,1} = \boldsymbol{X}_1, \boldsymbol{Y}_{2,1} = 0, \boldsymbol{Y}_{1,2} = 0, \boldsymbol{Y}_{2,2} = \boldsymbol{X}_2 + \boldsymbol{W}_{2,2}, \boldsymbol{W}_{2,2} \sim \mathcal{N}(0, 1),\\ &c_1 = k^2\boldsymbol{U}_{1,1}^2, c_2 = 0, c_3 = \boldsymbol{X}_3^2, \quad \text{no information sharing between agents.}\end{aligned}$$

Then, using the parameters k and σ_0 specified in Proposition .3, we construct a JCCO problem $\mathcal{D}$ with $n=2, H=2$, with the system dynamics being defined as

$$\boldsymbol{X}_1 \sim \mathcal{N}(0,\sigma_0^2), \boldsymbol{X}_2 = \boldsymbol{X}_1 + \boldsymbol{U}_{1,1}, \boldsymbol{X}_3 = \boldsymbol{X}_2 - \boldsymbol{U}_{2,2},$$
$$\boldsymbol{Y}_{1,1} = \boldsymbol{X}_1, \boldsymbol{Y}_{2,1} = 0, \boldsymbol{Y}_{1,2} = 0, \boldsymbol{Y}_{2,2} = \boldsymbol{X}_2 + \boldsymbol{W}_{2,2}, \boldsymbol{W}_{2,2} \sim \mathcal{N}(0,1).$$

For the communication part, suppose the baseline sharing is null and $\mathcal{M}_{i,1} = \{0,1\}, \mathcal{M}_{1,2} = \{0,1\}^3, \mathcal{M}_{2,2} = \{0,1\}^2$, where $\boldsymbol{M}_{i,1}$ represents whether agent i shares $\boldsymbol{Y}_{i,1}$, each digit of $\boldsymbol{M}_{1,2}$ represents whether agent 1 shares $\boldsymbol{Y}_{1,1}, \boldsymbol{U}_{1,1}, \boldsymbol{Y}_{1,2}$, and each digit of $\boldsymbol{M}_{2,2}$ represents whether agent 2 shares $\boldsymbol{Y}_{2,1}, \boldsymbol{Y}_{2,2}$. For control costs, we set $c_1 = k^2\boldsymbol{U}_{1,1}^2, c_2 = 0, c_3 = \boldsymbol{X}_3^2$ where k is specified as in Proposition .3; for communication costs, we set $\kappa_h = (\sigma_0^2+1)\mathbb{1}[\boldsymbol{Z}_h^a \neq \emptyset], \forall h \in [2]$. The information available to each agent at each timestep is defined as: $\forall i \in [2], \boldsymbol{I}_{i,1^-} = \{\boldsymbol{Y}_{i,1}\}, \boldsymbol{I}_{i,1^+} = \boldsymbol{I}_{i,1^-} \cup \boldsymbol{Z}_1^a$, and $\boldsymbol{I}_{1,2^-} = \boldsymbol{I}_{1,1^+} \cup \{\boldsymbol{U}_{1,1}, \boldsymbol{Y}_{1,2}\}, \boldsymbol{I}_{2,2^-} = \boldsymbol{I}_{2,1^+} \cup \{\boldsymbol{Y}_{2,2}\}, \boldsymbol{I}_{i,2^+} = \boldsymbol{I}_{i,2^-} \cup \boldsymbol{Z}_2^a, \boldsymbol{I}_{i,3^-} = \boldsymbol{I}_{i,2^+}$. Then, we can verify that:

- $\mathcal{D}$ **satisfies Assumption II.1.**
- $\mathcal{D}$ **satisfies Assumption III.2:** Only $B_{2,1} = 0$ and $\boldsymbol{U}_{2,1}$ does not appear in $\boldsymbol{I}_{2^-}$ or $\boldsymbol{I}_{2^+}$.
- $\mathcal{D}$ **satisfies Assumption III.4:** It holds that $\text{rank}(E_{1,2}B_{2,1}) = \text{rank}(B_{2,1}) = 0, \text{rank}(E_{2,2}B_{1,1}) = \text{rank}(B_{1,1}) = 1$.
- $\mathcal{D}$ **does not have PN IS:** If there is no additional sharing, agent $(1,1)$ influences agent $(2,2)$, but $\boldsymbol{I}_{1,1^-} \nsubseteq \boldsymbol{I}_{2,2^-}$.

For any communication strategy $g_{1:2}^m$, if it chooses any $\boldsymbol{M}_{i,h}$ with any digit non-zero (namely, sharing *some* information via additional sharing), then it cannot achieve the optimum, since it will suffer from a communication cost of $\kappa_h \geq \sigma_0^2+1$, which is larger than the total expected cost when each agent i chooses $\boldsymbol{U}_{1,1} = \boldsymbol{U}_{2,2} = 0$, and they do not have any additional sharing. Therefore, the optimal communication strategy $g_{1:2}^{m,*}$ must yield no additional sharing. Then, any solution to $\mathcal{D}$ can lead to a solution of the decentralized LQG control problem in Proposition .3, whose optimal control strategy is nonlinear. This completes the proof. ■

2) Proof of Lemma III.3: *Proof:* This proof consists of two **Parts**. **Part 1:** we construct a JCCO problem $\mathcal{D}_1$ whose optimal control strategy is nonlinear; **Part 2:** we construct another JCCO problem $\mathcal{D}_2$ whose team-optimal strategy does not exist.

Part 1: Consider an $H=2, n=2$ JCCO problem $\mathcal{D}_1$ with the system dynamics being defined as

$$\forall h \in [2], \boldsymbol{X}_h \in \mathbb{R}^2, \boldsymbol{U}_{1,1} \in \mathbb{R}, \boldsymbol{U}_{1,2}, \boldsymbol{U}_{2,1}, \boldsymbol{U}_{2,2} \in \mathbb{R}^2, \boldsymbol{X}_2 = \boldsymbol{X}_1, \boldsymbol{X}_3 = \boldsymbol{X}_2 - \boldsymbol{U}_{2,2},$$
$$\boldsymbol{X}_1 \sim \mathcal{N}(\boldsymbol{0}, \mathbb{I}), \boldsymbol{Y}_{1,1} = \boldsymbol{X}_1, \boldsymbol{Y}_{1,2} = \boldsymbol{Y}_{2,1} = \boldsymbol{Y}_{2,2} = 0, c_1 = c_2 = 0, c_3 = \boldsymbol{X}_3^\top \boldsymbol{X}_3.$$

For the communication part, suppose the baseline sharing is null and $\mathcal{M}_{i,1} = \{0,1\}, \mathcal{M}_{i,2} = \{0,1\}^3, \forall i \in [2]$, where $\boldsymbol{M}_{i,1}$ represents whether agent i shares $\boldsymbol{Y}_{i,1}$ and each digit of $\boldsymbol{M}_{i,2}$ represents whether agent i shares $\boldsymbol{Y}_{i,1}, \boldsymbol{U}_{i,1}$, and $\boldsymbol{Y}_{i,2}$, respectively. For communication costs, we set $\kappa_1 = \mathbb{1}[\boldsymbol{M}_{1,1} = 1 \text{ or } \boldsymbol{M}_{2,1} = 1], \kappa_2 = \mathbb{1}[\boldsymbol{M}_{1,2} \neq (0,0,0) \text{ or } (0,1,0) \text{ or } \boldsymbol{M}_{2,2} \neq (0,0,0)]$. Such communication costs mean that if each agent share any information except $\boldsymbol{U}_{1,1}$ via additional sharing, then the communication cost will be 1. The information available to each agent at each timestep is defined as $\forall i \in [2], \boldsymbol{I}_{i,1^-} = \{\boldsymbol{Y}_{i,1}\}, \boldsymbol{I}_{i,1^+} = \boldsymbol{I}_{i,1^-} \cup \boldsymbol{Z}_1^a$, and $\boldsymbol{I}_{1,2^-} = \boldsymbol{I}_{1,1^+} \cup \{\boldsymbol{U}_{1,1}, \boldsymbol{Y}_{1,2}\}, \boldsymbol{I}_{2,2^-} = \boldsymbol{I}_{2,1^+} \cup \{\boldsymbol{U}_{2,1}, \boldsymbol{Y}_{2,2}\}, \boldsymbol{I}_{i,2^+} = \boldsymbol{I}_{i,2^-} \cup \boldsymbol{Z}_2^a, \boldsymbol{I}_{i,3^-} = \boldsymbol{I}_{i,2^+}$.

Then, we can verify that:

- $\mathcal{D}_1$ **satisfies Assumption II.1.**
- $\mathcal{D}_1$ **has PN IS:** If there is no additional sharing, then for any $i_1, i_2 \in [2]$, $\boldsymbol{I}_{i_1,1^-} \subseteq \boldsymbol{I}_{i_2,2^-}$ if $i_1 = i_2$, and otherwise agent $(i_1,1)$ does not influence agent $(i_2,2)$.
- $\mathcal{D}_1$ **satisfies Assumption III.4:** For any $i \in [2]$, $\text{rank}(E_{-i,2}B_{i,1}) = \text{rank}(B_{i,1}) = 0$.
- $\mathcal{D}_1$ **does not satisfy Assumption III.2:** $B_{1,1} = B_{2,1} = 0$, but $\boldsymbol{U}_{1,1} \in \boldsymbol{I}_{1,2^-}$ and $\boldsymbol{U}_{2,1} \in \boldsymbol{I}_{2,2^-}$.

Now, we aim to show that the optimal control strategy is nonlinear. Firstly, we can construct a strategy $(g_{1:2}^{m,*}, g_{1:2}^{a,*})$ as

$$g_{1,1}^{m,*} = g_{2,1}^{m,*} = 0, g_{1,2}^{m,*} = (0,1,0), g_{2,2}^{m,*} = (0,0,0), g_{1,1}^{a,*}(\boldsymbol{I}_{1,1^+}) = \varphi_{bi}(\boldsymbol{Y}_{1,1}), g_{2,2}^{a,*}(\boldsymbol{I}_{2,2^+}) = \varphi_{bi}^{-1}(\boldsymbol{U}_{1,1}),$$

where φ_{bi} can be any Borel measurable bijection mapping from $\mathbb{R}^2$ to $\mathbb{R}^1$, and $g_{1,2}^{a,*}, g_{2,1}^{a,*}$ can be any strategy, choosing which does not matter. Under such a construction, one can verify that $\mathbb{E}[\sum_{h=1}^2 \kappa_h + c_h \,|\, g_{1:2}^{m,*}, g_{1:2}^{a,*}] = 0$ since $\mathbb{P}(\boldsymbol{X}_3 = \boldsymbol{0} \,|\, g_{1:2}^{m,*}, g_{1:2}^{a,*}) = 1$. Therefore, $(g_{1:2}^{m,*}, g_{1:2}^{a,*})$ is a team-optimal strategy. Note that a Borel measurable bijection mapping from $\mathbb{R}^2$ to $\mathbb{R}^1$ always exists, but can only be a nonlinear function. This is because for any linear function $\mathbb{T}: \mathbb{R}^2 \to \mathbb{R}$, it holds that $\text{nullity}(\mathbb{T}) = \dim(\mathbb{R}^2) - \text{rank}(\mathbb{T}) \geq 1$ due to the *rank–nullity theorem* [21], and then $\mathbb{T}$ cannot a be bijection mapping.

However, we will show that for any strategy $(g_{1:2}^m, g_{1:2}^a)$ such that $g_{1:2}^a$ is linear, $(g_{1:2}^m, g_{1:2}^a)$ cannot be team-optimal, i.e., $J_{\mathcal{D}_1}(g_{1:2}^m, g_{1:2}^a) > 0$. We prove it by contradiction, and suppose that $(g_{1:2}^{m,'}, g_{1:2}^{a,'})$ satisfies that $J_{\mathcal{D}_1}(g_{1:2}^{m,'}, g_{1:2}^{a,'}) = 0$ and $g_{1:2}^{a,'}$ is linear. Then, firstly, it must hold that $g_{1,1}^{m,'} = g_{2,1}^{m,'} = 0, g_{2,2}^{m,'} = (0,0,0), g_{1,2}^{m,'} = (0,0,0)$ or $(0,1,0)$, since otherwise the communication cost is at least 1. Secondly, if $g_{1,2}^{m,'} = (0,0,0)$, we can change it to be $(0,1,0)$ and it is still a team-optimal strategy, since additionally sharing $\boldsymbol{U}_{1,1}$ enlarges $\boldsymbol{I}_{2,2^+}$ but incurs no communication cost. Then, we can assume $g_{1,2}^{m,'} =$

$(0,1,0)$, and under such a communication strategy we have $\boldsymbol{I}_{i,1^+} = \{\boldsymbol{Y}_{i,1}\}, \boldsymbol{I}_{2,2^+} = \{\boldsymbol{Y}_{2,1}, \boldsymbol{Y}_{2,2}, \boldsymbol{U}_{1,1}, \boldsymbol{U}_{2,1}\}, \forall i \in [2]$. Thirdly, since $g^{a,'}_{1:2}$ is linear, we can write:

$$\boldsymbol{U}_{1,1} = G^{1,1,y}_{1,1}\boldsymbol{Y}_{1,1}, \quad \boldsymbol{U}_{2,1} = G^{2,1,y}_{2,1}\boldsymbol{Y}_{2,1}, \quad \boldsymbol{U}_{2,2} = G^{2,1,y}_{2,2}\boldsymbol{Y}_{2,1} + G^{2,2,y}_{2,2}\boldsymbol{Y}_{2,2} + G^{1,1,u}_{2,2}\boldsymbol{U}_{1,1} + G^{2,1,u}_{2,2}\boldsymbol{U}_{2,1},$$

for some matrices $G^{1,1,y}_{1,1}, G^{2,1,y}_{2,1}, G^{2,1,y}_{2,2}, G^{2,2,y}_{2,2}, G^{1,1,u}_{2,2}, G^{2,1,u}_{2,2}$ with proper dimensions. Then, we can write $\boldsymbol{U}_{2,2}$ as $\boldsymbol{U}_{2,2} = G^{1,1,u}_{2,2}G^{1,1,y}_{1,1}\boldsymbol{Y}_{1,1}$. Then, we can write $J_{\mathcal{D}_1}(g^{m,'}_{1:2}, g^{a,'}_{1:2}) = \mathbb{E}[((\mathbb{I} - G^{1,1,u}_{2,2}G^{1,1,y}_{1,1})\boldsymbol{X}_1)^\top(\mathbb{I} - G^{1,1,u}_{2,2}G^{1,1,y}_{1,1})\boldsymbol{X}_1)] = \mathrm{Tr}((\mathbb{I} - G^{1,1,u}_{2,2}G^{1,1,y}_{1,1})^\top(\mathbb{I} - G^{1,1,u}_{2,2}G^{1,1,y}_{1,1}))$. Note that $G^{1,1,u}_{2,2}$ is a 1×2 matrix and $G^{1,1,y}_{1,1}$ is a 2×1 matrix. Therefore, we have rank$(G^{1,1,u}_{2,2}G^{1,1,y}_{1,1}) \leq 1$, and then $\mathrm{Tr}((\mathbb{I} - G^{1,1,u}_{2,2}G^{1,1,y}_{1,1})^\top(\mathbb{I} - G^{1,1,u}_{2,2}G^{1,1,y}_{1,1})) \geq 1$ due to Eckart-Young Theorem [22]. This means that $J_{\mathcal{D}_1}(g^{m,'}_{1:2}, g^{a,'}_{1:2}) > 0$ and leads to the contradiction. Hence, we know that there exists a team-optimal strategy of $\mathcal{D}_1$ and the control strategy of any team-optimal strategy is nonlinear, which completes the proof of **Part 1**.

Part 2: Consider an $n = 2, H = 2$ JCCO problem $\mathcal{D}_2$ with the system dynamics being defined as

$$\forall h \in [2], i \in [2], \boldsymbol{X}_h, \boldsymbol{X}_3, \boldsymbol{Y}_{i,h}, \boldsymbol{U}_{i,h} \in \mathbb{R}, \boldsymbol{X}_2 = \boldsymbol{X}_1, \boldsymbol{X}_3 = \boldsymbol{X}_2 - \boldsymbol{U}_{2,2},$$
$$\boldsymbol{Y}_{1,1} = \boldsymbol{X}_1, \boldsymbol{Y}_{1,2} = \boldsymbol{Y}_{2,1} = \boldsymbol{Y}_{2,2} = 0, \boldsymbol{X}_1 \sim \mathcal{N}(0, \frac{1}{4}), c_1 = \boldsymbol{U}^2_{1,1}, c_2 = 0, c_3 = \boldsymbol{X}^2_3.$$

For the communication part, similarly as **Part 1**, suppose the baseline sharing is null and $\mathcal{M}_{i,1} = \{0,1\}, \mathcal{M}_{i,2} = \{0,1\}^3, \forall i \in [2]$, where $\boldsymbol{M}_{i,1}$ represents whether agent i shares $\boldsymbol{Y}_{i,1}$ and each digit of $\boldsymbol{M}_{i,2}$ represents whether agent i shares $\boldsymbol{Y}_{i,1}, \boldsymbol{U}_{i,1}$, and $\boldsymbol{Y}_{i,2}$, respectively. For communication cost $\kappa_1 = \mathbb{1}[\boldsymbol{Z}^a_1 \neq \emptyset], \kappa_2 = \mathbb{1}[\boldsymbol{Z}^a_1 \backslash \{\boldsymbol{U}_{1,1}\} \neq \emptyset]$. This communication cost means that if any information except $\boldsymbol{U}_{1,1}$ is shared, then all agents will incur a communication cost of 1. The information available to each agent at each timestep is defined as $\forall i \in [2], \boldsymbol{I}_{i,1^-} = \{\boldsymbol{Y}_{i,1}\}, \boldsymbol{I}_{i,1^+} = \boldsymbol{I}_{i,1^-} \cup \boldsymbol{Z}^a_1, \boldsymbol{I}_{i,2^-} = \boldsymbol{I}_{i,1^+} \cup \{\boldsymbol{U}_{i,1}, \boldsymbol{Y}_{i,2}\}, \boldsymbol{I}_{i,2^+} = \boldsymbol{I}_{i,2^-} \cup \boldsymbol{Z}^a_2, \boldsymbol{I}_{i,3^-} = \boldsymbol{I}_{i,2^+}$.

Then, we can verify that:

- $\mathcal{D}_2$ **satisfies Assumption II.1.**
- $\mathcal{D}_2$ **has PN IS:** If there is no additional sharing, then for any $i_1, i_2 \in [2]$, $\boldsymbol{I}_{i_1,1^-} \subseteq \boldsymbol{I}_{i_2,2^-}$ if $i_1 = i_2$, otherwise agent $(i_1, 1)$ does not influence agent $(i_2, 2)$.
- $\mathcal{D}_2$ **satisfies Assumption III.4:** For any $i \in [2]$, rank$(E_{-i,2}B_{i,1})$ = rank$(B_{i,1}) = 0$.
- $\mathcal{D}_2$ **does not satisfy Assumption III.2:** $B_{1,1} = B_{2,1} = 0$, but $\boldsymbol{U}_{1,1} \in \boldsymbol{I}_{1,2^-}$ and $\boldsymbol{U}_{2,1} \in \boldsymbol{I}_{2,2^-}$.

We prove it by contradiction. We assume $(g^{m,*}_{1:2}, g^{a,*}_{1:2})$ is the team-optimal strategy of $\mathcal{D}$. Firstly, it holds that $g^{m,*}_{1,1} = g^{m,*}_{2,1} = 0, g^{m,*}_{2,2} = (0,0,0), g^{m,*}_{1,2} = (0,0,0)$ or $(0,1,0)$, since otherwise $J_{\mathcal{D}_2}(g^{m,*}_{1:2}, g^{a,*}_{1:2}) \geq \mathbb{E}[\kappa_1 + \kappa_2 \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}] \geq 1$; however, if we consider strategy $(g^m_{1:2}, g^a_{1:2})$ that shares nothing and chooses $\boldsymbol{U}_{i,h} = 0, \forall i \in [2], h \in [2]$, then $J_{\mathcal{D}_2}(g^m_{1:2}, g^a_{1:2}) = \mathbb{E}[\boldsymbol{X}^2_3 \,|\, g^m_{1:2}, g^a_{1:2}] = \mathbb{E}[\boldsymbol{X}^2_1] = \frac{1}{4}$. Secondly, we can assume $g^{m,*}_{1,2} = (0,1,0)$, otherwise we can change it to be $(0,1,0)$ and it is still a team-optimal strategy, since additionally sharing $\boldsymbol{U}_{1,1}$ enlarges the $\boldsymbol{I}_{2,2^+}$ but incurs no communication cost. Therefore, we know that under $g^{m,*}_{1:2}$, we have $\boldsymbol{I}_{i,1^+} = \{\boldsymbol{Y}_{i,1}\}, \forall i \in [2], \boldsymbol{I}_{2,2^+} = \{\boldsymbol{Y}_{2,1}, \boldsymbol{U}_{2,1}, \boldsymbol{Y}_{2,2}, \boldsymbol{U}_{1,1}\}$. Thirdly, if $\mathbb{P}(\boldsymbol{U}_{1,1} = 0 \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}) = 1$, then we consider the realization $I_{2,2^+} = \{\boldsymbol{Y}_{2,1} = 0, \boldsymbol{U}_{2,1} = g^{a,*}_{2,1}(0), \boldsymbol{Y}_{2,2} = 0, \boldsymbol{U}_{1,1} = 0\}$, and let $U_{2,2} = g^{a,*}_{2,2}(I_{2,2^+})$ be the realization of $\boldsymbol{U}_{2,2}$ when $g^{a,*}_{2,2}$ taking realization $I_{2,2^+}$ as input. Then, it holds that $\mathbb{P}(\boldsymbol{U}_{2,2} = U_{2,2} \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}) = 1$, and we have

$$J_{\mathcal{D}}(g^{m,*}_{1:2}, g^{a,*}_{1:2}) \geq \mathbb{E}[\boldsymbol{X}^2_3 \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}] = \mathbb{E}[(\boldsymbol{X}_1 - \boldsymbol{U}_{2,2})^2 \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}] = \mathbb{E}[(\boldsymbol{X}_1 - U_{2,2})^2 \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}] \geq \mathbb{E}[\boldsymbol{X}^2_1] \geq \frac{1}{4}.$$

However, we consider the strategy $(g^{m,*}_{1:2}, g^{a,'}_{1:2})$ with $g^{a,'}_{1:2}$ being defined as $g^{a,'}_{1,1}(\boldsymbol{I}_{1,1^+}) = \frac{\boldsymbol{Y}_{1,1}}{2}, g^{a,'}_{2,2}(\boldsymbol{I}_{2,2^+}) = 2\boldsymbol{U}_{1,1}$ if $\boldsymbol{U}_{1,1} \in \boldsymbol{I}_{2,2^+}$ and otherwise 0, and choose arbitrary $g^{a,'}_{2,1}, g^{a,'}_{1,2}$. Then, we can verify that

$$J_{\mathcal{D}}(g^{m,*}_{1:2}, g^{a,'}_{1:2}) = \mathbb{E}[c_1 + c_3 \,|\, g^{m,*}_{1:2}, g^{a,'}_{1:2}] = \mathbb{E}[\boldsymbol{U}^2_{1,1} + \boldsymbol{X}^2_3 \,|\, g^{m,*}_{1:2}, g^{a,'}_{1:2}] = \mathbb{E}[\frac{1}{4}\boldsymbol{Y}^2_{1,1} + (\boldsymbol{X}_1 - 2 * \frac{\boldsymbol{Y}_{1,1}}{2})^2] = \mathbb{E}[\frac{1}{4}\boldsymbol{Y}^2_{1,1}] = \frac{1}{16}.$$

Therefore, we know that it holds that $\mathbb{P}(\boldsymbol{U}_{1,1} = 0 \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}) < 1$, and then $\mathbb{E}[\boldsymbol{U}^2_{1,1} \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}] > 0$. Lastly, we can construct a new strategy $g^a_{1:2}$ based on $g^{a,*}_{1:2}$ as

$$g^a_{1,1}(\boldsymbol{I}_{1,1^+}) = \frac{1}{2}g^{a,*}_{1,1}(\boldsymbol{I}_{1,1^+}),$$
$$g^a_{2,2}(\boldsymbol{I}_{2,2^+}) = g^a_{2,2}(\boldsymbol{Y}_{2,1}, \boldsymbol{U}_{2,1}, \boldsymbol{Y}_{2,2}, \boldsymbol{U}_{1,1}) = g^{a,*}_{2,2}(\boldsymbol{Y}_{2,1}, \boldsymbol{U}_{2,1}, \boldsymbol{Y}_{2,2}, 2\boldsymbol{U}_{1,1}) \text{ if } \boldsymbol{I}_{2,2^+} = \{\boldsymbol{Y}_{2,1}, \boldsymbol{U}_{2,1}, \boldsymbol{Y}_{2,2}, \boldsymbol{U}_{1,1}\}, \text{ otherwise } 0.$$

This construction means, $g^a_{1,1}$ will choose $\boldsymbol{U}_{1,1}$ to be half of the $\boldsymbol{U}_{1,1}$ chosen by $g^{a,*}_{1,1}$, but $g^{a,*}_{2,2}, g^a_{2,2}$ choose the same $\boldsymbol{U}_{2,2}$. Therefore, we can verify that

$$\begin{aligned}
J_{\mathcal{D}}(g^{m,*}_{1:2}, g^a_{1:2}) &= \mathbb{E}[\boldsymbol{U}^2_{1,1} \,|\, g^{m,*}_{1:2}, g^a_{1:2}] + \mathbb{E}[(\boldsymbol{X}_1 - \boldsymbol{U}_{2,2})^2 \,|\, g^{m,*}_{1:2}, g^a_{1:2}] \\
&= \mathbb{E}[(\boldsymbol{U}_{1,1}/2)^2 \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}] + \mathbb{E}[(\boldsymbol{X}_1 - \boldsymbol{U}_{2,2})^2 \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}] \\
&< \mathbb{E}[\boldsymbol{U}^2_{1,1} \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}] + \mathbb{E}[(\boldsymbol{X}_1 - \boldsymbol{U}_{2,2})^2 \,|\, g^{m,*}_{1:2}, g^{a,*}_{1:2}] = J_{\mathcal{D}}(g^{m,*}_{1:2}, g^{a,*}_{1:2}),
\end{aligned}$$

where the inequality is because $\mathbb{E}[\boldsymbol{U}_{1,1}^2 \mid g_{1:2}^{m,*}, g_{1:2}^{a,*}] > 0$ as proved before. This means that for any team-optimal strategy $g_{1:2}^{m,*}, g_{1:2}^{a,*}$, we can construct a strategy $(g_{1:2}^{m,*}, g_{1:2}^{a})$ that is strictly better than it. Therefore, the team-optimal strategy does not exist. ■

3) Proof of Lemma III.5: *Proof:* Consider a JCCO problem $\mathcal{D}$ with $n = 2, H = 2$, parameters k, σ_0 specified in Proposition .3, and the system dynamics being defined as

$$\forall h \in [2], i \in [2], \boldsymbol{X}_h, \boldsymbol{X}_3, \boldsymbol{Y}_{i,h}, \boldsymbol{U}_{i,h} \in \mathbb{R}, \boldsymbol{X}_2 = \boldsymbol{X}_1 + \boldsymbol{U}_{1,1}, \boldsymbol{X}_3 = \boldsymbol{X}_2 - \boldsymbol{U}_{2,2},$$
$$\boldsymbol{Y}_{1,1} = \boldsymbol{X}_1, \boldsymbol{Y}_{1,2} = \boldsymbol{X}_2 + \boldsymbol{W}_{2,2}, \boldsymbol{Y}_{2,1} = \boldsymbol{Y}_{2,2} = 0, \boldsymbol{X}_1 \sim \mathcal{N}(0, \sigma_0^2), \boldsymbol{W}_{2,2} \sim \mathcal{N}(0, 1).$$

For the communication part, baseline sharing is null and

$$\mathcal{M}_{i,1} = \{0,1\}, \mathcal{M}_{1,2} = \{0,1\}^3, \mathcal{M}_{2,2} = \{0,1\}^2,$$

where $\boldsymbol{M}_{i,1}$ represents whether agent i shares $\boldsymbol{Y}_{i,1}$. Each digit of $\boldsymbol{M}_{1,2}$ represents whether agent 1 shares $\boldsymbol{Y}_{1,1}, \boldsymbol{U}_{1,1}$, and $\boldsymbol{Y}_{1,2}$. Each digit of $\boldsymbol{M}_{2,2}$ represents whether agent 2 shares $\boldsymbol{Y}_{2,1}$ and $\boldsymbol{Y}_{2,2}$.

For the cost, we set $\forall h \in [2], i \in [2], c_1 = k^2 \boldsymbol{U}_{1,1}^2, c_2 = 0, c_3 = \boldsymbol{X}_3^2$; for communication cost $\kappa_1 = (\sigma_0^2 + 1)\mathbb{1}[\boldsymbol{Z}_1^a \neq \emptyset], \kappa_2 = (\sigma_0^2 + 1)\mathbb{1}[\boldsymbol{Z}_2^a \backslash \{\boldsymbol{Y}_{1,2}\} \neq \emptyset]$. This communication cost means if any information except $\boldsymbol{Y}_{1,2}$ is shared through additional sharing, it will incur a communication cost of $\sigma_0^2 + 1$. The information available to each agent at each timestep is defined as: $\forall i \in [2], \boldsymbol{I}_{i,1^-} = \{\boldsymbol{Y}_{i,1}\}, \boldsymbol{I}_{i,1^+} = \boldsymbol{I}_{i,1^-} \cup \boldsymbol{Z}_1^a$, and $\boldsymbol{I}_{1,2^-} = \boldsymbol{I}_{1,1^+} \cup \{\boldsymbol{U}_{1,1}, \boldsymbol{Y}_{1,2}\}, \boldsymbol{I}_{2,2^-} = \boldsymbol{I}_{2,1^+} \cup \{\boldsymbol{Y}_{2,2}\}, \boldsymbol{I}_{i,2^+} = \boldsymbol{I}_{i,2^-} \cup \boldsymbol{Z}_2^a, \boldsymbol{I}_{1,3^-} = \boldsymbol{I}_{1,2^+}, \boldsymbol{I}_{2,3^-} = \boldsymbol{I}_{2,2^+}$. Then, we can verify that:

- $\mathcal{D}$ **satisfies Assumption II.1.**
- $\mathcal{D}$ **has PN IS:** If there is no additional sharing, then for any $i_1, i_2 \in [2]$, $\boldsymbol{I}_{i_1,1^-} \subseteq \boldsymbol{I}_{i_2,2^-}$ if $i_1 = i_2$, otherwise agent $(i_1, 1)$ does not influence agent $(i_2, 2)$.
- $\mathcal{D}$ **satisfies Assumption III.2:** Only $B_{2,1} = 0$ and $\boldsymbol{U}_{2,1}$ does not appear in $\boldsymbol{I}_{2^-}$ and $\boldsymbol{I}_{2^+}$.
- $\mathcal{D}$ **does not satisfy Assumption III.4:** It holds that $\text{rank}(E_{2,2}B_{1,1}) = 0 \neq 1 = \text{rank}(B_{1,1})$.

First, agents can never choose to share any information other than $\boldsymbol{Y}_{1,2}$ through additional sharing, since otherwise it will incur a communication cost $\kappa_h \geq \sigma_0^2 + 1$, which is larger than the total cost when each agent i chooses $\boldsymbol{U}_{i,h} = 0, \forall h \in [2]$ and shares nothing via additional sharing.

Second, if the communication strategy yields that agents share nothing through additional sharing, i.e., $\boldsymbol{Z}_1^a = \boldsymbol{Z}_2^a = \emptyset$, then $\boldsymbol{I}_{2,2^-} = \{\boldsymbol{Y}_{2,1}, \boldsymbol{Y}_{2,2}\}$, which are always zero. Then, the optimal control strategy under such a communication strategy becomes the Witsenhausen's counterexample [20], where the second agent does not use her observation, and thus cannot achieve the optimum.

Therefore, we know that the optimal communication strategy must lead to $\boldsymbol{Z}_1^a = \emptyset, \boldsymbol{Z}_2^a = \{\boldsymbol{Y}_{1,2}\}$. Then, finding a team-optimal strategy of $\mathcal{D}$ can be reduced to finding an optimal strategy of the Decentralized LQG problem in Proposition .3, where the optimal control strategy is nonlinear. ■

4) Proof of Theorem III.6: *Proof:* Fix any communication strategy $g_{1:H}^m \in \mathcal{G}_{1:H}^m$, let $M_{1:H}$ be the communication action chosen by $g_{1:H}^m \in \mathcal{G}_{1:H}^m$. Then, $\mathcal{D}$ becomes a decentralized LQG problem denoted as $\mathcal{D}(g_{1:H}^m)$, with the same system dynamics as $\mathcal{D}$ and the following information evolution: for any $h \in [H]$

(a) The common information in $\mathcal{D}(g_{1:H}^m)$ evolves as $\bar{\boldsymbol{C}}_h = \cup_{t=1}^h \bar{\boldsymbol{Z}}_t$, and $\bar{\boldsymbol{Z}}_h = \bar{\chi}_h(\bar{\boldsymbol{P}}_{h-1}, \bar{\boldsymbol{U}}_{h-1}, \bar{\boldsymbol{Y}}_h)$ for some fixed projection function $\bar{\chi}_h$. Here, $\bar{\chi}_h$ is defined as

$$\bar{\chi}_h(\bar{\boldsymbol{P}}_{h-1}, \bar{\boldsymbol{U}}_{h-1}, \bar{\boldsymbol{Y}}_h) = \chi_h(\bar{\boldsymbol{P}}_{h-1}, \bar{\boldsymbol{U}}_{h-1}, \bar{\boldsymbol{Y}}_h) \cup \phi_h(M_h, \zeta_h(\bar{\boldsymbol{P}}_{h-1}, \bar{\boldsymbol{U}}_{h-1}, \bar{\boldsymbol{Y}}_h)).$$

Recall that χ_h, ζ_h, ϕ_h are defined in Assumption II.1. Such a $\bar{\chi}_h$ is a fixed projection function since $\chi_h, \{\phi_{i,h}(M_{i,h}, \cdot)\}_{i \in [n]}$ are fixed projection functions, and for any $P_{h^-} \in \mathcal{P}_{h^-}, \phi_h(M_h, P_{h^-}) = \cup_{i=1}^n \phi_{i,h}(M_{i,h}, P_{i,h^-})$.

(b) The private information in $\mathcal{D}(g_{1:H}^m)$ evolves as follows. For any $i \in [n], \bar{\boldsymbol{P}}_{i,h} = \bar{\zeta}_{i,h}(\bar{\boldsymbol{P}}_{i,h-1}, \bar{\boldsymbol{U}}_{i,h-1}, \bar{\boldsymbol{Y}}_{i,h})$ for some fixed projection function $\bar{\zeta}_{i,h}$ and thus $\bar{\boldsymbol{P}}_h = \bar{\zeta}_h(\bar{\boldsymbol{P}}_{h-1}, \bar{\boldsymbol{U}}_{h-1}, \bar{\boldsymbol{Y}}_h)$ for some fixed projection function $\bar{\zeta}_h$. Here, $\bar{\zeta}_{i,h}$ is defined as

$$\bar{\zeta}_{i,h}(\bar{\boldsymbol{P}}_{i,h-1}, \bar{\boldsymbol{U}}_{i,h-1}, \bar{\boldsymbol{Y}}_{i,h}) = \zeta_{i,h}(\bar{\boldsymbol{P}}_{i,h-1}, \bar{\boldsymbol{U}}_{i,h-1}, \bar{\boldsymbol{Y}}_{i,h}) \backslash \phi_{i,h}(M_{i,h}, \zeta_{i,h}(\bar{\boldsymbol{P}}_{i,h-1}, \bar{\boldsymbol{U}}_{i,h-1}, \bar{\boldsymbol{Y}}_{i,h})).$$

Recall that $\zeta_{i,h}, \phi_{i,h}$ are defined in Assumption II.1. Such a $\zeta_{i,h}$ is a fixed projection function since $\zeta_{i,h}, \phi_{i,h}(M_{i,h}, \cdot)$ are fixed projection functions.

(c) For each agent $i \in [n]$, $\bar{\boldsymbol{I}}_{i,h} \subseteq \bar{\boldsymbol{I}}_{i,h+1}$, and $\bar{\boldsymbol{Y}}_{i,h} \in \bar{\boldsymbol{I}}_{i,h}$ still holds, since additional sharing only moves some private information into common information but does not remove any information.

Then, we know that $\mathcal{D}(g_{1:H}^m)$ is a decentralized LQG with some information sharing due to both the baseline sharing and the communication actions $M_{1:H}$. Now, we aim to show that $\mathcal{D}(g_{1:H}^m)$ has a PN IS. We use $\check{\cdot}$ to denote the elements in $\check{\mathcal{D}}$ (decentralized LQG induced by $\mathcal{D}$ §A). For any $i_1, i_2 \in [n], 1 \leq h_1 < h_2 \leq H$, if $\bar{\boldsymbol{U}}_{i_1,h_1}$ influences $\bar{\boldsymbol{I}}_{i_2,h_2}$ in problem $\mathcal{D}(g_{1:H}^m)$, then $B_{i_1,h_1} \neq \mathbf{0}$, since otherwise, $\bar{\boldsymbol{U}}_{i_1,h_1}$ cannot influence $\bar{\boldsymbol{X}}_{h_1+1}$, and from Assumption III.2, for any

$h' > h_1, \boldsymbol{U}_{i_1,h_1} \notin \boldsymbol{I}_{h'}$, and then $\boldsymbol{U}_{i_1,h_1} \notin \boldsymbol{Z}^a_{h'}$, which further implies that $\bar{\boldsymbol{U}}_{i_1,h_1} = \boldsymbol{U}_{i_1,h_1}$ cannot influence $\bar{\boldsymbol{I}}_{i_2,h_2}$. Therefore, from $B_{i_1,h_1} \neq \mathbf{0}$ and Assumption III.4, there exists some $j \neq i_1$ such that $\check{\boldsymbol{U}}_{i_1,h_1}$ influences $\check{\boldsymbol{Y}}_{j,h_1+1}$ in problem $\check{\mathcal{D}}$. Also from Assumption II.1(e), we know that $\check{\boldsymbol{Y}}_{j,h_1+1} \in \check{\boldsymbol{I}}_{j,h_1+1}$. Then, $\check{\boldsymbol{U}}_{i_1,h_1}$ influences $\check{\boldsymbol{I}}_{j,h_1+1}$ in $\check{\mathcal{D}}$. From the PN IS of $\mathcal{D}$, we know that $\check{\boldsymbol{I}}_{i_1,h_1} \subseteq \check{\boldsymbol{I}}_{j,h_1+1}$. Also due to $j \neq i_1$, we know that $\check{\boldsymbol{I}}_{i_1,h_1} \subseteq \check{\boldsymbol{C}}_{h_1+1} \subseteq \check{\boldsymbol{C}}_{h_2} \subseteq \check{\boldsymbol{I}}_{i_2,h_2}$. From the fact that any information shared through additional sharing is known by all the agents, we know $\bar{\boldsymbol{I}}_{i_1,h_1} \backslash \check{\boldsymbol{I}}_{i_1,h_1} \subseteq \bar{\boldsymbol{C}}_{h_1} \subseteq \bar{\boldsymbol{C}}_{h_2} \subseteq \bar{\boldsymbol{I}}_{i_2,h_2}$. Therefore, we conclude that $\bar{\boldsymbol{I}}_{i_1,h_1} \subseteq \bar{\boldsymbol{I}}_{i_2,h_2}$, and thus $\mathcal{D}(g^m_{1:H})$ has the PN IS. Therefore, the optimal strategy of $\mathcal{D}(g^m_{1:H})$ is linear. Due to the arbitrary choice of the communication strategies $g^m_{1:H}$, an optimal control strategy $g^{a,*}_{1:H}$ corresponding to the optimal $g^{m,*}_{1:H}$ is thus also linear. ■

C. Deferred Details of §IV

1) Proof of Lemma IV.1: *Proof:* From the construction, the system dynamics and control cost are the same for both decentralized LQG problems, and it holds that $\bar{\boldsymbol{I}}_{i,h} \subseteq \widetilde{\boldsymbol{I}}_{i,h}$ for any $i \in [n], h \in [H]$. Therefore, the agents in $\widetilde{\mathcal{D}}(g^m_{1:H})$ have larger strategy spaces than those in $\mathcal{D}(g^m_{1:H})$, and then $\min_{\widetilde{g}^*_{1:H} \in \widetilde{\mathcal{G}}_{1:H}} J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}^*_{1:H}) \leq \min_{\bar{g}^*_{1:H} \in \bar{\mathcal{G}}_{1:H}} J_{\mathcal{D}(g^m_{1:H})}(\bar{g}^*_{1:H})$.

Now, we want to show that for any optimal $\widetilde{g}^*_{1:H}$ of problem $\widetilde{\mathcal{D}}(g^m_{1:H})$, we can construct $\bar{g}^*_{1:H} = \varphi(\widetilde{g}^*_{1:H}, \mathcal{D}(g^m_{1:H}))$ to be a linear optimal strategy of $\mathcal{D}(g^m_{1:H})$, and $J_{\mathcal{D}(g^m_{1:H})}(\bar{g}^*_{1:H}) = J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}^*_{1:H})$. Let $\widetilde{g}^*_{1:H} \in \operatorname{argmin}_{\widetilde{g}_{1:H} \in \widetilde{\mathcal{G}}_{1:H}} J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}_{1:H})$ be any optimal linear strategy for problem $\widetilde{\mathcal{D}}(g^m_{1:H})$, we construct a strategy $\bar{g}^*_{1:H} = \varphi(\widetilde{g}^*_{1:H}, \mathcal{D}(g^m_{1:H}))$ of $\mathcal{D}(g^m_{1:H})$ in a recursive way. For $h = 1$ and any $i \in [n]$, from the construction of $\widetilde{\mathcal{D}}(g^m_{1:H})$, we know that $\widetilde{\boldsymbol{I}}_{i,1} = \bar{\boldsymbol{I}}_{i,1}$ always holds. Then we can define $\widetilde{g}^*_{i,1}(\widetilde{I}_{i,1}) = \bar{g}^*_{i,1}(\widetilde{I}_{i,1})$ for any $i \in [n], \widetilde{I}_{i,1} \in \widetilde{\mathcal{I}}_{i,1} = \bar{\mathcal{I}}_{i,1}$. Then, for any $i \in [n]$, $\widetilde{g}^*_{i,1}$ and $\bar{g}^*_{i,1}$ output the same control action, namely, $\widetilde{\boldsymbol{U}}_{i,1} = \bar{\boldsymbol{U}}_{i,1}$, and $\bar{g}^*_{i,1}$ is linear.

Now, for any $h \in [H], h > 2$, assume that we already constructed linear $\bar{g}^*_{1:t-1}$ from $\widetilde{g}_{1:t-1}$ such that for any $i \in [n], t \leq h, \bar{g}_{i,t}, \widetilde{g}_{i,t}$ output the same actions. Now, we aim to construct $\bar{g}^*_h$ from $\widetilde{g}^*_h$. From the construction of problem $\widetilde{\mathcal{D}}(g^m_{1:H})$ from $\mathcal{D}(g^m_{1:H})$, for any $i \in [n], \bar{\boldsymbol{I}}_{i,h} \subseteq \widetilde{\boldsymbol{I}}_{i,h}$ and there are only some additional actions $\widetilde{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h} \backslash \bar{\boldsymbol{I}}_{i,h}, j \in [n], t < h$. Furthermore, if $\widetilde{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h} \backslash \bar{\boldsymbol{I}}_{i,h}$, then it means that $\bar{B}_{j,t} \neq \mathbf{0}$, and $\bar{\boldsymbol{I}}_{j,t} \subseteq \bar{\boldsymbol{C}}_h$. Since $\bar{g}^*_{1:h-1}$ outputs the same actions as $\widetilde{g}^*_{1:h-1}$, then we can write $\widetilde{\boldsymbol{U}}_{j,t} = \bar{\boldsymbol{U}}_{j,t} = \bar{g}^*_{j,t}(\bar{\boldsymbol{I}}_{j,t})$, which means we can obtain $\widetilde{\boldsymbol{U}}_{j,t}$ from $\bar{g}^*_{j,t}$ and $\bar{\boldsymbol{I}}_{j,t}$. Since $\bar{g}^*_{j,t}$ is linear, we denote $\bar{g}^*_{j,t}$ as: $\bar{g}^*_{j,t}(\bar{\boldsymbol{I}}_{j,t}) = \sum_{\bar{\boldsymbol{Y}}_{k,s} \in \bar{\boldsymbol{I}}_{j,t}} \bar{G}^{k,s,y}_{j,t} \bar{\boldsymbol{Y}}_{k,s} + \sum_{\bar{\boldsymbol{U}}_{k,s} \in \bar{\boldsymbol{I}}_{j,t}} \bar{G}^{k,s,u}_{j,t} \bar{\boldsymbol{U}}_{k,s}$ for any $j \in [n], t < h$. Note that here we denote by $\sum_{\bar{\boldsymbol{Y}}_{k,s} \in \bar{\boldsymbol{I}}_{j,t}}$ enumerating all possible $k \in [n], s \in [H]$ such that $\overline{\boldsymbol{Y}}_{k,s} \in \overline{\boldsymbol{I}}_{j,t}$, and similarly for others. Meanwhile, from the linearity of $\widetilde{g}^*_{i,h}$ for any $i \in [n]$, we can write $\widetilde{g}^*_{i,h}(\widetilde{\boldsymbol{I}}_{i,h}) = \sum_{\widetilde{\boldsymbol{Y}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,y}_{i,h} \widetilde{\boldsymbol{Y}}_{j,t} + \sum_{\widetilde{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,u}_{i,h} \widetilde{\boldsymbol{U}}_{j,t}$ for some matrices $\widetilde{G}^{j,t,y}_{i,h}, \widetilde{G}^{j,t,u}_{i,h}$. Then, we have

$$\begin{aligned}
\widetilde{g}^*_{i,h}(\widetilde{\boldsymbol{I}}_{i,h}) &= \sum_{\widetilde{\boldsymbol{Y}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,y}_{i,h} \widetilde{\boldsymbol{Y}}_{j,t} + \sum_{\widetilde{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,u}_{i,h} \widetilde{\boldsymbol{U}}_{j,t} = \sum_{\bar{\boldsymbol{Y}}_{j,t} \in \bar{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,y}_{i,h} \bar{\boldsymbol{Y}}_{j,t} + \sum_{\bar{\boldsymbol{U}}_{j,t} \in \bar{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,u}_{i,h} \bar{\boldsymbol{U}}_{j,t} + \sum_{\bar{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h} \backslash \bar{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,u}_{i,h} \bar{\boldsymbol{U}}_{j,t} \\
&= \sum_{\bar{\boldsymbol{Y}}_{j,t} \in \bar{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,y}_{i,h} \bar{\boldsymbol{Y}}_{j,t} + \sum_{\bar{\boldsymbol{U}}_{j,t} \in \bar{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,u}_{i,h} \bar{\boldsymbol{U}}_{j,t} + \sum_{\bar{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h} \backslash \bar{\boldsymbol{I}}_{i,h}} \widetilde{G}^{j,t,u}_{i,h} \left(\sum_{\bar{\boldsymbol{Y}}_{k,s} \in \bar{\boldsymbol{I}}_{j,t}} \bar{G}^{k,s,y}_{j,t} \bar{\boldsymbol{Y}}_{k,s} + \sum_{\bar{\boldsymbol{U}}_{k,s} \in \bar{\boldsymbol{I}}_{j,t}} \bar{G}^{k,s,u}_{j,t} \bar{\boldsymbol{U}}_{k,s} \right).
\end{aligned}$$

The second equality in the first line is because, $\widetilde{g}^*_{1:h-1}$ and $\bar{g}^*_{1:h-1}$ output the same control actions, and the system dynamics of both problems are the same, so the observations are the same before agents take actions at timestep h. The equality between the first and the second lines is because $\widetilde{\boldsymbol{U}}_{j,t} = \bar{g}^*_{j,t}(\bar{\boldsymbol{I}}_{j,t})$ and the plugging-in of the explicit form of $\bar{g}^*_{j,t}$. Also, if $\widetilde{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h} \backslash \bar{\boldsymbol{I}}_{i,h}$, then $\bar{\boldsymbol{I}}_{j,t} \subseteq \bar{\boldsymbol{I}}_{i,h}$, and any $\bar{\boldsymbol{Y}}_{k,s} \in \bar{\boldsymbol{I}}_{j,t}, \bar{\boldsymbol{U}}_{k,s} \in \bar{\boldsymbol{I}}_{j,t}$ will also be included in $\bar{\boldsymbol{I}}_{i,h}$. Therefore, we have

$$\begin{aligned}
\widetilde{g}^*_{i,h}(\widetilde{\boldsymbol{I}}_{i,h}) = &\sum_{\bar{\boldsymbol{Y}}_{j,t} \in \bar{\boldsymbol{I}}_{i,h}} \left(\widetilde{G}^{i,t,y}_{i,h} + \sum_{\bar{\boldsymbol{U}}_{k,s} \in \widetilde{\boldsymbol{I}}_{i,h} \backslash \bar{\boldsymbol{I}}_{k,s}} \widetilde{G}^{k,s,u}_{i,h} \mathbb{1}[\bar{\boldsymbol{Y}}_{j,t} \in \bar{\boldsymbol{I}}_{k,s}] \bar{G}^{j,t,y}_{k,s} \right) \bar{\boldsymbol{Y}}_{j,t} \\
&+ \sum_{\bar{\boldsymbol{U}}_{j,t} \in \bar{\boldsymbol{I}}_{i,h}} \left(\widetilde{G}^{i,t,u}_{i,h} + \sum_{\bar{\boldsymbol{U}}_{k,s} \in \widetilde{\boldsymbol{I}}_{i,h} \backslash \bar{\boldsymbol{I}}_{k,s}} \widetilde{G}^{k,s,u}_{i,h} \mathbb{1}[\bar{\boldsymbol{U}}_{j,t} \in \bar{\boldsymbol{I}}_{k,s}] \bar{G}^{j,t,u}_{k,s} \right) \bar{\boldsymbol{U}}_{j,t}.
\end{aligned}$$

Note that in the equation above, if $\bar{\boldsymbol{Y}}_{j,t} \notin \bar{\boldsymbol{I}}_{k,s}$, then it means that $\bar{\boldsymbol{U}}_{k,s}$ is not based on $\bar{\boldsymbol{Y}}_{j,t}$, and we can just assign $\bar{G}^{j,t,y}_{k,s} = \mathbf{0}$.

Also, if $\bar{\boldsymbol{U}}_{j,t} \notin \bar{\boldsymbol{I}}_{k,s}$, then we assign $\bar{G}^{j,t,u}_{k,s} = \mathbf{0}$. Therefore, we can construct $\bar{g}^*_{i,h}$ for any $i \in [n]$ as

$$\bar{g}^*_{i,h}(\bar{\boldsymbol{I}}_{i,h}) = \sum_{\bar{\boldsymbol{Y}}_{j,t} \in \bar{\boldsymbol{I}}_{i,h}} \bar{G}^{j,t,y}_{i,h} \bar{\boldsymbol{Y}}_{j,t} + \sum_{\bar{\boldsymbol{U}}_{j,t} \in \bar{\boldsymbol{I}}_{i,h}} \bar{G}^{j,t,u}_{i,h} \bar{\boldsymbol{U}}_{j,t},$$

$$\forall j \in [n], t \leq h, \bar{G}^{j,t,y}_{i,h} := \widetilde{G}^{i,t,y}_{i,h} + \sum_{\bar{\boldsymbol{U}}_{k,s} \in \widetilde{\boldsymbol{I}}_{i,h} \setminus \bar{\boldsymbol{I}}_{k,s}} \widetilde{G}^{k,s,u}_{i,h} \mathbb{1}[\bar{\boldsymbol{Y}}_{j,t} \in \bar{\boldsymbol{I}}_{k,s}] \bar{G}^{j,t,y}_{k,s}$$

$$\forall j \in [n], t \leq h, \bar{G}^{j,t,u}_{i,h} := \widetilde{G}^{i,t,u}_{i,h} + \sum_{\bar{\boldsymbol{U}}_{k,s} \in \widetilde{\boldsymbol{I}}_{i,h} \setminus \bar{\boldsymbol{I}}_{k,s}} \widetilde{G}^{k,s,u}_{i,h} \mathbb{1}[\bar{\boldsymbol{U}}_{j,t} \in \bar{\boldsymbol{I}}_{k,s}] \bar{G}^{j,t,u}_{k,s}.$$

Then, $\bar{g}^*_h$ is linear, and outputs the same actions as $\widetilde{g}^*_h$. Recursively, we can construct $\bar{g}^*_h$ from $h = 1$ to H based on $\widetilde{g}^*_{1:H}$ and the problem $\mathcal{D}(g^m_{1:H})$ (, and the problem $\widetilde{\mathcal{D}}(g^m_{1:H})$, which can be constructed by $\mathcal{D}(g^m_{1:H})$), and we define such a construction process as some function ϕ, denoting $\bar{g}^*_{1:H} = \varphi(\widetilde{g}^*_{1:H}, \mathcal{D}(g^m_{1:H}))$. Since the dynamics and control costs are the same in the two problems, and $\bar{g}^*_{1:H}$ and $\widetilde{g}^*_{1:H}$ output the same actions, then $J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}^*_{1:H}) = J_{\mathcal{D}(g^m_{1:H})}(\bar{g}^*_{1:H})$. Also, we know that

$$J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}^*_{1:H}) = \min_{\widetilde{g}^*_{1:H} \in \widetilde{\mathcal{G}}_{1:H}} J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}^*_{1:H}) \leq \min_{\bar{g}^*_{1:H} \in \bar{\mathcal{G}}_{1:H}} J_{\mathcal{D}(g^m_{1:H})}(\bar{g}^*_{1:H}) \leq J_{\mathcal{D}(g^m_{1:H})}(\bar{g}^*_{1:H}).$$

Therefore, $\bar{g}^*_{1:H}$ is a linear optimal strategy of $\mathcal{D}(g^m_{1:H})$. ■

2) Proof of Theorem IV.2: *Proof:* The proof consists of three **Parts**: **Part 1:** $\widetilde{\mathcal{D}}(g^m_{1:H})$ is PN; **Part 2:** $\widetilde{\mathcal{D}}(g^m_{1:H})$ has the information evolution rules; **Part 3:** $\widetilde{\mathcal{D}}(g^m_{1:H})$ satisfies the SI-CIB condition.

Part 1: We want to prove that $\widetilde{\mathcal{D}}(g^m_{1:H})$ is PN.
For any $i_1, i_2 \in [n], h \in [H]$, if $\widetilde{B}_{i_1,h_1} = \mathbf{0}$, then from Assumption III.4, for any $h' > h_1, \boldsymbol{U}_{i_1,h_1} \notin \boldsymbol{I}_{(h')^+}$, then $\bar{\boldsymbol{U}}_{i_1,h_1} \notin \bar{\boldsymbol{I}}_{h'}$, and then $\widetilde{\boldsymbol{U}}_{i_1,h_1} \notin \widetilde{\boldsymbol{I}}_{h'}$ from the construction of $\widetilde{\mathcal{D}}(g^m_{1:H})$. Therefore, $\widetilde{\boldsymbol{U}}_{i_1,h_1}$ does not influence system dynamics, and does not appear in any information $\widetilde{\boldsymbol{I}}_{i,h'}$ for any $h' > h_1, i \in [n]$. Then, $\widetilde{\boldsymbol{U}}_{i_1,h_1}$ does not influence any $\widetilde{\boldsymbol{I}}_{i,h'}$ for any $h' > h_1, i \in [n]$.

If $\widetilde{B}_{i_1,h_1} = \bar{B}_{i_1,h_1} \neq \mathbf{0}$. From Assumption III.4, there exists $i \neq i_1$ such that $\bar{\boldsymbol{U}}_{i_1,h_1}$ influences $\bar{\boldsymbol{Y}}_{i_1,h_1}$. From Assumption II.1 (e), $\bar{\boldsymbol{Y}}_{i,h_1+1} \in \bar{\boldsymbol{I}}_{i,h_1+1} \subseteq \bar{\boldsymbol{I}}_{i,h_1+1} \subseteq \bar{\boldsymbol{I}}_{i,h_2}$, which means that $\bar{\boldsymbol{U}}_{i_1,h_1}$ influences $\bar{\boldsymbol{I}}_{i,h_2}$. Also, from Theorem III.6, we know that $\mathcal{D}(g^m_{1:H})$ is PN, then $\bar{\boldsymbol{I}}_{i_1,h_1} \subseteq \bar{\boldsymbol{I}}_{i,h_2}$. Since $i \neq i_1$, this can only happen when $\bar{\boldsymbol{I}}_{i_1,h_1} \subseteq \bar{\boldsymbol{C}}_{h_2}$, and then $\bar{\boldsymbol{I}}_{i_1,h_1} \subseteq \bar{\boldsymbol{I}}_{i_2,h_2}$. Also, we know $\widetilde{\boldsymbol{I}}_{i_1,h_1} \setminus \bar{\boldsymbol{I}}_{i_1,h_1} \subseteq \widetilde{\boldsymbol{C}}_{h_1} \subseteq \widetilde{\boldsymbol{C}}_{h_2} \subseteq \widetilde{\boldsymbol{I}}_{i_2,h_2}$. Therefore, $\widetilde{\boldsymbol{I}}_{i_1,h_1} \subseteq \widetilde{\boldsymbol{I}}_{i_2,h_2}$. This means that $\widetilde{\mathcal{D}}(g^m_{1:H})$ is a decentralized LQG with PN IS.

Part 2: We want to prove $\widetilde{\mathcal{D}}(g^m_{1:H})$ has the information evolution rules. From Theorem III.6, we know that the problem $\mathcal{D}(g^m_{1:H})$ has the information evolution rules with some projection functions $\{\bar{\chi}_h\}_{h \in [H+1]}, \{\bar{\zeta}_{i,h}\}_{i \in [n], h \in [H+1]}$.

For any $i \in [n], h \in [H]$, since which random variables in $\widetilde{\boldsymbol{Z}}_h, \widetilde{\boldsymbol{P}}_{i,h}$ are fixed, it suffices to prove that $\widetilde{\boldsymbol{Z}}_h \subseteq \widetilde{\boldsymbol{P}}_{h-1} \cup \{\widetilde{\boldsymbol{U}}_{h-1}, \widetilde{\boldsymbol{Y}}_h\}, \widetilde{\boldsymbol{P}}_{i,h} \subseteq \widetilde{\boldsymbol{P}}_{i,h} \cup \{\widetilde{\boldsymbol{U}}_{i,h}, \widetilde{\boldsymbol{Y}}_{i,h+1}\}$.

Firstly, we show that the common information of $\widetilde{\mathcal{D}}(g^m_{1:H})$ has the evolution rule. For any $h \in [H]$, consider the $\widetilde{\boldsymbol{Z}}_h$. If any $\widetilde{\boldsymbol{Y}}_{i,t} \in \widetilde{\boldsymbol{Z}}_h, i \in [n], t \leq h$, then $\widetilde{\boldsymbol{Y}}_{i,t} \in \widetilde{\boldsymbol{C}}_h$ and $\widetilde{\boldsymbol{Y}}_{i,t} \notin \widetilde{\boldsymbol{C}}_{h-1}$. Then $\bar{\boldsymbol{Y}}_{i,t} \in \bar{\boldsymbol{C}}_h$ and $\bar{\boldsymbol{Y}}_{i,t} \notin \bar{\boldsymbol{C}}_{h-1}$ since the construction does not change any observation in the information. Therefore, $\bar{\boldsymbol{Y}}_{i,t} \in \bar{\boldsymbol{Z}}_h \subseteq \bar{\boldsymbol{P}}_{h-1} \cup \{\bar{\boldsymbol{Y}}_h\}$ due to the evolution rule of problem $\mathcal{D}(g^m_{1:H})$, and then $\widetilde{\boldsymbol{Y}}_{i,t} \in \widetilde{\boldsymbol{P}}_{h-1} \cup \{\widetilde{\boldsymbol{Y}}_h\}$. If any $\widetilde{\boldsymbol{U}}_{i,t} \in \widetilde{\boldsymbol{Z}}_h$, then $\bar{B}_{i,t} \neq \mathbf{0}$ from Assumption III.4 and construction of $\widetilde{\mathcal{D}}(g^m_{1:H})$. If $t < h-1$, we know $\bar{\boldsymbol{U}}_{i,t}$ influences $\bar{\boldsymbol{X}}_{t+1}$ and thus influences $\bar{\boldsymbol{Y}}_{j,t+1}$ for some $j \neq i$ due to Assumption III.4. Then, from Theorem III.6, we know that $\bar{\boldsymbol{U}}_{i,t} \in \bar{\boldsymbol{C}}_{t+1}$. Therefore, $\widetilde{\boldsymbol{U}}_{i,t} \in \widetilde{\boldsymbol{C}}_{t+1}$ since the construction does not change any observation in the information. From $t < h-1$ we know $\widetilde{\boldsymbol{U}}_{i,t} \notin \widetilde{\boldsymbol{Z}}_h$, which leads to a contradiction. Hence, if $\widetilde{\boldsymbol{U}}_{i,t} \in \widetilde{\boldsymbol{Z}}_h$, it is only possible that $t = h-1$. In conclusion, $\widetilde{\boldsymbol{Z}}_h \subseteq \widetilde{\boldsymbol{P}}_{h-1} \cup \{\widetilde{\boldsymbol{U}}_{h-1}, \widetilde{\boldsymbol{Y}}_h\}$.

Secondly, we show that the private information of $\widetilde{\mathcal{D}}(g^m_{1:H})$ satisfies the evolution rule. For any $i \in [n], h \in [H]$, consider the $\widetilde{\boldsymbol{P}}_{i,h}$. If any $\widetilde{\boldsymbol{Y}}_{i,t} \in \widetilde{\boldsymbol{P}}_{i,h}, t \leq h$, then $\bar{\boldsymbol{Y}}_{i,t} \in \bar{\boldsymbol{P}}_{i,h}$ since the construction does not change any observation in the information. Therefore, $\bar{\boldsymbol{Y}}_{i,t} \in \bar{\boldsymbol{P}}_{i,h-1} \cup \{\bar{\boldsymbol{Y}}_{i,h}\}$ due to the evolution rule of problem $\mathcal{D}(g^m_{1:H})$, and then $\widetilde{\boldsymbol{Y}}_{i,t} \in \widetilde{\boldsymbol{P}}_{i,h-1} \cup \{\widetilde{\boldsymbol{Y}}_{i,h}\}$. If any $\widetilde{\boldsymbol{U}}_{i,t} \in \widetilde{\boldsymbol{P}}_{i,h}$, then $\bar{B}_{i,t} \neq \mathbf{0}$ from Assumption III.4, since otherwise it will be removed from $\bar{\boldsymbol{P}}_{i,h}$ and $\widetilde{\boldsymbol{P}}_{i,h} \subseteq \bar{\boldsymbol{P}}_{i,h}$. Then, we know $\bar{\boldsymbol{U}}_{i,t}$ influences $\bar{\boldsymbol{X}}_{t+1}$ and thus influences $\bar{\boldsymbol{Y}}_{j,t+1}$ for some $j \neq i$ due to Assumption III.4. Also, from Theorem III.6, we know that $\bar{\boldsymbol{U}}_{i,t} \in \bar{\boldsymbol{C}}_{t+1}$. Therefore, $\bar{\boldsymbol{U}}_{i,t} \notin \bar{\boldsymbol{P}}_{i,h}$ and thus $\widetilde{\boldsymbol{U}}_{i,t} \notin \widetilde{\boldsymbol{P}}_{i,h}$, which leads to a contradiction. In conclusion, $\widetilde{\boldsymbol{P}}_{i,h} \subseteq \widetilde{\boldsymbol{P}}_{i,h-1} \cup \{\widetilde{\boldsymbol{Y}}_h\}$.

Part 3: $\mathcal{D}(g^m_{1:H})$ satisfies the SI-CIB condition.
For each $h \in [n]$, consider the space $\widetilde{\mathcal{S}}_h = \widetilde{\mathcal{X}} \times \widetilde{\mathcal{P}}_{1,h} \times \cdots \times \widetilde{\mathcal{P}}_{n,h}$, it is a finite-dimensional vector space since the elements in each $\boldsymbol{P}_{i,h}, i \in [n]$ are fixed. Then, for any fixed cube $\mathcal{Q}_h = \mathcal{Q}^x_h \times \mathcal{Q}^1_h \times \cdots \times \mathcal{Q}^n_h$ such that $\mathcal{Q}_h \subseteq \widetilde{\mathcal{S}}_h, \mathcal{Q}^x_h \subseteq \widetilde{\mathcal{X}}, \mathcal{Q}^i_h \subseteq \widetilde{\mathcal{P}}_{i,h}, \forall i \in [n]$. Define the probability $\mathbb{P}_1 = \mathbb{P}(\widetilde{\boldsymbol{S}}_h \in \mathcal{Q}_h \,|\, \widetilde{\boldsymbol{C}}_h = \widetilde{C}_h, \widetilde{g}_{1:h-1}), \mathbb{P}_2 = \mathbb{P}(\widetilde{\boldsymbol{S}}_h \in \mathcal{Q}_h \,|\, \widetilde{\boldsymbol{C}}_h = \widetilde{C}_h, \widetilde{g}'_{1:h-1})$ for any fixed $C_h \in \mathcal{C}_h$, and any two control strategies $\widetilde{g}_{1:h-1}, \widetilde{g}'_{1:h-1}$ that can reach $\widetilde{C}_h$. It suffices to prove that $\mathbb{P}_1 = \mathbb{P}_2$. Without loss of generality, we can assume $\widetilde{g}_{1:h-1}, \widetilde{g}'_{1:h-1}$ only differ in $\widetilde{g}_{i_1,h_1}, \widetilde{g}'_{i_1,h_1}$ for some $i_1 \in [n], h_1 < h$.
If $\widetilde{B}_{i_1,h_1} \neq \mathbf{0}$, then $\bar{\boldsymbol{U}}_{i_1,h_1}$ influences $\bar{\boldsymbol{X}}_{h_1+1}$ in problem $\mathcal{D}(g^m_{1:H})$. From Assumption III.4, there exists $i_2 \neq i_1, \bar{\boldsymbol{U}}_{i_1,h_1}$

influences $\bar{\boldsymbol{Y}}_{i_2,h_1+1}$ in problem $\mathcal{D}(g^m_{1:H})$. Then from the PN IS of $\mathcal{D}(g^m_{1:H})$, we know $\bar{\boldsymbol{I}}_{i_1,h_1} \subseteq \bar{\boldsymbol{I}}_{i_2,h_1+1}$ and thus $\bar{\boldsymbol{I}}_{i_1,h_1} \subseteq \bar{\boldsymbol{C}}_{h_1+1}$. From the construction of $\widetilde{\mathcal{D}}(g^m_{1:H})$, we know that $\widetilde{\boldsymbol{U}}_{i_1,h_1} \subseteq \widetilde{\boldsymbol{C}}_{h_1+1} \subseteq \widetilde{\boldsymbol{C}}_h$. Also, it holds that $\widetilde{\boldsymbol{I}}_{i_1,h_1} \backslash \bar{\boldsymbol{I}}_{i_1,h_1} \subseteq \widetilde{\boldsymbol{C}}_{h_1}$ and $\bar{\boldsymbol{I}}_{i_1,h_1} \subseteq \bar{\boldsymbol{C}}_{h_1+1} \subseteq \widetilde{\boldsymbol{C}}_{h_1+1} \subseteq \widetilde{\boldsymbol{C}}_h$, so $\widetilde{\boldsymbol{I}}_{i_1,h_1} \subseteq \widetilde{\boldsymbol{C}}_h$. Since the input and output of $\widetilde{g}_{i_1,h_1}$ and $\widetilde{g}'_{i_1,h_1}$ are $\widetilde{\boldsymbol{I}}_{i_1,h_1}$ and $\widetilde{\boldsymbol{U}}_{i_1,h_1}$, which are contained in $\widetilde{\boldsymbol{C}}_h$, then we can conclude that $\mathbb{P}_1 = \mathbb{P}_2$.

If $\widetilde{B}_{i_1,h_1} = \mathbf{0}$, then $\widetilde{\boldsymbol{U}}_{i_1,h_1}$ will never influence the underlying system. Also, from Assumption III.2, $\bar{\boldsymbol{U}}_{i_1,h_1}$ will never appear in $\bar{\boldsymbol{C}}_{h'}$ and $\bar{\boldsymbol{P}}_{i,h'}$ for any $h' \in [H], i \in [n]$. From construction of $\widetilde{\mathcal{D}}(g^m_{1:H})$, $\widetilde{\boldsymbol{U}}_{i_1,h_1}$ will never appear in $\widetilde{\boldsymbol{C}}_{h'}$ and $\widetilde{\boldsymbol{P}}_{i,h'}$ for any $h' \in [H], i \in [n]$. Then, $\widetilde{\boldsymbol{U}}_{i_1,h_1}$ will never appear in the input of strategy $\widetilde{g}_{i,h'}$ for any $h' \in [H], i \in [n]$, and thus does not influence how it chooses $\widetilde{g}_{i,h'}$. Therefore, the system and information of each agent i at any timestep h' are the same no matter how agent i chooses $\widetilde{\boldsymbol{U}}_{i_1,h_1}$. Thus, we have $\mathbb{P}_1 = \mathbb{P}_2$, which completes the proof. ■

3) Proof of Lemma IV.3: This lemma is introduced in [8]. Note that there is slight difference between the settings of [8] and ours, where we allow the noises $\boldsymbol{W}_{0:n,1:h}$ (as defined in Eq. (II.1)) without full support. Therefore, we provide the proof for our setting.

Proof: The proof consists of two **Parts**.

Part 1: we aim to show that $\widetilde{\boldsymbol{B}}_h$ admits a Gaussian distribution. It suffices to show that, for any $h \in [H]$, strategy $\widetilde{g}_{1:h-1} \in \widetilde{\mathcal{G}}_{1:h-1}$ (not necessary to be linear), and realization $\widetilde{C}_h \in \widetilde{\mathcal{C}}_h$ that is reachable under $\widetilde{g}_{1:h-1}$, the belief $\Pi_h(\widetilde{C}_h, \widetilde{g}_{1:h-1})$ is a Gaussian distribution.

Since $\widetilde{C}_h$ is reachable under $\widetilde{g}_{1:h-1}$, there exists realizations $\widetilde{Y}_{1:h}, \widetilde{U}_{1:h-1}$ such that $(\widetilde{Y}_{1:h}, \widetilde{U}_{1:h-1}, \widetilde{C}_h)$ is reachable under $\widetilde{g}_{1:h-1}$. Then, we construct affine strategies $\widetilde{g}'_{1:h-1} \in \widetilde{\mathcal{G}}_{1:h-1}$ as: $\forall i \in [n], t < h, \forall \widetilde{I}_{i,t} \in \widetilde{\mathcal{I}}_{i,t}, \widetilde{g}'_{i,t}(\widetilde{I}_{i,t}) \equiv \widetilde{U}_{i,t}$. Then, $(\widetilde{Y}_{1:h}, \widetilde{U}_{1:h-1}, \widetilde{C}_h)$ is also reachable under $\widetilde{g}'_{1:h-1}$. Also, since $\widetilde{g}'_{1:h-1}$ are affine strategies and system dynamics are linear, then $\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_h, \widetilde{\boldsymbol{C}}_h$ are jointly Gaussian. Then, the distribution of $\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_h$ conditioned on $\widetilde{\boldsymbol{C}}_h$ is jointly Gaussian with covariance independent to $\widetilde{\boldsymbol{C}}_h$. Finally, from the SI-CIB condition, it holds that $\Pi_h(\widetilde{C}_h, \widetilde{g}_{1:h-1}) = \Pi_h(\widetilde{C}_h, \widetilde{g}'_{1:h-1})$, which is a Gaussian distribution and the covariance independent to $\widetilde{C}_h$.

Part 2: we aim to show that the conditional mean $\widetilde{\boldsymbol{\Theta}}_h$ and $\widetilde{\Sigma}_h$ evolves as:

$$\widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Psi}^1_h(\widetilde{\boldsymbol{\Theta}}_{h-1}, \widetilde{\boldsymbol{Z}}_h), \qquad \widetilde{\Sigma}_h = \widetilde{\Psi}^2_h(\widetilde{\Sigma}_{h-1}),$$

for some functions $\widetilde{\Psi}^1_h, \widetilde{\Psi}^2_h$.

Firstly, we want to show that the belief $\widetilde{\boldsymbol{B}}_h$ evolves as:

$$\widetilde{\boldsymbol{B}}_h = \widetilde{\Phi}^1_h(\widetilde{\boldsymbol{B}}_{h-1}, \widetilde{\boldsymbol{Z}}_h, \widetilde{\gamma}_{h-1}),$$

for some function $\widetilde{\Phi}^1_h$. This is because for any Borel set $\widetilde{\mathscr{S}_h} \subseteq \widetilde{\mathcal{S}}_h$, we have

$$\begin{aligned}
&\mathbb{P}[(\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_h) \in \widetilde{\mathscr{S}_h} \,|\, \widetilde{\boldsymbol{C}}_h] = \int_{\widetilde{\mathscr{S}_h}} \varrho_h(\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_h \,|\, \widetilde{\boldsymbol{C}}_h) \mathrm{d}\widetilde{\boldsymbol{X}}_h \mathrm{d}\widetilde{\boldsymbol{P}}_h \\
&= \int_{\widetilde{\mathcal{W}}_{1:n,h}} \Xi_{1:n,h}(\widetilde{\boldsymbol{W}}_{1:n,h}) \int_{\widetilde{\mathcal{W}}_{0,h-1}} \Xi_{0,h-1}(\widetilde{\boldsymbol{W}}_{0,h-1}) \int_{\widetilde{\mathcal{X}} \times \widetilde{\mathcal{P}}_{h-1}} \varrho_{h-1}(\widetilde{\boldsymbol{X}}_{h-1}, \widetilde{\boldsymbol{P}}_{h-1} \,|\, \widetilde{\boldsymbol{C}}_{h-1}) \\
&\quad \int_{\widetilde{\mathcal{X}} \times \widetilde{\mathcal{P}}_h} \mathbb{1}[(\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_h) \in \widetilde{\mathscr{S}_h}] \mathbb{1}[\widetilde{\boldsymbol{X}}_h = \widetilde{A}_h \widetilde{\boldsymbol{X}}_{h-1} + \widetilde{B}_h \widetilde{\gamma}_{h-1}(\widetilde{\boldsymbol{P}}_{h-1}) + \widetilde{\boldsymbol{W}}_{0,h-1}] \\
&\quad \mathbb{1}[\widetilde{\boldsymbol{Z}}_h = \widetilde{\chi}_h(\widetilde{\boldsymbol{P}}_{h-1}, \widetilde{\gamma}_{h-1}(\widetilde{\boldsymbol{P}}_{h-1}), \widetilde{E}_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{W}}_{1:n,h})] \\
&\quad \mathbb{1}[\widetilde{\boldsymbol{P}}_h = \widetilde{\zeta}_h(\widetilde{\boldsymbol{P}}_{h-1}, \widetilde{\gamma}_{h-1}(\widetilde{\boldsymbol{P}}_{h-1}), \widetilde{E}_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{W}}_{1:n,h})] \mathrm{d}\widetilde{\boldsymbol{W}}_{1:n,h} \mathrm{d}\widetilde{\boldsymbol{W}}_{0,h-1} \mathrm{d}\widetilde{\boldsymbol{X}}_{h-1} \mathrm{d}\widetilde{\boldsymbol{P}}_{h-1} \mathrm{d}\widetilde{\boldsymbol{X}}_h \mathrm{d}\widetilde{\boldsymbol{P}}_h
\end{aligned}$$

where ϱ_{h-1}, ϱ_h are the density functions of $\widetilde{\boldsymbol{B}}_{h-1}, \widetilde{\boldsymbol{B}}_h$, $\widetilde{\mathcal{W}_{1:n,h}}$ is the space of Gaussian noise $\widetilde{\boldsymbol{W}}_{1:n,h}$, $\widetilde{\mathcal{W}_{0,h-1}}$ is the space of $\widetilde{\boldsymbol{W}}_{0,h-1}$, $\widetilde{\Xi}_{0,h-1}$ is the density function of Gaussian noise $\widetilde{\boldsymbol{W}}_{0,h-1}$, $\widetilde{\Xi}_{1:n,h}(\widetilde{\boldsymbol{W}}_{1:n,h})$ is the joint density function of Gaussian noise $\widetilde{\boldsymbol{W}}_{1:n,h}$. Here, we denote $\widetilde{\boldsymbol{W}}_{1:n,h} = \begin{bmatrix} \widetilde{\boldsymbol{W}}_{1,h} \\ \cdots \\ \widetilde{\boldsymbol{W}}_{n,h} \end{bmatrix}$, and $\widetilde{\gamma}_{h-1}(\widetilde{\boldsymbol{P}}_{h-1}) = \begin{bmatrix} \widetilde{\gamma}_{1,h-1}(\widetilde{\boldsymbol{P}}_{1,h-1}) \\ \cdots \\ \widetilde{\gamma}_{n,h-1}(\widetilde{\boldsymbol{P}}_{n,h-1}) \end{bmatrix}$. Therefore, the right-hand side of the equation only depends on $\varrho_{h-1}, \widetilde{\gamma}_{h-1}, \widetilde{\boldsymbol{Z}}_h$ and this means $\widetilde{\boldsymbol{B}}_h = \widetilde{\Phi}^1_h(\widetilde{\boldsymbol{B}}_{h-1}, \widetilde{\gamma}_{h-1}, \widetilde{\boldsymbol{Z}}_h)$ for some $\widetilde{\Phi}^1_h$.

Secondly, from **Part 1**, the covariance $\widetilde{\Sigma}_h$ does not depend on $\widetilde{\boldsymbol{C}}_h$. Furthermore, since $\widetilde{\mathcal{D}}(g^m_{1:H})$ satisfies SI-CIB condition, $\widetilde{\Sigma}_h$ does not depend on $\widetilde{g}_{1:h-1}$, either. Therefore, $\widetilde{\Sigma}_h$ is pre-defined and satisfies $\widetilde{\Sigma}_h = \widetilde{\Psi}^2_h(\widetilde{\Sigma}_{h-1})$ for some function $\widetilde{\Psi}^2_h$. Meanwhile, since $\widetilde{\boldsymbol{B}}_h$ is a Gaussian distribution, which can be characterized by the mean $\widetilde{\boldsymbol{\Theta}}_h$ and $\widetilde{\Sigma}_h$, then there exists some function $\widetilde{\Phi}^3_h$ such that $(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\Sigma}_h) = \widetilde{\Phi}^3_h(\widetilde{\boldsymbol{\Theta}}_{h-1}, \widetilde{\Sigma}_{h-1}, \widetilde{\boldsymbol{Z}}_h, \widetilde{\gamma}_{h-1})$. Also, since $\widetilde{\mathcal{D}}(g^m_{1:H})$ satisfies SI-CIB condition, then $(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\Sigma}_h)$ cannot depend on $\widetilde{g}_{h-1}$ and $\widetilde{\gamma}_h$. Therefore, together with the fact that $\widetilde{\Sigma}_{h-1}$ is pre-defined, it holds that $\widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Psi}^1_h(\widetilde{\boldsymbol{\Theta}}_{h-1}, \widetilde{\boldsymbol{Z}}_h)$ for some function $\widetilde{\Psi}^1_h$, which completes the proof. ■

4) Proof of Theorem IV.4: *Proof:* **Part 1:** we aim to prove that $\mathcal{D}^{\ddagger}(g^m_{1:H})$ is a Markov decision problem. From the construction of $\mathcal{D}^{\ddagger}(g^m_{1:H})$, the cost function $c^{\ddagger}_h$ is only a function of state $\widetilde{\boldsymbol{\Theta}}_h$ and action $\widetilde{\gamma}_h$ for any $h \in [H]$, and $c^{\ddagger}_{H+1}$ is only a function of state $\widetilde{\boldsymbol{\Theta}}_{H+1}$. Also, from Lemma IV.3, we know that for any $h \in [H]$, state of $\mathcal{D}^{\ddagger}(g^m_{1:H})$ evolves as $\widetilde{\boldsymbol{\Theta}}_{h+1} = \widetilde{\Psi}^1_h(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\boldsymbol{Z}}_{h+1})$. Therefore, in order to show $\widetilde{\boldsymbol{\Theta}}_{h+1}$ only depends on $\widetilde{\boldsymbol{\Theta}}_h$ and $\widetilde{\gamma}_h$, it suffices to show that $\widetilde{\boldsymbol{Z}}_{h+1}$ only depends on $\widetilde{\boldsymbol{\Theta}}_h$ and $\widetilde{\gamma}_h$, independent of $\widetilde{\boldsymbol{C}}_h$, $\widetilde{\boldsymbol{\Theta}}_{1:h-1}$, and $\widetilde{\gamma}_{1:h-1}$. From Theorem IV.2, we know that $\widetilde{\boldsymbol{Z}}_{h+1} = \widetilde{\chi}_{h+1}(\widetilde{\boldsymbol{P}}_h, \widetilde{\boldsymbol{U}}_h, \widetilde{\boldsymbol{Y}}_{h+1})$ for some projection function $\widetilde{\chi}_{h+1}$. Then, for any arbitrary Borel set $\widetilde{\mathscr{Z}}_{h+1} \subseteq \widetilde{\mathcal{Z}}_{h+1}$, we can compute

$$
\begin{aligned}
&\mathbb{P}(\widetilde{\boldsymbol{Z}}_{h+1} \in \widetilde{\mathscr{Z}}_{h+1} \,|\, \widetilde{\boldsymbol{C}}_h, \widetilde{\boldsymbol{\Theta}}_{1:h}, \widetilde{\gamma}_{1:h-1}) = \int_{\widetilde{\mathscr{Z}}_{h+1}} \int_{\widetilde{\mathcal{X}}_h \times \widetilde{\mathcal{P}}_h} \int_{\widetilde{\mathcal{Y}}_{h+1} \times \widetilde{\mathcal{X}}_{h+1}} \mathbb{1}[\widetilde{Z}_{h+1} = \widetilde{\chi}_{h+1}(\widetilde{P}_h, \widetilde{\gamma}_h(\widetilde{P}_h), \widetilde{Y}_{h+1})] \\
&\quad \mathbb{P}(\widetilde{X}_{h+1} \,|\, \widetilde{\boldsymbol{X}}_h = \widetilde{X}_h, \widetilde{\gamma}_h(\widetilde{P}_h)) \mathbb{P}(\widetilde{Y}_{h+1} \,|\, \widetilde{\boldsymbol{X}}_{h+1} = \widetilde{X}_{h+1}) \mathbb{P}(\widetilde{X}_h, \widetilde{P}_h \,|\, \widetilde{\boldsymbol{\Theta}}_h) \mathrm{d}(\widetilde{Z}_{h+1}, \widetilde{X}_h, \widetilde{P}_h, \widetilde{Y}_{h+1}, \widetilde{X}_{h+1}) \\
&= \int_{\widetilde{\mathscr{Z}}_{h+1}} \int_{\widetilde{\mathcal{X}}_h \times \widetilde{\mathcal{P}}_h} \int_{\mathcal{W}_{0,h}} \int_{\mathcal{W}_{h+1}} \mathbb{1}[\widetilde{Z}_{h+1} = \widetilde{\chi}_{h+1}\Big(\widetilde{P}_h, \widetilde{\gamma}_h(\widetilde{P}_h), \widetilde{E}_h(A_h \widetilde{X}_h + B_h \widetilde{\gamma}_h(\widetilde{P}_h) + W_{0,h}) + W_{h+1}\Big)] \\
&\quad \mathbb{P}(\widetilde{X}_h, \widetilde{P}_h \,|\, \widetilde{\boldsymbol{\Theta}}_h) \mathrm{d}(\widetilde{Z}_{h+1}, \widetilde{X}_h, \widetilde{P}_h, W_{0,h}, W_{h+1}),
\end{aligned}
\tag{.3}
$$

where $A_h, B_h, E_h = \begin{bmatrix} E_{1,h} \\ \cdots \\ E_{n,h} \end{bmatrix}$ are the system dynamics of original problem $\mathcal{D}$ (and the problem $\widetilde{\mathcal{D}}(g_{1:H})$), and $W_{0,h}, W_{h+1} = \begin{bmatrix} W_{1,h} \\ \cdots \\ W_{n,h} \end{bmatrix}$ are noises which follow Gaussian distributions with mean zero and pre-defined covariance. Since from Lemma IV.3, $[\widetilde{\boldsymbol{X}}_h^\top \quad \widetilde{\boldsymbol{P}}_h^\top]^\top$ follows a Gaussian distribution with mean $\widetilde{\boldsymbol{\Theta}}_h$ and covariance $\widetilde{\Sigma}_h$ that is independent of $\widetilde{\boldsymbol{\Theta}}_{1:h-1}, \widetilde{\gamma}_{1:h-1}$. Therefore, the right-hand side of Equation (.3) only depends on $\widetilde{\Theta}_h$ and $\widetilde{\gamma}_h$, then we prove **Part 1**.

Part 2: We aim to prove that for any strategy $\widetilde{g}_{1:H} \in \widetilde{\mathcal{G}}_{1:H}, J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}_{1:H}) = J_{\mathcal{D}^{\ddagger}(g^m_{1:H})}(g^{\ddagger}_{1:H})$, where $g^{\ddagger}_h = \varsigma_h(\widetilde{g}_h), \forall h \in [H]$. This is because it holds for any $h \in [H]$ that

$$
\begin{aligned}
&\mathbb{E}[c^{\ddagger}_h \,|\, g^{\ddagger}_{1:H}] = \mathbb{E}\Big[\mathbb{E}[c^{\ddagger}_h(\widetilde{\boldsymbol{\Theta}}_h, g^{\ddagger}_h(\widetilde{\boldsymbol{C}}_h)) \,|\, \widetilde{\boldsymbol{C}}_h, \widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Psi}_3(\widetilde{\boldsymbol{C}}_h)] \,\Big|\, g^{\ddagger}_{1:h}\Big] && (.4) \\
&= \mathbb{E}\left[\mathbb{E}\left[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \begin{bmatrix} \widetilde{\gamma}_{1,h}(\widetilde{\boldsymbol{P}}_{1,h})^\top & \cdots & \widetilde{\gamma}_{n,h}(\widetilde{\boldsymbol{P}}_{n,h})^\top \end{bmatrix} \widetilde{Q}^2_h \begin{bmatrix} \widetilde{\gamma}_{1,h}(\widetilde{\boldsymbol{P}}_{1,h}) \\ \cdots \\ \widetilde{\gamma}_{n,h}(\widetilde{\boldsymbol{P}}_{n,h}) \end{bmatrix} \,\middle|\, \widetilde{\boldsymbol{C}}_h, \widetilde{\gamma}_h = g^{\ddagger}_h(\widetilde{\boldsymbol{C}}_h)\right] \,\middle|\, g^{\ddagger}_{1:h}\right] && (.5) \\
&= \mathbb{E}\left[\mathbb{E}\left[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \begin{bmatrix} \widetilde{\gamma}_{1,h}(\widetilde{\boldsymbol{P}}_{1,h})^\top & \cdots & \widetilde{\gamma}_{n,h}(\widetilde{\boldsymbol{P}}_{n,h})^\top \end{bmatrix} \widetilde{Q}^2_h \begin{bmatrix} \widetilde{\gamma}_{1,h}(\widetilde{\boldsymbol{P}}_{1,h}) \\ \cdots \\ \widetilde{\gamma}_{n,h}(\widetilde{\boldsymbol{P}}_{n,h}) \end{bmatrix} \,\middle|\, \widetilde{\boldsymbol{C}}_h, \widetilde{\gamma}_h(\cdot) = \widetilde{g}_h(\widetilde{\boldsymbol{C}}_h, \cdot)\right] \,\middle|\, g^{\ddagger}_{1:h}\right] && (.6) \\
&= \mathbb{E}\left[\mathbb{E}\left[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \begin{bmatrix} \widetilde{\gamma}_{1,h}(\widetilde{\boldsymbol{P}}_{1,h})^\top & \cdots & \widetilde{\gamma}_{n,h}(\widetilde{\boldsymbol{P}}_{n,h})^\top \end{bmatrix} \widetilde{Q}^2_h \begin{bmatrix} \widetilde{\gamma}_{1,h}(\widetilde{\boldsymbol{P}}_{1,h}) \\ \cdots \\ \widetilde{\gamma}_{n,h}(\widetilde{\boldsymbol{P}}_{n,h}) \end{bmatrix} \,\middle|\, \widetilde{\boldsymbol{C}}_h, \widetilde{\gamma}_h(\cdot) = \widetilde{g}_h(\widetilde{\boldsymbol{C}}_h, \cdot)\right] \,\middle|\, \widetilde{g}_{1:h}\right] && (.7) \\
&= \mathbb{E}[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}_h^\top \widetilde{Q}^2_h \widetilde{\boldsymbol{U}}_h \,|\, \widetilde{g}_{1:H}]. && (.8)
\end{aligned}
$$

Eq. (.4) and Eq. (.8) are due to the tower property; Eq. (.5) is from the definition of $c^{\ddagger}_h$; Eq. (.6) is from the definition of ς_h; Eq. (.7) is because $\widetilde{\boldsymbol{C}}_h$ has the same distribution under both strategies $\widetilde{g}_{1:h}$ and $g^{\ddagger}_{1:h}$, i.e., for any Borel set $\widetilde{\mathscr{C}}_h \subseteq \widetilde{\mathcal{C}}_h$, it holds that

$$
\begin{aligned}
&\mathbb{P}(\widetilde{\boldsymbol{C}}_h \in \widetilde{\mathscr{C}}_h \,|\, g^{\ddagger}_{1:h}) = \mathbb{E}[\mathbb{1}[\widetilde{\boldsymbol{C}}_h \in \widetilde{\mathscr{C}}_h] \,|\, g^{\ddagger}_{1:h}] = \mathbb{E}[\mathbb{E}\Big[\mathbb{1}[\widetilde{\boldsymbol{C}}_h \in \widetilde{\mathscr{C}}_h] \,|\, \widetilde{\boldsymbol{C}}_{h-1}, \widetilde{\gamma}_{h-1} = g^{\ddagger}_{h-1}(\widetilde{\boldsymbol{C}}_{h-1})\Big] \,|\, g^{\ddagger}_{1:h-1}] \\
&= \mathbb{E}[\mathbb{E}\Big[\cdots \mathbb{E}\Big[\mathbb{E}\Big[\mathbb{1}[\widetilde{\boldsymbol{C}}_h \in \widetilde{\mathscr{C}}_h] \,|\, \widetilde{\boldsymbol{C}}_{h-1}, \widetilde{\gamma}_{h-1} = g^{\ddagger}_{h-1}(\widetilde{\boldsymbol{C}}_{h-1})\Big] \,|\, \widetilde{\boldsymbol{C}}_{h-2}, \widetilde{\gamma}_{h-2} = g^{\ddagger}_{h-2}(\widetilde{\boldsymbol{C}}_{h-2})\Big] \cdots \,|\, \widetilde{\boldsymbol{C}}_1, \widetilde{\gamma}_1 = g^{\ddagger}_1(\widetilde{\boldsymbol{C}}_1)\Big]] \\
&= \mathbb{E}[\mathbb{E}\Big[\cdots \mathbb{E}\Big[\mathbb{E}\Big[\mathbb{1}[\widetilde{\boldsymbol{C}}_h \in \widetilde{\mathscr{C}}_h] \,|\, \widetilde{\boldsymbol{C}}_{h-1}, \widetilde{\gamma}_{h-1}(\cdot) = \widetilde{g}_{h-1}(\widetilde{\boldsymbol{C}}_{h-1}, \cdot)\Big] \,|\, \widetilde{\boldsymbol{C}}_{h-2}, \widetilde{\gamma}_{h-2}(\cdot) = \widetilde{g}_{h-2}(\widetilde{\boldsymbol{C}}_{h-2}, \cdot)\Big] \cdots \,|\, \widetilde{\boldsymbol{C}}_1, \widetilde{\gamma}_1(\cdot) = \widetilde{g}_1(\widetilde{\boldsymbol{C}}_1, \cdot)\Big]] \\
&= \mathbb{E}[\mathbb{1}[\widetilde{\boldsymbol{C}}_h \in \widetilde{\mathscr{C}}_h] \,|\, \widetilde{g}_{1:h}] = \mathbb{P}(\widetilde{\boldsymbol{C}}_h \in \widetilde{\mathscr{C}}_h \,|\, \widetilde{g}_{1:h}).
\end{aligned}
$$

Similarly, we can show that $\mathbb{E}[c^{\ddagger}_{H+1} \,|\, g^{\ddagger}_{1:H}] = \mathbb{E}[\widetilde{\boldsymbol{X}}_{H+1}^\top \widetilde{Q}_H \widetilde{\boldsymbol{X}}_{H+1} \,|\, \widetilde{g}_{1:H}]$. Therefore, we have $J_{\mathcal{D}^{\ddagger}(g^m_{1:H})}(g^{\ddagger}_{1:H}) = \mathbb{E}[\sum_{h=1}^{H+1} c^{\ddagger}_h \,|\, g^{\ddagger}_{1:H}] = \sum_{h=1}^{H+1} \mathbb{E}[c^{\ddagger}_h \,|\, g^{\ddagger}_{1:H}] = \sum_{h=1}^{H} \mathbb{E}[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}_h^\top \widetilde{Q}^2_h \widetilde{\boldsymbol{U}}_h \,|\, \widetilde{g}_{1:H}] + \mathbb{E}[\widetilde{\boldsymbol{X}}_{H+1}^\top \widetilde{Q}_H \widetilde{\boldsymbol{X}}_{H+1} \,|\, \widetilde{g}_{1:H}] = J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}_{1:H})$, which completes the proof. ■

5) Proof of Lemma IV.5: *Proof:* For convenience, given any matrices $\{\widetilde{\mathcal{E}}_{i,h}\}_{i\in[n],h\in[H]}$ and $\{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n],h\in[H]}$ with proper dimensions, we denote by $\widetilde{g}_{1:H} = \Lambda(\{\widetilde{\mathcal{E}}_{i,h}\}_{i\in[n],h\in[H]}, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n],h\in[H]})$ be the linear strategy generated by $\{\widetilde{\mathcal{E}}_{i,h}\}_{i\in[n],h\in[H]}, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n],h\in[H]}$, namely for any $i \in [n], h \in [H], \widetilde{g}_{i,h}(\widetilde{\boldsymbol{C}}_h, \widetilde{\boldsymbol{P}}_{i,h}) = \widetilde{\mathcal{E}}_{i,h}\widetilde{\boldsymbol{C}}_h + \widetilde{\mathcal{F}}_{i,h}\widetilde{\boldsymbol{P}}_{i,h}$. From Theorem IV.2, we know that $\widetilde{\mathcal{D}}(g_{1:H})$ has PN IS, and then there exists an optimal linear strategy $\widetilde{g}^*_{1:H}$. Since $\widetilde{g}^*_{1:H}$ is linear, then there exist $\{\widetilde{\mathcal{E}}^*_{i,h}\}_{i\in[n],h\in[H]}, \{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]}$ such that $\widetilde{g}^*_{1:H} = \Lambda(\{\widetilde{\mathcal{E}}^*_{i,h}\}_{i\in[n],h\in[H]}, \{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]})$. If we fix $\{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]}$, then $\{\mathcal{E}^*_{i,h}\}_{i\in[n],h\in[H]}$ are the optimal matrices that optimize $J_{\widetilde{\mathcal{D}}(g^m_{1:H})}$ with respect to $\{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]}$, i.e.,

$$\{\widetilde{\mathcal{E}}^*_{i,h}\}_{i\in[n],h\in[H]} \in \underset{\{\widetilde{\mathcal{E}}_{i,h}\}_{i\in[n],h\in[H]}}{\operatorname{argmin}} J_{\widetilde{\mathcal{D}}(M_{1:H})}\left(\Lambda(\{\widetilde{\mathcal{E}}_{i,h}\}_{i\in[n],h\in[H]}, \{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]})\right).$$

From Theorem IV.2, we know that for any $h \in [H]$

$$\widetilde{\boldsymbol{Z}}_h = \widetilde{\chi}_h(\widetilde{\boldsymbol{P}}_{h-1}, \widetilde{\boldsymbol{U}}_{h-1}, \widetilde{\boldsymbol{Y}}_h), \forall i \in [n], \widetilde{\boldsymbol{P}}_h = \widetilde{\zeta}_h(\widetilde{\boldsymbol{P}}_{h-1}, \widetilde{\boldsymbol{U}}_{h-1}, \widetilde{\boldsymbol{Y}}_h),$$

where $\widetilde{\chi}_h, \widetilde{\zeta}_h$ are fixed projection functions. Therefore, there exist matrices such that

$$\widetilde{\boldsymbol{Z}}_h = \widetilde{\mathcal{T}}^1_h\widetilde{\boldsymbol{P}}_{h-1} + \widetilde{\mathcal{T}}^2_h\widetilde{\boldsymbol{U}}_{h-1} + \widetilde{\mathcal{T}}^3_h\widetilde{\boldsymbol{Y}}_h, \forall i \in [n], \widetilde{\boldsymbol{P}}_h = \widetilde{\mathcal{L}}^1_h\widetilde{\boldsymbol{P}}_{h-1} + \widetilde{\mathcal{L}}^2_h\widetilde{\boldsymbol{U}}_{h-1} + \widetilde{\mathcal{L}}^3_h\widetilde{\boldsymbol{Y}}_h,$$

for some matrices $\widetilde{\mathcal{T}}^1_h, \widetilde{\mathcal{T}}^2_h, \widetilde{\mathcal{T}}^3_h, \widetilde{\mathcal{L}}^1_h, \widetilde{\mathcal{L}}^2_h, \widetilde{\mathcal{L}}^3_h$.

As shown in [14], if we fix $\{\mathcal{F}^*_{i,h}\}_{i\in[n],h\in[H]}$, we can construct an H-step coordinated centralized linear control problem $\breve{\mathcal{D}}$ based on $\widetilde{\mathcal{D}}(g^m_{1:H})$, where $\breve{\mathcal{D}}$ is an LQG system with linear dynamics, linear observations, quadratic costs, and Gaussian noises. For completeness, we include the construction below.

Specifically, we can construct an H-step coordinated centralized linear control problem $\breve{\mathcal{D}}$ based on $\widetilde{\mathcal{D}}(g^m_{1:H})$ and fixed $\{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]}$, where the state $\breve{\boldsymbol{X}}_h \in \breve{\mathcal{X}}_h$, action $\breve{\boldsymbol{U}}_h \in \breve{\mathcal{U}}_h$ and observation $\breve{\boldsymbol{Y}}_h \in \breve{\mathcal{Y}}_h$ evolve following the dynamics

$$\begin{aligned}
\forall h \in [H], \breve{\mathcal{X}}_h = \widetilde{\mathcal{X}} \times \widetilde{\mathcal{P}}_h,\ \breve{\mathcal{U}}_h = \widetilde{\mathcal{U}}_h,\ \breve{\mathcal{Y}}_h = \widetilde{\mathcal{Z}}_h, \breve{\boldsymbol{X}}_{h+1} &= \breve{A}_h\breve{\boldsymbol{X}}_h + \breve{B}_h\breve{\boldsymbol{U}}_h + \breve{\boldsymbol{W}}_h,\\
\breve{\boldsymbol{Y}}_h = \breve{E}_h\breve{\boldsymbol{X}}_{h-1} + \breve{F}_h\breve{\boldsymbol{U}}_{h-1} &+ \breve{\boldsymbol{V}}_h (\text{for } h > 1), \breve{\boldsymbol{Y}}_1 = \breve{\boldsymbol{V}}_1\\
\breve{\boldsymbol{X}}_{H+1} = \widetilde{\boldsymbol{X}}_{H+1}, \breve{\boldsymbol{X}}_h = \begin{bmatrix}\widetilde{\boldsymbol{X}}_h\\ \widetilde{\boldsymbol{P}}_h\end{bmatrix}, \breve{\boldsymbol{U}}_h &= \widetilde{\boldsymbol{U}}_h - \operatorname{diag}(\widetilde{\mathcal{F}}^*_{1,h}, \cdots, \widetilde{\mathcal{F}}^*_{n,h})\widetilde{\boldsymbol{P}}_h, \breve{\boldsymbol{Y}}_h = \widetilde{\boldsymbol{Z}}_h.
\end{aligned}$$

Define $\widetilde{\mathcal{F}}^*_h = \operatorname{diag}(\widetilde{\mathcal{F}}^*_{1,h}, \cdots, \widetilde{\mathcal{F}}^*_{n,h})$, then the matrices $\breve{A}_h, \breve{B}_h, \breve{E}_h, \breve{F}_h$ and noises $\breve{\boldsymbol{W}}_h, \breve{\boldsymbol{V}}_h$ are therefore defined as

$$\begin{aligned}
&\forall h \in [H-1], \breve{A}_h = \begin{bmatrix}\widetilde{A}_h & \widetilde{B}_h\widetilde{\mathcal{F}}^*_h\\ \widetilde{\mathcal{L}}^3_{h+1}\widetilde{\mathcal{E}}_{h+1}\widetilde{A}_h & \widetilde{\mathcal{L}}^2_{h+1}\widetilde{\mathcal{F}}^*_h + \widetilde{\mathcal{L}}^3_{h+1}\widetilde{\mathcal{E}}_{h+1}\widetilde{B}_h\widetilde{\mathcal{F}}^*_h\end{bmatrix}, \breve{B}_h = \begin{bmatrix}\widetilde{B}_h\\ \widetilde{\mathcal{L}}^2_{h+1} + \widetilde{\mathcal{L}}^3_{h+1}\widetilde{E}_{h+1}\widetilde{B}_h\end{bmatrix};\\
&\forall h \in [2:H], \breve{E}_h = \begin{bmatrix}\widetilde{\mathcal{T}}^3_h\widetilde{E}_h\widetilde{A}_{h-1} & \widetilde{\mathcal{T}}^1_{h+1} + \widetilde{\mathcal{T}}^2_h\widetilde{\mathcal{F}}^*_{h-1} + \widetilde{\mathcal{T}}^3_h\widetilde{E}_h\widetilde{B}_{h-1}\widetilde{\mathcal{F}}^*_{h-1}\end{bmatrix}, \breve{F}_h = \begin{bmatrix}\widetilde{\mathcal{T}}^2_h + \widetilde{\mathcal{T}}^3_h\widetilde{E}_h\widetilde{B}_{h-1}\end{bmatrix};\\
&\forall h \in [H-1], \breve{\boldsymbol{W}}_h = \begin{bmatrix}\widetilde{\boldsymbol{W}}_{0,h}\\ \widetilde{\mathcal{L}}^3_{h+1}(\widetilde{E}_{h+1}\widetilde{\boldsymbol{W}}_{0,h} + \widetilde{\boldsymbol{W}}_{1:n,h+1})\end{bmatrix}; \forall h \in [2:H], \breve{\boldsymbol{V}}_h = \widetilde{\mathcal{T}}^3_{h+1}(\widetilde{E}_{h+1}\widetilde{\boldsymbol{W}}_{0,h} + \widetilde{\boldsymbol{W}}_{1:n,h+1});\\
&\breve{A}_H = \begin{bmatrix}\widetilde{A}_H & \widetilde{B}_H\widetilde{\mathcal{F}}^*_H\end{bmatrix}, \breve{B}_H = \widetilde{B}_H, \breve{\boldsymbol{W}}_H = \widetilde{\boldsymbol{W}}_{0,H}, \breve{\boldsymbol{V}}_1 = \widetilde{\mathcal{T}}^3_1(\widetilde{E}_1\widetilde{\boldsymbol{X}}_1 + \widetilde{\boldsymbol{W}}_{1:n,1}),
\end{aligned}$$

where we recall that $\forall h \in [H], \widetilde{\boldsymbol{W}}_{1:n,h+1} = \begin{bmatrix}\widetilde{\boldsymbol{W}}_{1,h+1}\\ \cdots\\ \widetilde{\boldsymbol{W}}_{n,h+1}\end{bmatrix}$. The cost of each stage $h \in [H]$ is $\breve{c}_h = \begin{bmatrix}\breve{\boldsymbol{X}}_h^\top & \breve{\boldsymbol{U}}_h^\top\end{bmatrix}^\top \begin{bmatrix}\breve{Q}^1_h & \breve{N}_h\\ \breve{N}_h^\top & \breve{Q}^2_h\end{bmatrix}\begin{bmatrix}\breve{\boldsymbol{X}}_h\\ \breve{\boldsymbol{U}}_h\end{bmatrix}$, and $\breve{c}_{H+1} = \breve{\boldsymbol{X}}_{H+1}^\top\breve{Q}_{H+1}\breve{\boldsymbol{X}}_{H+1}$, where the matrices $\breve{Q}^1_h, \breve{Q}^2_h, \breve{N}_h, \breve{Q}_{H+1}$ are defined as

$$\forall h \in [H], \breve{Q}^1_h = \begin{bmatrix}\widetilde{Q}^1_h & 0\\ 0 & (\widetilde{\mathcal{F}}^*_h)^\top\widetilde{Q}^2_h\widetilde{\mathcal{F}}^*_h\end{bmatrix}, \breve{Q}^2_h = \widetilde{Q}^2_h, \breve{N}_h = \begin{bmatrix}0\\ (\widetilde{\mathcal{F}}^*_h)^\top\widetilde{Q}^2_h\end{bmatrix}, \breve{Q}_{H+1} = \widetilde{Q}_{H+1}.$$

The strategy of $\breve{\mathcal{D}}$ at each stage $h \in [H]$ is defined as $\breve{g}_h : \Pi^h_{t=1}\breve{\mathcal{Y}}_t \to \breve{\mathcal{U}}_h$ and the objective is defined as $J_{\breve{\mathcal{D}}}(\breve{g}_{1:H}) = \mathbb{E}[\sum_{h=1}^{H+1}\breve{c}_h \mid \breve{g}_{1:H}]$. Then, the optimal strategy $\breve{g}^*_{1:H} \in \operatorname{argmin}_{\breve{g}_{1:H}} J_{\breve{\mathcal{D}}}(\breve{g}_{1:H})$.

Since $\breve{\mathcal{D}}$ is an LQG problem, from standard results in linear stochastic control [23], we know that there exist matrices $\breve{K}_h$ such that $\widehat{\breve{\boldsymbol{X}}}_h = \mathbb{E}[\breve{\boldsymbol{X}}_h \mid \breve{\boldsymbol{Y}}_{1:h}] = \breve{K}_h\breve{\boldsymbol{Y}}_{1:h}$ for any $h \in [H]$, and such $\breve{K}_h$ can be computed through Kalman Filter. Meanwhile, there exists an optimal control strategy $\breve{g}^*_{1:H}$ with the form $\breve{g}^*_h(\breve{\boldsymbol{Y}}_h) = \breve{\mathcal{E}}^*_h\widehat{\breve{\boldsymbol{X}}}_h = \breve{\mathcal{E}}^*_h\breve{K}_h\breve{\boldsymbol{Y}}_{1:h}$ for some $\breve{\mathcal{E}}_h, h \in [H]$. Also, one

can verify that for any linear strategy $\breve{g}_{1:H}$ with the form $\breve{g}_{1:H}(\breve{\boldsymbol{Y}}_h) = \breve{\mathcal{E}}_h \breve{\boldsymbol{Y}}_h, \forall h \in [H]$ and $\breve{\mathcal{E}}_h = \begin{bmatrix} \breve{\mathcal{E}}_{1,h} \\ \cdots \\ \breve{\mathcal{E}}_{n,h} \end{bmatrix}$, we can construct a linear strategy of $\widetilde{\mathcal{D}}(g^m_{1:H})$ as $\widetilde{g}_{1:H} := \Lambda(\{\breve{\mathcal{E}}_{i,h}\}_{i\in[n],h\in[H]}, \{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]})$ such that $J_{\breve{\mathcal{D}}}(\breve{g}_{1:H}) = J_{\widetilde{\mathcal{D}}(g^m_{1:H})}(\widetilde{g}_{1:H})$, and vice versa. Therefore, $\widetilde{g}^*_{1:H} := \Lambda(\{\breve{\mathcal{E}}^*_{i,h}\breve{\mathcal{K}}_h\}_{i\in[n],h\in[H]}, \{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]})$ is an optimal strategy of $\widetilde{\mathcal{D}}(g^m_{1:H})$, where $\breve{\mathcal{E}}^*_h = \begin{bmatrix} \breve{\mathcal{E}}^*_{1,h} \\ \cdots \\ \breve{\mathcal{E}}^*_{n,h} \end{bmatrix}$ for any $h \in [H]$.

From the construction of $\breve{\mathcal{D}}$, we know that $\breve{\boldsymbol{X}}_h = \begin{bmatrix} \widetilde{\boldsymbol{X}}_h \\ \widetilde{\boldsymbol{P}}_h \end{bmatrix}, \breve{\boldsymbol{Y}}_h = \widetilde{\boldsymbol{Z}}_h$ for any $h \in [H]$, then we know that $\widehat{\breve{\boldsymbol{X}}}_h = \mathbb{E}\left[\breve{\boldsymbol{X}}_h \,|\, \breve{\boldsymbol{Y}}_{1:h}\right] = \mathbb{E}\left[\begin{bmatrix} \widetilde{\boldsymbol{X}}_h \\ \widetilde{\boldsymbol{P}}_h \end{bmatrix} \middle| \, \widetilde{\boldsymbol{Z}}_{1:h}\right] = \begin{bmatrix} \mathbb{E}[\widetilde{\boldsymbol{X}}_h \,|\, \widetilde{\boldsymbol{C}}_h] \\ \mathbb{E}[\widetilde{\boldsymbol{P}}_h \,|\, \widetilde{\boldsymbol{C}}_h] \end{bmatrix} = \widetilde{\boldsymbol{\Theta}}_h$, and $\widetilde{\Psi}^3_h(\widetilde{\boldsymbol{C}}_h) = \breve{K}_h \widetilde{\boldsymbol{C}}_h$. Therefore, we obtain the such $\widetilde{g}^*_{1:H}$, and matrices $\{\widetilde{\mathcal{E}}^*_{i,h}\}_{i\in[n],h\in[H]}, \{\widetilde{\mathcal{F}}^*_{i,h}\}_{i\in[n],h\in[H]}$ as above such that for each $\widetilde{\Theta}_h \in \widetilde{\mathcal{S}}_h$, $\widetilde{\gamma}^*_{i,h}(\cdot) = \widetilde{\mathcal{E}}^*_{i,h}\widetilde{\Theta}_h + \widetilde{\mathcal{F}}^*_{i,h}\cdot$, $\widetilde{\gamma}^*_h$ minimizes the $\mathbb{E}[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}_h^\top \widetilde{Q}^2_h \widetilde{\boldsymbol{U}}_h + \widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) \,|\, \widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Theta}_h, \widetilde{\gamma}_h]$, and $\widetilde{g}^*_{i,h}(\widetilde{C}_h, \widetilde{P}_{i,h}) = \widetilde{\mathcal{E}}^*_{i,h}\widetilde{\Psi}^3_h(\widetilde{C}_h) + \widetilde{\mathcal{F}}^*_{i,h}\widetilde{P}_{i,h}$ for any $i \in [n], h \in [H], \widetilde{C}_h \in \widetilde{\mathcal{C}}_h, \widetilde{P}_{i,h} \in \widetilde{\mathcal{P}}_{i,h}$. ■

6) Proof of Theorem IV.6: *Proof:* We prove this by induction. For $h = H+1$, we know that $\widetilde{V}_{H+1}(\widetilde{\boldsymbol{\Theta}}_{H+1}) = \mathbb{E}[\widetilde{\boldsymbol{X}}_{H+1}^\top \widetilde{Q}^1_{H+1} \widetilde{\boldsymbol{X}}_{H+1}] = \widetilde{\boldsymbol{\Theta}}_{H+1}^\top \widetilde{Q}^1_{H+1} \widetilde{\boldsymbol{\Theta}}_{H+1} + \mathrm{Tr}(\Sigma_{H+1}\mathbb{I}^\top_{x,H+1}\widetilde{Q}^1_{H+1}\mathbb{I}_{x,H+1})$, for some matrix $\mathbb{I}_{x,H+1}$ projection mapping $\widetilde{\boldsymbol{S}}_{H+1}$ to $\widetilde{\boldsymbol{X}}_{h+1}$. Then, we know that $\widetilde{R}_{H+1} = \widetilde{Q}^1_{H+1}, \widetilde{c}_{H+1} = \mathrm{Tr}(\Sigma_{H+1}\mathbb{I}^\top_{x,H+1}\widetilde{Q}^1_{H+1}\mathbb{I}_{x,H+1})$, and the induction hypothesis holds for $h = H+1$.

Now, for any given $h \in [H]$, we assume that $\widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) = \widetilde{\boldsymbol{\Theta}}_{h+1}^\top \widetilde{R}_{h+1}\widetilde{\boldsymbol{\Theta}}_{h+1} + \widetilde{c}_{h+1}$ for some $\widetilde{R}_{h+1}$ and $\widetilde{c}_{h+1}$. From Lemma IV.3, we know that the CIB belief $\widetilde{\boldsymbol{B}}_h$ is a Gaussian distribution $\mathcal{N}(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\Sigma}_h)$, so we can write $\widetilde{\boldsymbol{S}}_h = \widetilde{\boldsymbol{\Theta}}_h + \widetilde{\omega}_h$, where $\widetilde{\omega}_h \sim \mathcal{N}(\boldsymbol{0}, \widetilde{\Sigma}_h)$. From the definition of $\widetilde{\boldsymbol{S}}_h$, we can write $\widetilde{\boldsymbol{X}}_h = \mathbb{I}_{x,h}\widetilde{\boldsymbol{S}}_h, \widetilde{\boldsymbol{P}}_h = \mathbb{I}_{p,h}\widetilde{\boldsymbol{S}}_h$ for some transformation matrix $\mathbb{I}_{x,h}, \mathbb{I}_{p,h}$. Also, for any $i \in [n]$, the action of agent i has the form $\widetilde{\boldsymbol{U}}_{i,h} = \widetilde{\mathcal{E}}_{i,h}\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\mathcal{F}}_{i,h}\widetilde{\boldsymbol{P}}_{i,h}$, then we can write the joint control action as $\widetilde{\boldsymbol{U}}_h = \widetilde{\mathcal{E}}_h\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\mathcal{F}}_h\widetilde{\boldsymbol{P}}_h$ with $\widetilde{\mathcal{E}}_h := \begin{bmatrix} \widetilde{\mathcal{E}}_{1,h} \\ \cdots \\ \widetilde{\mathcal{E}}_{n,h} \end{bmatrix}$ and $\widetilde{\mathcal{F}}_h := \mathrm{diag}(\widetilde{\mathcal{F}}_{1,h}, \cdots, \widetilde{\mathcal{F}}_{n,h})$. Therefore, it holds $\widetilde{\boldsymbol{U}}_h = \widetilde{\mathcal{E}}_h\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\mathcal{F}}_h\mathbb{I}_{p,h}(\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\omega}_h) = (\widetilde{\mathcal{E}}_h + \widetilde{\mathcal{F}}_h\mathbb{I}_{p,h})\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\mathcal{F}}_h\mathbb{I}_{p,h}\widetilde{\omega}_h$. Now, we define $\widehat{\mathcal{E}}_h := \widetilde{\mathcal{E}}_h + \widetilde{\mathcal{F}}_h\mathbb{I}_{p,h}$, and then optimizing $(\widetilde{\mathcal{E}}_h, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n]})$ is equivalent to optimizing $(\widetilde{\mathcal{E}}_h, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n]})$ with respect to the corresponding spaces. This is because for any $(\widetilde{\mathcal{E}}_h, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n]})$, we can find the corresponding $(\widehat{\mathcal{E}}_h, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n]})$, and vice versa. Then, we can write the objective as

$$
\begin{aligned}
&\min_{\widetilde{\mathcal{E}}_h, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n]}} \mathbb{E}[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}_h^\top \widetilde{Q}^2_h \widetilde{\boldsymbol{U}}_h \,|\, \widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Theta}_h, \widetilde{\mathcal{E}}_h, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n]}] \\
&= \min_{\widetilde{\mathcal{E}}_h, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n]}} \mathbb{E}[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}^1_h \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}_h^\top \widetilde{Q}^2_h \widetilde{\boldsymbol{U}}_h \,|\, \widetilde{\boldsymbol{\Theta}}_h = \widetilde{\Theta}_h, \widehat{\mathcal{E}}_h, \widetilde{\mathcal{F}}_h] \\
&= \min_{\widetilde{\mathcal{E}}_h, \{\widetilde{\mathcal{F}}_{i,h}\}_{i\in[n]}} \widetilde{\Theta}_h^\top \mathbb{I}^\top_{x,h}\widetilde{Q}^1_h\mathbb{I}_{x,h}\widetilde{\Theta}_h + \widetilde{\Theta}_h^\top \widehat{\mathcal{E}}_h^\top \widetilde{Q}^2_h \widehat{\mathcal{E}}_h \widetilde{\Theta}_h + \mathrm{Tr}(\mathbb{I}_{x,h}\widetilde{\Sigma}_h\mathbb{I}^\top_{x,h}\widetilde{Q}^1_h) + \mathrm{Tr}(\widetilde{\mathcal{F}}_h\mathbb{I}_{p,h}\widetilde{\Sigma}_h\mathbb{I}^\top_{p,h}\widetilde{\mathcal{F}}_h^\top\widetilde{Q}^2_h).
\end{aligned}
$$

From Theorem IV.2, we know that the information evolves as $\widetilde{\boldsymbol{Z}}_{h+1} = \widetilde{\chi}_{h+1}(\widetilde{\boldsymbol{P}}_h, \widetilde{\boldsymbol{U}}_h, \widetilde{\boldsymbol{Y}}_{h+1})$, and from Lemma IV.3 $\widetilde{\boldsymbol{\Theta}}_{h+1} = \widetilde{\Psi}^1_{h+1}(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\boldsymbol{Z}}_{h+1})$. Then, we can write $\widetilde{\boldsymbol{\Theta}}_{h+1} = \widetilde{\Psi}^4_{h+1}(\widetilde{\boldsymbol{\Theta}}_h, \widetilde{\boldsymbol{P}}_h, \widetilde{\boldsymbol{U}}_h, \widetilde{\boldsymbol{Y}}_{h+1})$ for some linear function $\widetilde{\Psi}^4_{h+1}$. Therefore, we can write $\widetilde{\boldsymbol{\Theta}}_{h+1} = K^1_{h+1}\widetilde{\boldsymbol{\Theta}}_h + K^2_{h+1}\widetilde{\boldsymbol{P}}_h + K^3_{h+1}\widetilde{\boldsymbol{U}}_h + K^4_{h+1}\widetilde{\boldsymbol{Y}}_{h+1}$ for some matrices $K^1_{h+1}, K^2_{h+1}, K^3_{h+1}, K^4_{h+1}$. Also, we know that $\widetilde{\boldsymbol{P}}_h = \mathbb{I}_{p,h}(\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\omega}_h), \widetilde{\boldsymbol{U}}_h = (\widetilde{\mathcal{E}}_h + \widetilde{\mathcal{F}}_h\mathbb{I}_{p,h})\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\mathcal{F}}_h\mathbb{I}_{p,h}\widetilde{\omega}_h, \widetilde{\boldsymbol{Y}}_{h+1} = \widetilde{E}_{h+1}\widetilde{\boldsymbol{X}}_{h+1} + \widetilde{\boldsymbol{W}}_{h+1} = \widetilde{E}_{h+1}(\widetilde{A}_h\widetilde{\boldsymbol{X}}_h + \widetilde{B}_h\widetilde{\boldsymbol{U}}_h + \widetilde{\boldsymbol{W}}_{0,h}) + \widetilde{\boldsymbol{W}}_{1:n,h+1}$, where $\widetilde{E}_{h+1} = \begin{bmatrix} \widetilde{E}_{1,h+1} \\ \cdots \\ \widetilde{E}_{n,h+1} \end{bmatrix}, \widetilde{\boldsymbol{W}}_{1:n,h+1} = \begin{bmatrix} \widetilde{\boldsymbol{W}}_{1,h+1} \\ \cdots \\ \widetilde{\boldsymbol{W}}_{n,h+1} \end{bmatrix}$. Therefore, we can write $\widetilde{\boldsymbol{\Theta}}_{h+1}$ as

$$
\begin{aligned}
\widetilde{\boldsymbol{\Theta}}_{h+1} &= K^1_{h+1}\widetilde{\boldsymbol{\Theta}}_h + K^2_{h+1}\mathbb{I}_{p,h}(\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\omega}_h) + K^3_{h+1}(\widehat{\mathcal{E}}_h\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\mathcal{F}}_h\mathbb{I}_{p,h}\widetilde{\omega}_h) \\
&\quad + K^4_{h+1}(\widetilde{E}_{h+1}(\widetilde{A}_h\mathbb{I}_{x,h}(\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\omega}_h) + \widetilde{B}_h(\widehat{\mathcal{E}}_h\widetilde{\boldsymbol{\Theta}}_h + \widetilde{\mathcal{F}}_h\mathbb{I}_{p,h}\widetilde{\omega}_h) + \widetilde{\boldsymbol{W}}_{0,h}) + \widetilde{\boldsymbol{W}}_{1:n,h+1}), \\
&= (K^1_{h+1} + K^2_{h+1}\mathbb{I}_{p,h} + K^3_{h+1}\widehat{\mathcal{E}}_h + K^4_{h+1}\widetilde{E}_{h+1}\widetilde{A}_h\mathbb{I}_{x,h} + K^4_{h+1}\widetilde{E}_{h+1}\widetilde{B}_h\widehat{\mathcal{E}}_h)\widetilde{\boldsymbol{\Theta}}_h + K^4_{h+1}\widetilde{\boldsymbol{W}}_{1:n,h+1} \\
&\quad + (K^2_{h+1}\mathbb{I}_{p,h} + K^3_{h+1}\widetilde{\mathcal{F}}_h\mathbb{I}_{p,h} + K^4_{h+1}\widetilde{E}_{h+1}(\widetilde{A}_h\mathbb{I}_{x,h} + \widetilde{B}_h\widetilde{\mathcal{F}}_h\mathbb{I}_{p,h}))\widetilde{\omega}_h + K^4_{h+1}\widetilde{E}_{h+1}\widetilde{\boldsymbol{W}}_{0,h}.
\end{aligned}
$$

Since $\widetilde{\omega}_h$ only depends on the randomness before h, it is independent to $\widetilde{\boldsymbol{W}}_{0,h}$ and $\widetilde{\boldsymbol{W}}_{1:n,h+1}$. Then, we can define the

parameters as

$$\widetilde{L}_h^1 := K_{h+1}^3 + K_{h+1}^4 \widetilde{E}_{h+1}\widetilde{B}_h, \quad \widetilde{L}_h^2 := K_{h+1}^1 + K_{h+1}^2 \mathbb{I}_{p,h} + K_{h+1}^4 \widetilde{E}_{h+1}\widetilde{A}_h \mathbb{I}_{x,h},$$
$$\widetilde{L}_h^3 := K_{h+1}^3 + K_{h+1}^4 \widetilde{E}_{h+1}\widetilde{B}_h, \quad \widetilde{L}_h^4 := K_{h+1}^2 \mathbb{I}_{p,h} + K_{h+1}^4 \widetilde{E}_{h+1}\widetilde{A}_h \mathbb{I}_{x,h},$$

and have

$$\widetilde{\boldsymbol{\Theta}}_{h+1} = (\widetilde{L}_h^1 \widehat{\mathcal{E}}_h + \widetilde{L}_h^2)\widetilde{\boldsymbol{\Theta}}_h + (\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} + \widetilde{L}_h^4)\widetilde{\omega}_h + K_{h+1}^4 \widetilde{E}_{h+1}\widetilde{\boldsymbol{W}}_{0,h} + K_{h+1}^4 \widetilde{\boldsymbol{W}}_{1:n,h+1}.$$

Thus, from the induction hypothesis, we have

$$\begin{aligned}
&\mathbb{E}[\widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) \,|\, \widetilde{\boldsymbol{\Theta}}_h, \widetilde{\mathcal{E}}_h, \widetilde{\mathcal{F}}_h] = \mathbb{E}[\widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) \,|\, \widetilde{\boldsymbol{\Theta}}_h, \widehat{\mathcal{E}}_h, \widetilde{\mathcal{F}}_h] = \mathbb{E}[\widetilde{\boldsymbol{\Theta}}_{h+1}^\top \widetilde{R}_{h+1}\widetilde{\boldsymbol{\Theta}}_{h+1} + \widetilde{c}_{h+1} \,|\, \widetilde{\boldsymbol{\Theta}}_h, \widetilde{\mathcal{E}}_h, \widetilde{\mathcal{F}}_h] \\
&\quad = \widetilde{\boldsymbol{\Theta}}_h^\top (\widetilde{L}_h^1 \widehat{\mathcal{E}}_h + \widetilde{L}_h^2)^\top \widetilde{R}_{h+1}(\widetilde{L}_h^1 \widehat{\mathcal{E}}_h + \widetilde{L}_h^2)\widetilde{\boldsymbol{\Theta}}_h + \mathrm{Tr}((\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} + \widetilde{L}_h^4)\widetilde{\Sigma}_h(\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} + \widetilde{L}_h^4)^\top \widetilde{R}_{h+1}) \\
&\qquad + \mathrm{Tr}(K_{h+1}^4 \widetilde{E}_{h+1}\Sigma_{0,h}\widetilde{E}_{h+1}^\top (K_{h+1}^4)^\top \widetilde{R}_{h+1}) + \mathrm{Tr}(K_{h+1}^4 \mathrm{diag}(\widetilde{\Sigma}_{1,h+1}, \cdots, \widetilde{\Sigma}_{n,h+1})(K_{h+1}^4)^\top \widetilde{R}_{h+1}) + \widetilde{c}_{h+1}.
\end{aligned}$$

If we define the following quantities:

$$\begin{aligned}
&J_h^1(\widetilde{\boldsymbol{\Theta}}_h) := \widetilde{\boldsymbol{\Theta}}_h^\top (\widetilde{Q}_h^1 + \widehat{\mathcal{E}}_h^\top \widetilde{Q}_h^2 \widehat{\mathcal{E}}_h + (\widetilde{L}_h^1 \widehat{\mathcal{E}}_h + \widetilde{L}_h^2)^\top \widetilde{R}_{h+1}(\widetilde{L}_h^1 \widehat{\mathcal{E}}_h + \widetilde{L}_h^2))\widetilde{\boldsymbol{\Theta}}_h, \\
&J_h^2 := \mathrm{Tr}(\widetilde{\mathcal{F}}_h \mathbb{I}_{p,h}\widetilde{\Sigma}_h \mathbb{I}_{p,h}^\top \widetilde{\mathcal{F}}_h^\top \widetilde{Q}_h^2) + \mathrm{Tr}((\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} + \widetilde{L}_h^4)\widetilde{\Sigma}_h(\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} + \widetilde{L}_h^4)^\top \widetilde{R}_{h+1}), \\
&J_h^3 := \mathrm{Tr}(\mathbb{I}_{x,h}\widetilde{\Sigma}_h \mathbb{I}_{x,h}^\top \widetilde{Q}_h^1) + \mathrm{Tr}(K_{h+1}^4 \widetilde{E}_{h+1}\Sigma_{0,h}\widetilde{E}_{h+1}^\top (K_{h+1}^4)^\top \widetilde{R}_{h+1}) + \mathrm{Tr}(K_{h+1}^4 \mathrm{diag}(\widetilde{\Sigma}_{1,h+1}, \cdots, \widetilde{\Sigma}_{n,h+1})(K_{h+1}^4)^\top \widetilde{R}_{h+1}) + \widetilde{c}_{h+1},
\end{aligned}$$

then we can write $\mathbb{E}[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}_h^1 \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}_h^\top \widetilde{Q}_h^2 \widetilde{\boldsymbol{U}}_h + \widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) \,|\, \widetilde{\boldsymbol{\Theta}}_h, \widetilde{\mathcal{E}}_h, \widetilde{\mathcal{F}}_h] = J_h^1(\widetilde{\boldsymbol{\Theta}}_h) + J_h^2 + J_h^3$. Note that $J_h^1(\widetilde{\boldsymbol{\Theta}}_h)$ only depends on $\widetilde{\boldsymbol{\Theta}}_h$ and $\widehat{\mathcal{E}}_h$, J_h^2 only depends on $\widetilde{\mathcal{F}}_h$, and J_h^3 is just a constant. Therefore, if $\widehat{\mathcal{E}}_h^*, \widetilde{\mathcal{F}}_h$ satisfy

$$\widehat{\mathcal{E}}_h^* \in \operatorname*{argmin}_{\widehat{\mathcal{E}}_h} J_h^1(\widetilde{\boldsymbol{\Theta}}_h), \quad \forall \widetilde{\boldsymbol{\Theta}}_h \in \widetilde{\mathcal{S}}_h, \qquad \widetilde{\mathcal{F}}_h^* \in \operatorname*{argmin}_{\widetilde{\mathcal{F}}_h} J_h^2,$$

then $\widetilde{\mathcal{E}}_h^* := \widehat{\mathcal{E}}_h^* - \widetilde{\mathcal{F}}_h^* \mathbb{I}_{p,h}$ and $\widetilde{\mathcal{F}}_h^*$ will satisfy

$$\widetilde{\mathcal{E}}_h^*, \widetilde{\mathcal{F}}_h^* \in \operatorname*{argmin}_{\widetilde{\mathcal{E}}_h, \widetilde{\mathcal{F}}_h} \mathbb{E}[\widetilde{\boldsymbol{X}}_h^\top \widetilde{Q}_h^1 \widetilde{\boldsymbol{X}}_h + \widetilde{\boldsymbol{U}}_h^\top \widetilde{Q}_h^2 \widetilde{\boldsymbol{U}}_h + \widetilde{V}_{h+1}(\widetilde{\boldsymbol{\Theta}}_{h+1}) \,|\, \widetilde{\boldsymbol{\Theta}}_h, \widetilde{\mathcal{E}}_h, \widetilde{\mathcal{F}}_h].$$

Firstly, we intend to minimize $J_h^1(\widetilde{\boldsymbol{\Theta}}_h)$ and find optimal $\widehat{\mathcal{E}}_h^*$. Since $\widetilde{Q}_h^1, \widetilde{Q}_h^2, \widetilde{R}_{h+1} \succeq 0$, we can write

$$J_h^1(\widetilde{\boldsymbol{\Theta}}_h) = ||(\widetilde{Q}_h^1)^{\frac{1}{2}} \widetilde{\boldsymbol{\Theta}}_h||_2^2 + ||(\widetilde{Q}_h^2)^{\frac{1}{2}} \widehat{\mathcal{E}}_h \widetilde{\boldsymbol{\Theta}}_h||_2^2 + ||\widetilde{R}_h^{\frac{1}{2}}(\widetilde{L}_h^1 \widehat{\mathcal{E}}_h + \widetilde{L}_h^2)\widetilde{\boldsymbol{\Theta}}_h||_2^2,$$

which is a convex function of $\widehat{\mathcal{E}}_h$ given $\widetilde{\boldsymbol{\Theta}}_h$. Therefore, we can compute the optimal $\widehat{\mathcal{E}}_h$ via the first-order condition. Take the gradient of $J_h^1(\widetilde{\boldsymbol{\Theta}}_h)$ with respect to $\widehat{\mathcal{E}}_h$, we get

$$\nabla_{\widehat{\mathcal{E}}_h} J_h^1(\widetilde{\boldsymbol{\Theta}}_h) = (2(\widetilde{Q}_h^2 + (\widetilde{L}_h^1)^\top \widetilde{R}_{h+1}\widetilde{L}_h^1)\widehat{\mathcal{E}}_h + 2(\widetilde{L}_h^1)^\top \widetilde{R}_{h+1}\widetilde{L}_h^2)\widetilde{\boldsymbol{\Theta}}_h \widetilde{\boldsymbol{\Theta}}_h^\top.$$

Since it holds that $\ker(\widetilde{Q}_h^2 + (\widetilde{L}_h^1)^\top \widetilde{R}_{h+1}\widetilde{L}_h^1) \subseteq \ker((\widetilde{L}_h^2)^\top \widetilde{R}_{h+1}\widetilde{L}_h^1)$ due to Lemma .2, and $\widetilde{Q}_h^2 + (\widetilde{L}_h^1)^\top \widetilde{R}_{h+1}\widetilde{L}_h^1 \succeq 0$, we know that this gradient is zero if we choose

$$\widehat{\mathcal{E}}_h^* = -(\widetilde{Q}_h^2 + (\widetilde{L}_h^1)^\top \widetilde{R}_{h+1}\widetilde{L}_h^1)^\dagger (\widetilde{L}_h^1)^\top \widetilde{R}_{h+1}\widetilde{L}_h^2.$$

Then, we know that such a $\widehat{\mathcal{E}}_h^*$ minimizes $J_h^1(\widetilde{\boldsymbol{\Theta}}_h)$, and we can compute

$$\widetilde{R}_h = \widetilde{Q}_h^1 + (\widehat{\mathcal{E}}_h^*)^\top \widetilde{Q}_h^2 \widehat{\mathcal{E}}_h^* + (\widetilde{L}_h^1 \widehat{\mathcal{E}}_h^* + \widetilde{L}_h^2)^\top \widetilde{R}_{h+1}(\widetilde{L}_h^1 \widehat{\mathcal{E}}_h^* + \widetilde{L}_h^2).$$

Note that the use of pseudo-inverse here precisely corresponds to the proof of *generalized* Riccati Equations in [24], [25].

Secondly, we intend to minimize J_h^2 and find optimal $\widetilde{\mathcal{F}}_h^*$. Hence, the optimal $\mathcal{F}_h^*$ can be found via the following optimization problem

$$\begin{aligned}
\widetilde{\mathcal{F}}_h^* &\in \operatorname*{argmin}_{\widetilde{\mathcal{F}}_h = \mathrm{diag}(\widetilde{\mathcal{F}}_{1,h}, \cdots, \widetilde{\mathcal{F}}_{n,h})} J_h^2 \\
&\in \operatorname*{argmin}_{\widetilde{\mathcal{F}}_h = \mathrm{diag}(\widetilde{\mathcal{F}}_{1,h}, \cdots, \widetilde{\mathcal{F}}_{n,h})} \mathrm{Tr}(\widetilde{\mathcal{F}}_h \mathbb{I}_{p,h}\widetilde{\Sigma}_h \mathbb{I}_{p,h}^\top \widetilde{\mathcal{F}}_h^\top \widetilde{Q}_h^2) + \mathrm{Tr}((\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} + \widetilde{L}_h^4)\widetilde{\Sigma}_h(\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} + \widetilde{L}_h^4)^\top \widetilde{R}_{h+1}). \qquad (.9)
\end{aligned}$$

Then, plugging in the optimal $\widetilde{\mathcal{F}}_h^*$, we can get

$$\begin{aligned}
\widetilde{c}_h = &\mathrm{Tr}(\widetilde{\mathcal{F}}_h^* \mathbb{I}_{p,h}\widetilde{\Sigma}_h \mathbb{I}_{p,h}^\top (\widetilde{\mathcal{F}}_h^*)^\top \widetilde{Q}_h^2) + \mathrm{Tr}((\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h^* \mathbb{I}_{p,h} + \widetilde{L}_h^4)\widetilde{\Sigma}_h(\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h^* \mathbb{I}_{p,h} + \widetilde{L}_h^4)^\top \widetilde{R}_{h+1}) + \mathrm{Tr}(K_{h+1}^4 \widetilde{E}_{h+1}\Sigma_{0,h}\widetilde{E}_{h+1}^\top (K_{h+1}^4)^\top \widetilde{R}_{h+1}) \\
&+ \mathrm{Tr}(K_{h+1}^4 \mathrm{diag}(\widetilde{\Sigma}_{1,h+1}, \cdots, \widetilde{\Sigma}_{n,h+1})(K_{h+1}^4)^\top \widetilde{R}_{h+1}) + \mathrm{Tr}(\mathbb{I}_{x,h}\widetilde{\Sigma}_h \mathbb{I}_{x,h}^\top \widetilde{Q}_h^1) + \widetilde{c}_{h+1},
\end{aligned}$$

which completes the proof. ■

Solution of Eq. (.9)**:** From the fact that $\widetilde{\Sigma}_h, \widetilde{Q}_h^2, \widetilde{R}_{h+1} \succeq 0$, we can write

$$J_h^2 = ||(\widetilde{Q}_h^2)^{\frac{1}{2}} \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} (\widetilde{\Sigma}_h)^{\frac{1}{2}}||_F^2 + ||(\widetilde{R}_{h+1})^{\frac{1}{2}} (\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} + \widetilde{L}_h^4)(\widetilde{\Sigma}_h)^{\frac{1}{2}}||_F^2,$$

where $||X||_F$ denotes the Frobenius norm of X, for any matrix X. Therefore, we know that J_h^2 is a convex function with respect to $\widetilde{\mathcal{F}}_h$, and the optimal $\widetilde{\mathcal{F}}_h^*$ can be computed via the first-order condition.

Meanwhile, since $\widetilde{\Sigma}_h$ is the variance matrix of $\widetilde{\boldsymbol{S}}_h$, we can write it as $\widetilde{\Sigma}_h = \begin{bmatrix} \widetilde{\Sigma}_{xx} & \widetilde{\Sigma}_{xp} \\ \widetilde{\Sigma}_{px} & \widetilde{\Sigma}_{pp} \end{bmatrix}$. Then, we can simplify J_h^2 as

$$J_h^2 = \mathrm{Tr}(\widetilde{\mathcal{F}}_h \widetilde{\Sigma}_{pp} \widetilde{\mathcal{F}}_h^\top \widetilde{Q}_h^2) + \mathrm{Tr}(\widetilde{\mathcal{F}}_h \widetilde{\Sigma}_{pp} \widetilde{\mathcal{F}}_h^\top \widetilde{R}_{h+1}) + 2\,\mathrm{Tr}(\widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \begin{bmatrix} \widetilde{\Sigma}_{px} & \widetilde{\Sigma}_{xx} \end{bmatrix} (\widetilde{L}_h^4)^\top \widetilde{R}_{h+1}) + \mathrm{Tr}(\widetilde{L}_h^4 \widetilde{\Sigma}_h (\widetilde{L}_h^4)^\top \widetilde{R}_{h+1}).$$

By taking the gradient of J_h^2 with respect to $\widetilde{\mathcal{F}}_h$, we get

$$\nabla_{\widetilde{\mathcal{F}}_h} J_h^2 = 2(\widetilde{Q}_h^2 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} \widetilde{\Sigma}_h \mathbb{I}_{p,h}^\top + (\widetilde{L}_h^3)^\top \widetilde{R}_{h+1} \widetilde{L}_h^3 \widetilde{\mathcal{F}}_h \mathbb{I}_{p,h} \widetilde{\Sigma}_h \mathbb{I}_{p,h}^\top + (\widetilde{L}_h^3)^\top \widetilde{R}_{h+1} \widetilde{L}_h^4 \widetilde{\Sigma}_h \mathbb{I}_{p,h}^\top).$$

Then, we decouple $\nabla_{\widetilde{\mathcal{F}}_h} J_h^2$ into $n \times n$ parts, and set the block diagonal parts of it to zero, namely, let $[\nabla_{\widetilde{\mathcal{F}}_h} J_h^2]_{i,i} = \mathbf{0}$ for all $i \in [n]$. Here, $[X]_{i,j}$ means the block matrix at i^{th} row and j^{th} column of the matrix X with proper dimensions (i.e., the matrix has dimension $m_i \times m_j$, where m_k is the dimension of $\widetilde{\mathcal{P}}_{k,h}$, $\forall k \in [n]$). We denote the following matrices as $\widehat{\Sigma}_h = \mathbb{I}_{p,h} \widetilde{\Sigma}_h \mathbb{I}_{p,h}^\top, T_h^1 = (\widetilde{L}_h^3)^\top \widetilde{R}_{h+1} \widetilde{L}_h^3, T_h^2 = (\widetilde{L}_h^3)^\top \widetilde{R}_{h+1} \widetilde{L}_h^4 \widetilde{\Sigma}_h \mathbb{I}_{p,h}^\top$, here $\widehat{\Sigma}_h$ is the covariance matrix of $\widetilde{\boldsymbol{P}}_h$. Then, for any $i \in [n]$,

$$[\nabla_{\widetilde{\mathcal{F}}_h}]_{i,i} = \sum_{k=1}^{n} [\widetilde{Q}_h^2]_{i,k} \widetilde{\mathcal{F}}_{k,h} [\widehat{\Sigma}_h]_{k,i} + [T_h^1]_{i,k} \widetilde{\mathcal{F}}_{k,h} [\widehat{\Sigma}_h]_{k,i} + [T_h^2]_{i,i},$$

and we set $[\nabla_{\widetilde{\mathcal{F}}_h}]_{i,i} = \mathbf{0}$ for any $i \in [n]$. Define N_h to be the matrix such that $[N_h]_{i,j} = [\widehat{\Sigma}_h]_{j,i}^\top \otimes ([\widetilde{Q}_h^2]_{i,j} + [T_h^1]_{i,j})$, then we have

$$N_h \begin{bmatrix} \mathrm{vec}(\widetilde{\mathcal{F}}_{1,h}) \\ \cdots \\ \mathrm{vec}(\widetilde{\mathcal{F}}_{n,h}) \end{bmatrix} = - \begin{bmatrix} \mathrm{vec}([T_h^2]_{1,1}) \\ \cdots \\ \mathrm{vec}([T_h^2]_{n,n}) \end{bmatrix},$$

where vec() means the vectorization of a matrix. We can then solve it to obtain the optimal solutions $\{\widetilde{\mathcal{F}}_{i,h}^*\}_{i \in [n]}$ in a vectorization form as

$$\begin{bmatrix} \mathrm{vec}(\widetilde{\mathcal{F}}_{1,h}^*) \\ \cdots \\ \mathrm{vec}(\widetilde{\mathcal{F}}_{n,h}^*) \end{bmatrix} = -N_h^\dagger \begin{bmatrix} \mathrm{vec}([T_h^2]_{1,1}) \\ \cdots \\ \mathrm{vec}([T_h^2]_{n,n}) \end{bmatrix}.$$

D. Deferred Details of §V

Firstly, for any $h \in [H]$ and any realization $M_{1:h} \in \mathcal{M}_{1:h}$, we define the prescriptions of agent i as $\gamma_{i,h}^a \in \Gamma_{i,h}(M_{1:h}) := \{\gamma_{i,h}^a : \mathcal{P}_{i,h}(M_{1:h}) \to \mathcal{U}_i\}$. We denote by γ_h^a the joint prescriptions, and by $\Gamma_h^a(M_{1:h})$ the set of all possible joint prescriptions, under $M_{1:h}$. Then, we can have the following lemma.

Lemma .4: Let $\mathcal{D}$ be a JCCO problem satisfying Assumption II.1 and III.2, then the CIB belief evolves as follows

$$\boldsymbol{B}_{h^-} = \Phi_h^1(\boldsymbol{B}_{(h-1)^+}, \boldsymbol{Z}_h^b, \gamma_{h-1}^a), \boldsymbol{B}_{h^+} = \Phi_h^2(\boldsymbol{B}_{h^-}, \boldsymbol{Z}_h^a, \boldsymbol{M}_h), \tag{.10}$$

for some functions Φ_h^1, Φ_h^2.

Proof: Let $\boldsymbol{B}_{(h-1)^+}$ take values in $\mathcal{P}(\mathcal{X} \times \mathcal{P}_{(h-1)^+}(\boldsymbol{M}_{1:h-1}))$ for some $\boldsymbol{M}_{1:h-1}$, and then $\boldsymbol{B}_{h^-}$ takes value in $\mathcal{P}(\mathcal{X} \times \mathcal{P}_{h^-}(\boldsymbol{M}_{1:h-1}))$ for the same $\boldsymbol{M}_{1:h-1}$. We denote by $\varrho_{h^-}(X_h, P_{h^-} \,|\, \boldsymbol{C}_{h^-}, \boldsymbol{M}_{1:h-1}), \varrho_{h^+}(X_h, P_{h^+} \,|\, \boldsymbol{C}_{h^+}, \boldsymbol{M}_{1:h})$ the density function of distribution $\boldsymbol{B}_{h^-}, \boldsymbol{B}_{h^+}$, respectively. Then, considering any $\mathscr{S}_{h^-} \subseteq \mathcal{X} \times \mathcal{P}_{h^-}(\boldsymbol{M}_{1:h-1})$, we have

$$\begin{aligned}
&\mathbb{P}[(\boldsymbol{X}_h, \boldsymbol{P}_{h^-}) \in \mathscr{S}_{h^-} \,|\, \boldsymbol{C}_{h^-}, \boldsymbol{M}_{1:h-1}] = \int_{\mathscr{S}_{h^-}} \varrho_{h^-}(\boldsymbol{X}_h, \boldsymbol{P}_{h^-} \,|\, \boldsymbol{C}_{h^-}, \boldsymbol{M}_{1:h-1}) \mathrm{d}\boldsymbol{X}_h \mathrm{d}\boldsymbol{P}_{h^-} \\
&= \int_{\mathcal{W}_{0,h-1}} \Xi_{0,h-1}(\boldsymbol{W}_{0,h}) \int_{\mathcal{W}_{1:n,h}} \Xi_{1:n,h}(\boldsymbol{W}_{1:n,h}) \int_{\mathcal{X} \times \mathcal{P}_{(h-1)^+}(\boldsymbol{M}_{1:h-1})} \varrho_{(h-1)^+}(\boldsymbol{X}_{h-1}, \boldsymbol{P}_{(h-1)^+} \,|\, \boldsymbol{C}_{(h-1)^+}, \boldsymbol{M}_{1:h-1}) \\
&\quad \int_{\mathcal{X} \times \mathcal{P}_{h^-}(\boldsymbol{M}_{1:h-1})} \mathbb{1}[(\boldsymbol{X}_h, \boldsymbol{P}_{h^-}) \in \mathscr{S}_{h^-}] \mathbb{1}[\boldsymbol{X}_h = A_h \boldsymbol{X}_{(h-1)^+} + B_h \gamma_{h-1}^a(\boldsymbol{P}_{(h-1)^+}) + \boldsymbol{W}_{0,h-1}] \\
&\quad \mathbb{1}[\boldsymbol{Z}_h^b = \chi_h(\boldsymbol{P}_{(h-1)^+}, \gamma_{h-1}^a(\boldsymbol{P}_{(h-1)^+}), E_h \boldsymbol{X}_h + \boldsymbol{W}_{1:n,h})] \\
&\quad \mathbb{1}[\boldsymbol{P}_{h^-} = \zeta_h(\boldsymbol{P}_{(h-1)^+}, \gamma_{h-1}^a(\boldsymbol{P}_{(h-1)^+}), E_h \boldsymbol{X}_h + \boldsymbol{W}_{1:n,h})] \mathrm{d}\boldsymbol{W}_{1:n,h} \mathrm{d}\boldsymbol{W}_{0,h-1} \mathrm{d}\boldsymbol{X}_{h-1} \mathrm{d}\boldsymbol{P}_{(h-1)^+} \mathrm{d}\boldsymbol{X}_h \mathrm{d}\boldsymbol{P}_{h^-},
\end{aligned}$$

where $\mathcal{W}_{1:n,h}$ is the space of Gaussian noise $\boldsymbol{W}_{1:n,h}$, $\mathcal{W}_{0,h-1}$ is the space of $\boldsymbol{W}_{0,h-1}$, $\Xi_{0,h-1}$ is the density function of Gaussian noise $\boldsymbol{W}_{0,h-1}$, $\Xi_{1:n,h}(\boldsymbol{W}_{1:n,h})$ is the joint density function of Gaussian noise $\boldsymbol{W}_{1:n,h}$. Here, we denote $\boldsymbol{W}_{1:n,h} = \begin{bmatrix} \boldsymbol{W}_{1,h} \\ \cdots \\ \boldsymbol{W}_{n,h} \end{bmatrix}$, and $\gamma^a_{h-1}(\boldsymbol{P}_{(h-1)^+}) = \begin{bmatrix} \gamma^a_{1,h-1}(\boldsymbol{P}_{1,(h-1)^+}) \\ \cdots \\ \gamma^a_{n,h-1}(\boldsymbol{P}_{n,(h-1)^+}) \end{bmatrix}$. Therefore, the right-hand side of the equation only depends on $\varrho_{(h-1)^+}, \gamma^a_{h-1}, \boldsymbol{Z}^b_h$, where $\varrho_{(h-1)^+}$ is the density function of $\boldsymbol{B}_{(h-1)^+}$, and this means $\boldsymbol{B}_{h^-} = \Phi^1_h(\boldsymbol{B}_{(h-1)^+}, \gamma^a_{h-1}), \boldsymbol{Z}^b_h)$ for some Φ^1_h.

Similarly, for any $\mathscr{S}_{h^+} \subseteq \mathcal{X} \times \mathcal{P}_{h^+}(\boldsymbol{M}_{1:h})$, we have

$$\begin{aligned}\mathbb{P}[(\boldsymbol{X}_h, \boldsymbol{P}_{h^+}) \in \mathscr{S}_{h^+} \,|\, \boldsymbol{C}_{h^+}, \boldsymbol{M}_{1:h}] &= \int_{\mathscr{S}_{h^+}} \varrho_{h^-}(\boldsymbol{X}_h, \boldsymbol{P}_{h^+} \,|\, \boldsymbol{C}_{h^+}, \boldsymbol{M}_{1:h-1}) \mathrm{d}\boldsymbol{X}_h \mathrm{d}\boldsymbol{P}_{h^+} \\ &= \int_{\mathcal{X} \times \mathcal{P}_{h^-}(\boldsymbol{M}_{1:h-1})} \varrho_{h^-}(\boldsymbol{X}_h, \boldsymbol{P}_{h^-} \,|\, \boldsymbol{C}_{h^-}, \boldsymbol{M}_{1:h-1}) \int_{\mathcal{P}_{h^+}(\boldsymbol{M}_{1:h})} \mathbb{1}[(\boldsymbol{X}_h, \boldsymbol{P}_{h^+}) \in \mathscr{S}_{h^+}] \\ &\quad \mathbb{1}[\boldsymbol{Z}^a_h = \phi_h(\boldsymbol{M}_h, \boldsymbol{P}_{h^-}), \boldsymbol{P}_{h^+} = \boldsymbol{P}_{h^-} \backslash \boldsymbol{Z}^a_h] \mathrm{d}\boldsymbol{X}_h \mathrm{d}\boldsymbol{P}_{h^-} \mathrm{d}\boldsymbol{P}_{h^+}.\end{aligned}$$

Since the right-hand side of the equation only depends on $\varrho_{h^-}, \boldsymbol{Z}^a_h, \boldsymbol{M}_h$, where ϱ_{h^-} is the density function of $\boldsymbol{B}_{h^-}$, and this means $\boldsymbol{B}_{h^+} = \Phi^2_h(\boldsymbol{B}_{h^-}, \boldsymbol{Z}^a_h, \boldsymbol{M}_h)$ for some function Φ^2_h. ■

Given any JCCO problem $\mathcal{D}$ with closed-loop communication strategy, we can expand it into another JCCO problem $\widetilde{\mathcal{D}}$ (with notation system $\widetilde{\ }$). First, for any $h \in [H]$, we define the set $\Xi_{h^-} := \{(i,t) \,|\, i \in [n], t < h, \boldsymbol{I}_{i,t^+} \subseteq \boldsymbol{C}_{h^-}$ under any communication strategy, and $B_{i,t} \neq \boldsymbol{0}\}$ and $\Xi_{h^+} := \{(i,t) \,|\, i \in [n], t < h, \boldsymbol{I}_{i,t^+} \subseteq \boldsymbol{C}_{h^+}$ under any communication strategy, and $B_{i,t} \neq \boldsymbol{0}\}$. Then, we can expand $\mathcal{D}$ to $\widetilde{\mathcal{D}}$ as: for any $h \in [H]$

$$\begin{aligned}\widetilde{\boldsymbol{C}}_{h^-} &= \boldsymbol{C}_{h^-} \cup \{\widetilde{\boldsymbol{U}}_{i,t} \,|\, i \in [n], t < h, (i,t) \in \Xi_{h^-}\}, \widetilde{\boldsymbol{C}}_{h^+} = \boldsymbol{C}_{h^+} \cup \{\widetilde{\boldsymbol{U}}_{i,t} \,|\, i \in [n], t < h, (i,t) \in \Xi_{h^+}\}, \\ \forall i \in [n], \widetilde{\boldsymbol{P}}_{i,h^-} &= \boldsymbol{P}_{i,h^-} \backslash \{\widetilde{\boldsymbol{U}}_{i,t} \,|\, i \in [n], t < h, (i,t) \in \Xi_{h^-}\}, \widetilde{\boldsymbol{P}}_{i,h^+} = \boldsymbol{P}_{i,h^+} \backslash \{\widetilde{\boldsymbol{U}}_{i,t} \,|\, i \in [n], t < h, (i,t) \in \Xi_{h^+}\}.\end{aligned} \tag{.11}$$

Then, we have the following lemma.

Lemma .5: Let $\mathcal{D}$ be a JCCO problem with PN IS that satisfies Assumptions II.1, III.2, III.4, and V.1, and let $\widetilde{\mathcal{D}}$ be the JCCO expanded from $\mathcal{D}$ according to Eq. (.11). Then, $\widetilde{\mathcal{D}}$ satisfies Assumption II.1, and there exists a function φ^c such that for any team-optimal strategy $\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}$ of $\widetilde{\mathcal{D}}(g^m_{1:H})$, $(g^{m,*}_{1:H}, g^{a,*}_{1:H}) = \varphi^c(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}, \mathcal{D})$ is a team-optimal strategy of $\mathcal{D}(g^m_{1:H})$, and $J_{\mathcal{D}}(g^{m,*}_{1:H}, g^{a,*}_{1:H}) = J_{\widetilde{\mathcal{D}}}(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H})$.

Proof: This proof consists of two **Parts**: **Part 1**: $\widetilde{\mathcal{D}}$ satisfies Assumption II.1; **Part 2**: we can construct optimal strategies of $\mathcal{D}$ from the optimal strategies of $\widetilde{\mathcal{D}}$, and the optimal values of the two problem are the same.

Part 1: To begin with, we discuss a property of the problem $\widetilde{\mathcal{D}}$. For any $i \in [n], h \in [H]$, if $B_{i,h} = \boldsymbol{0}$, then from Assumption III.2, $\boldsymbol{U}_{i,h} \notin \boldsymbol{I}_{j,(h')^-}, \boldsymbol{U}_{i,h} \notin \boldsymbol{I}_{j,(h')^+}, \forall h' > h, j \in [n]$. Also from expansion, we know that $\widetilde{\boldsymbol{U}}_{i,h} = \boldsymbol{U}_{i,h}$ will never be added into common information. Therefore, $(\widetilde{\boldsymbol{U}}_{i,h}) \notin \widetilde{\boldsymbol{I}}_{j,(h')^-}, \widetilde{\boldsymbol{U}}_{i,h} \notin \widetilde{\boldsymbol{I}}_{j,(h')^+}, \forall h' > h, j \in [n]$.

If $B_{i,h} \neq \boldsymbol{0}$, then from Assumption III.4, there exists $j \neq i$ such that $\boldsymbol{U}_{i,h}$ influences $\boldsymbol{Y}_{j,h+1}$. Also, since $\mathcal{D}$ satisfies Assumption II.1 (e), $\boldsymbol{U}_{i,h}$ influences $\boldsymbol{I}_{j,(h+1)^-}$. Together with the fact that $\mathcal{D}$ is PN, we know that $\boldsymbol{I}_{i,h^-} \subseteq \boldsymbol{I}_{j,(h+1)^-}$, and this can happen only if $\boldsymbol{I}_{i,h^-} \subseteq \boldsymbol{C}_{(h+1)^-}$. Meanwhile, due to $\boldsymbol{I}_{i,h^+} \backslash \boldsymbol{I}_{i,h^-} = \boldsymbol{Z}^a_{i,h} \subseteq \boldsymbol{C}_{h^+} \subseteq \boldsymbol{C}_{(h+1)^-}$, we know $\boldsymbol{I}_{i,h^+} \subseteq \boldsymbol{C}_{(h+1)^-}$. Finally, we know that $(i,h) \in \Xi_{(h')^-}$ and $(i,h) \in \Xi_{(h')^+}$ for any $h' > h$. This means if $B_{i,h} \neq \boldsymbol{0}, \widetilde{\boldsymbol{U}}_{i,h}$ will be added into $\widetilde{\boldsymbol{C}}_{(h+1)^-}$ in $\widetilde{\mathcal{D}}$ immediately (if it is not shared via the baseline sharing in $\mathcal{D}$).

Then, based on the property discussed above, we show that $\widetilde{\mathcal{D}}$ satisfies Assumption II.1.

(a) Since $\mathcal{D}$ satisfies Assumption II.1, then $\boldsymbol{Z}^b_h = \chi_h(\boldsymbol{P}_{(h-1)^+}, \boldsymbol{U}_{h-1}, \boldsymbol{Y}_h)$ for a fixed projection function χ_h. We claim that $\widetilde{\boldsymbol{Z}}^b_h \subseteq \widetilde{\boldsymbol{P}}_{(h-1)^+} \cup \{\widetilde{\boldsymbol{U}}_{h-1}, \widetilde{\boldsymbol{Y}}_h\}$: This is because it holds that $\boldsymbol{Z}^b_h \subseteq \boldsymbol{P}_{(h-1)^-} \cup \{\boldsymbol{U}_{h-1}, \boldsymbol{Y}_h\}$; from the property discussed above, $\widetilde{\boldsymbol{Z}}^b_h \backslash \boldsymbol{Z}^b_h$ only consists of some control actions at timestep $h-1$; from expansion, $(\widetilde{\boldsymbol{P}}_{h-1} \backslash \boldsymbol{P}_{h-1}) \subseteq \widetilde{\boldsymbol{C}}_{h^-}$, so $(\widetilde{\boldsymbol{P}}_{h-1} \backslash \boldsymbol{P}_{h-1}) \cap \widetilde{\boldsymbol{Z}}^b_h = \emptyset$. Also, since χ_h is a projection function, and Ξ_{h^-}, Ξ_{h^+} is pre-defined and is not affected by the realization of each random variable, there exists a fixed projection function $\widetilde{\chi}_h$ such that $\widetilde{\boldsymbol{Z}}^b_h = \widetilde{\chi}_h(\widetilde{\boldsymbol{P}}_{(h-1)^+}, \widetilde{\boldsymbol{U}}_{h-1}, \widetilde{\boldsymbol{Y}}_h)$.

(b) From the property discussed above, we know that $\widetilde{\boldsymbol{Z}}^a_{i,h} \subseteq \boldsymbol{Z}^a_{i,h}$. Also, for any control action $\widetilde{\boldsymbol{U}}_{i,t} \in \boldsymbol{Z}^a_{i,h} \subseteq \widetilde{\boldsymbol{Z}}^a_h, \forall t < h$, such control must be already shared in $\widetilde{\mathcal{D}}$ before timestep h, i.e., $\widetilde{\boldsymbol{U}}_{i,t} \in \boldsymbol{C}_{h^-}$. Therefore, it holds that $\widetilde{\boldsymbol{Z}}^a_{i,h} \subseteq \widetilde{\boldsymbol{P}}_{i,h^-}$. Meanwhile, since for any realization $M_{i,h} \in \mathcal{M}_{i,h}, \phi_{i,h}(\cdot, M_{i,h})$ is a fixed projection functions, and which control action lies in $\boldsymbol{Z}^a_{i,h} \backslash \widetilde{\boldsymbol{Z}}^a_{i,h}$ is determined by Ξ_{h^-}, Ξ_{h^+}, which is pre-defined and does not affect by the realization of each random variable, then there exists a fixed projection function $\widetilde{\phi}_{i,h}(\cdot, \widetilde{M}_{i,h})$ such that $\widetilde{\boldsymbol{Z}}^a_{i,h} = \widetilde{\phi}_{i,h}(\widetilde{\boldsymbol{P}}_{i,h^-}, \widetilde{M}_{i,h})$. Therefore, there exists a function $\widetilde{\phi}_h$ such that $\widetilde{\boldsymbol{Z}}^a_h = \widetilde{\phi}_h(\widetilde{\boldsymbol{P}}_{h^-}, \widetilde{\boldsymbol{M}}_h)$.

(c) Since $\mathcal{D}$ satisfies Assumption II.1, then for any $i \in [n], \boldsymbol{P}_{i,h^-} = \zeta_{i,h}(\boldsymbol{P}_{i,(h-1)^+}, \boldsymbol{U}_{i,h-1}, \boldsymbol{Y}_{i,h})$ for a fixed projection function ζ_h. We claim that $\widetilde{\boldsymbol{P}}_{i,h^-} \subseteq \widetilde{\boldsymbol{P}}_{i,(h-1)^+} \cup \{\widetilde{\boldsymbol{U}}_{i,h-1}, \widetilde{\boldsymbol{Y}}_{i,h}\}$: This is because it holds that $\boldsymbol{P}_{i,h^-} \subseteq \boldsymbol{P}_{i,(h-1)^-} \cup \{\boldsymbol{U}_{i,h-1}, \boldsymbol{Y}_{i,h}\}$; from expansion, it holds that $\widetilde{\boldsymbol{P}}_{i,h^-} \subseteq \boldsymbol{P}_{i,h^-}$, and for any $t < h$ such $(i,t) \in \Xi_{(h-1)^+}$, it must hold that

$(i,t) \in \Xi_{h^-}$, which means if $\widetilde{\boldsymbol{U}}_{i,t} \in \boldsymbol{P}_{i,(h-1)^+} \backslash \widetilde{\boldsymbol{P}}_{i,(h-1)^+}$, then $\widetilde{\boldsymbol{U}}_{i,t} \in \boldsymbol{P}_{i,h^-} \backslash \widetilde{\boldsymbol{P}}_{i,h^-}$, so $\widetilde{\boldsymbol{U}}_{i,t} \notin \widetilde{\boldsymbol{P}}_{i,h^-}$. Also, since $\zeta_{i,h}$ is a projection function, and Ξ_{h^-}, Ξ_{h^+} is pre-defined and does not affect by the realization of each random variable, there exists a fixed projection function $\widetilde{\zeta}_{i,h}$ such that $\widetilde{\boldsymbol{P}}_{i,h^-} = \widetilde{\zeta}_{i,h}(\widetilde{\boldsymbol{P}}_{i,(h-1)^+}, \widetilde{\boldsymbol{U}}_{i,h-1}, \widetilde{\boldsymbol{Y}}_{i,h})$. Therefore, there exists a function $\widetilde{\zeta}_h$ such that $\widetilde{\boldsymbol{P}}_{h^-} = \widetilde{\zeta}_h(\widetilde{\boldsymbol{P}}_{(h-1)^+}, \widetilde{\boldsymbol{U}}_{h-1}, \widetilde{\boldsymbol{Y}}_h)$.

(d) For each $i \in [n], \widetilde{\boldsymbol{P}}_{i,h^+} = \widetilde{\boldsymbol{P}}_{i,h^-} \backslash \widetilde{\boldsymbol{Z}}^a_{i,h}$.

(e) For each $i \in [n]$, since construction of $\widetilde{\mathcal{D}}$ only adds some actions into $\widetilde{\boldsymbol{C}}_{h^-}, \widetilde{\boldsymbol{C}}_{h^+}$ and does not change observations, and it always holds that $\widetilde{\boldsymbol{C}}_{h^-} \subseteq \widetilde{\boldsymbol{C}}_{h^+} \subseteq \widetilde{\boldsymbol{C}}_{(h+1)^-}$. Therefore, $\widetilde{\boldsymbol{I}}_{i,h^-} \subseteq \widetilde{\boldsymbol{I}}_{i,h^+} \subseteq \widetilde{\boldsymbol{I}}_{i,(h+1)^-}$, and $\widetilde{\boldsymbol{Y}}_{i,h} \in \widetilde{\boldsymbol{I}}_{i,h^-}$.

Part 2: From the construction of $\widetilde{\mathcal{D}}$, we know that the system dynamics and communication cost are the same for both $\mathcal{D}$ and $\widetilde{\mathcal{D}}$, and it holds that $\boldsymbol{I}_{i,h} \subseteq \widetilde{\boldsymbol{I}}_{i,h}$ for any $i \in [n], h \in [H]$. So the agents in $\widetilde{\mathcal{D}}$ have larger strategy spaces than those in $\mathcal{D}$. Therefore, $\min_{g^m_{1:H}, g^a_{1:H}} J_{\mathcal{D}}(g^m_{1:H}, g^a_{1:H}) \geq \min_{\widetilde{g}^m_{1:H}, \widetilde{g}^a_{1:H}} J_{\widetilde{\mathcal{D}}}(\widetilde{g}^m_{1:H}, \widetilde{g}^a_{1:H})$.

Now we show that for any optimal strategies $(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H})$ of problem $\widetilde{\mathcal{D}}$, we can construct $(g^{m,*}_{1:H}, g^{a,*}_{1:H}) = \varphi^c(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}, \mathcal{D})$ of problem $\mathcal{D}$ such that $J_{\mathcal{D}}(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}) = J_{\widetilde{\mathcal{D}}}(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H})$. Let $(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}) \in \operatorname{argmin}_{\widetilde{g}^m_{1:H}, \widetilde{g}^a_{1:H}} J_{\widetilde{\mathcal{D}}}(\widetilde{g}^m_{1:H}, \widetilde{g}^a_{1:H})$, we construct strategies $(g^{m,*}_{1:H}, g^{a,*}_{1:H})$ in a recursive way. For $h = 1$ and any $i \in [n]$, note that $\boldsymbol{I}_{i,1^-} = \widetilde{\boldsymbol{I}}_{i,1^-}, \boldsymbol{I}_{i,1^+} = \widetilde{\boldsymbol{I}}_{i,1^+}$ always holds. Then, just construct $\widetilde{g}^{m,*}_{i,1} = g^{m,*}_{i,1}, \widetilde{g}^{a,*}_{i,1} = g^{a,*}_{i,1}$. It is immediate that these two strategies output the same communication and control actions.

Now we assume that we have got $\widetilde{g}^{m,*}_{1:h-1}, \widetilde{g}^{a,*}_{1:h-1}$ that output the same actions as $g^{m,*}_{1:h-1}, g^{a,*}_{1:h-1}$. There are only some actions $\widetilde{\boldsymbol{U}}_{j,t}, j \in [n], t < h$ in $\widetilde{\boldsymbol{C}}_{h^-} \backslash \boldsymbol{C}_{h^-}$, and if $\widetilde{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{C}}_{h^-} \backslash \boldsymbol{C}_{h^-}, \forall j \in [n]$, then from expansion, we know that $\boldsymbol{I}_{j,t^+} \subseteq \boldsymbol{C}_{h^-}$ will hold under any additional sharing. Meanwhile, since $g^{a,*}_{j,t}$ outputs the same control action as $\widetilde{g}^{a,*}_{j,t}$, we can recover $\widetilde{U}_{j,t} = g^{a,*}_{j,t}(\boldsymbol{I}_{j,t^+}, \boldsymbol{M}_{1:t})$. Therefore, we can construct $\widetilde{\boldsymbol{C}}_{h^-}$ from $\boldsymbol{C}_{h^-}, \boldsymbol{M}_{1:t}$ and $g^{a,*}_{1:h-1}$ by recovering all possible $\widetilde{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{C}}_{h^-} \backslash \boldsymbol{C}_{h^-}$, and then we construct $g^{m,*}_{i,h}(\boldsymbol{C}_{h^-}, \boldsymbol{M}_{1:h-1}) = \widetilde{g}^{m,*}_{i,h^-}(\widetilde{\boldsymbol{C}}_{h^-}, \boldsymbol{M}_{1:h-1})$, and these two strategies will output the same communication action.

Similarly, we assume that we have obtained $\widetilde{g}^{m,*}_{1:h}, \widetilde{g}^{a,*}_{1:h-1}$ that output the same actions as $g^{m,*}_{1:h}, \widetilde{g}^{a,*}_{1:h-1}$. For any i, there are only some actions $\widetilde{\boldsymbol{U}}_{j,t}, j \in [n], t < h$ in $\widetilde{\boldsymbol{I}}_{i,h^+} \backslash \boldsymbol{I}_{i,h^+}$. And if $\widetilde{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h^+} \backslash \boldsymbol{I}_{i,h^+}, \forall j \in [n]$, then we know that $B_{j,t} \neq \mathbf{0}$. Then, from Assumption III.4, then from expansion, we know that $\boldsymbol{I}_{j,t^+} \subseteq \boldsymbol{C}_{h^+}$ will hold under any additional sharing. Meanwhile, since $g^{a,*}_{j,t}$ outputs the same control action as $\widetilde{g}^{a,*}_{j,t}$, we can recover $\widetilde{U}_{j,t} = g^{a,*}_{j,t}(\boldsymbol{I}_{j,t^+}, \boldsymbol{M}_{1:t})$. Therefore, we can construct $\widetilde{\boldsymbol{I}}_{i,h^+}$ from $\boldsymbol{I}_{i,h^+}, \boldsymbol{M}_{1:t}$ and $g^{a,*}_{1:h-1}$ by recovering all possible $\widetilde{\boldsymbol{U}}_{j,t} \in \widetilde{\boldsymbol{I}}_{i,h^+} \backslash \boldsymbol{I}_{i,h^+}$, and then we construct $g^{a,*}_{i,h}(\boldsymbol{I}_{i,h^+}, \boldsymbol{M}_{1:h}) = \widetilde{g}^{a,*}_{i,h^+}(\widetilde{\boldsymbol{I}}_{i,h^+}, \boldsymbol{M}_{1:h})$, and these two strategies will output the same control action.

Recursively, we can construct $(g^{m,*}_h, g^{a,*}_h)$ from $h = 1$ to H based on $(\widetilde{g}^{m,*}_h, \widetilde{g}^{a,*}_h)$ and the problem $\mathcal{D}$ (, and the problem $\widetilde{\mathcal{D}}$, which can be constructed from $\mathcal{D}$). Then, we define such $(g^{m,*}_{1:H}, g^{a,*}_{1:H}) = \varphi^c(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}, \mathcal{D})$. Since the dynamics and cost are the same in two problems, and $(g^{m,*}_h, g^{a,*}_h)$ and $(\widetilde{g}^{m,*}_h, \widetilde{g}^{a,*}_h)$ will output the same communication and control actions for any $h \in [H]$. Therefore, we have $J_{\mathcal{D}}(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}) = J_{\widetilde{\mathcal{D}}}(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H})$. Also, we know that

$$J_{\mathcal{D}}(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}) = \min_{\widetilde{g}^m_{1:H} \in \widetilde{\mathcal{G}}^m_{1:H}, \widetilde{g}^a_{1:H} \in \widetilde{\mathcal{G}}^a_{1:H}} J_{\widetilde{\mathcal{D}}}(\widetilde{g}^m_{1:H}, \widetilde{g}^a_{1:H}) \leq \min_{g^m_{1:H} \in \mathcal{G}^m_{1:H}, g^a_{1:H} \in \mathcal{G}^a_{1:H}} J_{\mathcal{D}}(g^m_{1:H}, g^a_{1:H}) \leq J_{\widetilde{\mathcal{D}}}(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}).$$

Therefore, we know that $(g^{m,*}_{1:H}, g^{a,*}_{1:H})$ is an optimal strategy of the problem $\mathcal{D}$, completing the proof. ■

Now, we introduce the following theorem as a full version of Theorem V.3.

Theorem .6 (Full version of Theorem V.3): Let $\mathcal{D}$ be a JCCO problem with PN IS and satisfying Assumptions II.1, III.2, III.4, and V.1, and $\widetilde{\mathcal{D}}$ be the JCCO expanded from $\mathcal{D}$, then $\widetilde{\mathcal{D}}$ is a JCCO problem satisfying the SI-CIB condition, and Assumptions II.1, III.2, III.4, V.1. Moreover, for any $h \in [H]$, $\widetilde{\boldsymbol{B}}_{h^-}, \widetilde{\boldsymbol{B}}_{h^+}$ admit Gaussian distributions. Formally, $\widetilde{\boldsymbol{B}}_{h^-} = \mathcal{N}(\widetilde{\boldsymbol{\Theta}}_{h^-}, \widetilde{\Sigma}_{h^-}), \widetilde{\boldsymbol{B}}_{h^+} = \mathcal{N}(\widetilde{\boldsymbol{\Theta}}_{h^+}, \widetilde{\Sigma}_{h^+})$, and

$$\begin{aligned} \widetilde{\boldsymbol{\Theta}}_{h^-} &= \widetilde{\Psi}^1_{h^-}(\widetilde{\boldsymbol{\Theta}}_{(h-1)^+}, \widetilde{\boldsymbol{Z}}^b_h, \widetilde{\boldsymbol{M}}_{1:h-1}), & \widetilde{\boldsymbol{\Theta}}_{h^+} &= \widetilde{\Psi}^1_{h^+}(\widetilde{\boldsymbol{\Theta}}_{h^-}, \widetilde{\boldsymbol{Z}}^a_h, \widetilde{\boldsymbol{M}}_{1:h}), \\ \widetilde{\Sigma}_{h^-} &= \widetilde{\Psi}^2_{h^-}(\widetilde{\boldsymbol{M}}_{1:h-1}), & \widetilde{\Sigma}_{h^+} &= \widetilde{\Psi}^2_{h^+}(\widetilde{\boldsymbol{M}}_{1:h}), \end{aligned} \tag{.12}$$

for some functions $\widetilde{\Psi}^1_{h^-}, \widetilde{\Psi}^1_{h^+}, \widetilde{\Psi}^2_{h^-}, \widetilde{\Psi}^2_{h^+}$ that do not depend on the strategies $\widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h}$. Therefore, it holds that

$$\widetilde{\boldsymbol{\Theta}}_{h^-} = \widetilde{\Psi}^3_{h^-}(\widetilde{\boldsymbol{C}}_{h^-}, \widetilde{\boldsymbol{M}}_{1:h-1}), \widetilde{\boldsymbol{\Theta}}_{h^+} = \widetilde{\Psi}^3_{h^+}(\widetilde{\boldsymbol{C}}_{h^+}, \widetilde{\boldsymbol{M}}_{1:h}), \tag{.13}$$

for some functions $\widetilde{\Psi}^3_{h^-}, \widetilde{\Psi}^3_{h^+}$ that do not depend on the strategies $\widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h}$.

Proof: This proof consists of three **Parts**: **Part 1:** $\widetilde{\mathcal{D}}$ satisfies Assumptions II.1, III.2, III.4, and V.1; **Part 2:** $\widetilde{\mathcal{D}}$ satisfies the SI-CIB condition; **Part 3:** Beliefs $\widetilde{\boldsymbol{B}}_{h^-}, \widetilde{\boldsymbol{B}}_{h^+}$ admit Gaussian distributions, and the mean and covariance satisfies Eq. (.12) and (.13).

Part 1: From Lemma .5, we know that $\widetilde{\mathcal{D}}$ satisfies Assumption II.1; Since for any $i \in [n], h \in [H], t < h$, we add control actions $\widetilde{\boldsymbol{U}}_{j,t}$ into common information at timestep h in the expansion procedure only if $\widetilde{B}_{j,t} = B_{j,t} \neq \mathbf{0}$, $\widetilde{\mathcal{D}}$ satisfies Assumption III.2; Since expansion procedure does not change the system dynamics and $\widetilde{E}_h = E_{i,h}, \forall i \in [n], h \in [H]$,

$\widetilde{\mathcal{D}}$ satisfies Assumption III.4; After expansion, $\widetilde{g}^m_{i,h}, \forall i \in [n], h \in [H]$ can still only take $\widetilde{\boldsymbol{C}}_{h^-}$ as input, and $\widetilde{\mathcal{D}}$ satisfies Assumption V.1.

Part 2: Firstly, let $\widetilde{C}_{h^-} \in \widetilde{\mathcal{C}}_{h^-}, \widetilde{M}_{1:h-1} \in \widetilde{\mathcal{M}}_{1:h-1}$ and any two strategies $(\widetilde{g}^m_{1:h-1}, \widetilde{g}^a_{1:h-1})$ and $(\widetilde{g}^{m,'}_{1:h-1}, \widetilde{g}^{a,'}_{1:h-1})$ that can reach $\widetilde{C}_{h^-}$ and $\widetilde{M}_{1:h-1}$, we want to show that

$$\widetilde{\Pi}_{h^-}(\widetilde{C}_{h^-}, \widetilde{M}_{1:h-1}, \widetilde{g}^m_{1:h-1}, \widetilde{g}^a_{1:h-1}) = \widetilde{\Pi}_{h^-}(\widetilde{C}_{h^-}, \widetilde{M}_{1:h-1}, \widetilde{g}^{m,'}_{1:h-1}, \widetilde{g}^{a,'}_{1:h-1}).$$

Given $\widetilde{M}_{1:h-1} \in \widetilde{\mathcal{M}}_{1:h-1}$, we know that the belief $\widetilde{\boldsymbol{B}}_{h^-}$ takes value in $\mathcal{P}(\widetilde{\mathcal{X}} \times \widetilde{\mathcal{P}}_{h^-}(\widetilde{M}_{1:h-1}))$. Consider any cube $Q \subseteq \mathcal{P}(\widetilde{\mathcal{X}} \times \widetilde{\mathcal{P}}_{h^-}(\widetilde{M}_{1:h-1}))$, let probability $\mathbb{P}_1 = \mathbb{P}((\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_{h^-}) \in Q \,|\, \widetilde{\boldsymbol{C}}_{h^-}, \widetilde{\boldsymbol{M}}_{1:h-1}, \widetilde{g}^m_{1:h-1}, \widetilde{g}^a_{1:h-1}), \mathbb{P}_2 = \mathbb{P}((\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_{h^-}) \in Q \,|\, \widetilde{\boldsymbol{C}}_{h^-}, \widetilde{\boldsymbol{M}}_{1:h-1}, \widetilde{g}^{m,'}_{1:h-1}, \widetilde{g}^{a,'}_{1:h-1})$. It suffices to prove that $\mathbb{P}_1 = \mathbb{P}_2$. Without loss of generality, we can assume $(\widetilde{g}^m_{1:h-1}, \widetilde{g}^a_{1:h-1})$ and $(\widetilde{g}^{m,'}_{1:h-1}, \widetilde{g}^{a,'}_{1:h-1})$ only differ in $\widetilde{g}^m_{i_1,h_1}$ and $\widetilde{g}^{m,'}_{i_1,h_1}$, or differ in $\widetilde{g}^a_{i_1,h_1}$ and $\widetilde{g}^{a,'}_{i_1,h_1}$, for some $i_1 \in [n], h_1 < h$.

Case 1: they differ in $\widetilde{g}^m_{i_1,h_1}$ and $\widetilde{g}^{m,'}_{i_1,h_1}$. From Assumption V.1, the input and output of $\widetilde{g}^m_{i_1,h_1}$ are $\widetilde{\boldsymbol{C}}_{h_1^-}$ and $\widetilde{\boldsymbol{M}}_{i_1,h_1}$, and $\widetilde{\boldsymbol{C}}_{h_1^-} \subseteq \widetilde{\boldsymbol{C}}_{h^-}$ and $\widetilde{\boldsymbol{M}}_{i_1,h_1} \subseteq \widetilde{\boldsymbol{M}}_{1:h-1}$. Therefore, the probability does not depend on how agent i chooses $\widetilde{g}^m_{i_1,h_1}$ and therefore $\mathbb{P}_1 = \mathbb{P}_2$.

Case 2: they differ in $\widetilde{g}^a_{i_1,h_1}$ and $\widetilde{g}^{a,'}_{i_1,h_1}$. If $\widetilde{B}_{i_1,h_1} = \mathbf{0}$, then $\widetilde{\boldsymbol{U}}_{i_1,h_1}$ will never influence system dynamics. Also, from Assumption III.2, $\widetilde{\boldsymbol{U}}_{i_1,h_1}$ will never appear in $\widetilde{\boldsymbol{I}}_{j,h'}$ for any $h' > h, j \in [n]$. Therefore, the control action $\widetilde{\boldsymbol{U}}_{i_1,h_1}$ cannot influence $\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_{h^-}, \widetilde{\boldsymbol{C}}_{h^-}$ and $\widetilde{\boldsymbol{M}}_{1:h-1}$, and then how agent i chooses it based on $\widetilde{g}^a_{i_1,h_1}$ does not influence the probability $\mathbb{P}((\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_{h^-}) \in Q \,|\, \widetilde{\boldsymbol{C}}_{h^-}, \widetilde{\boldsymbol{M}}_{1:h-1})$, which means $\mathbb{P}_1 = \mathbb{P}_2$. If $\widetilde{B}_{i_1,h_1} \neq \mathbf{0}$, then since $\mathcal{D}$ satisfies Assumption III.4, the action $\boldsymbol{U}_{i_1,h_1}$ will influence $\boldsymbol{Y}_{j,h_1+1}$ for some $j \neq i_1$. Also, from Assumption II.1 (e), the action $\boldsymbol{U}_{i_1,h_1}$ will influence $\boldsymbol{I}_{j,(h_1+1)^-}$. Since $\mathcal{D}$ is PN and $j \neq i_1$, then $\boldsymbol{I}_{i_1,h_1^-} \subseteq \boldsymbol{C}_{(h_1+1)^-}$ when there is no additional sharing. Therefore, $\boldsymbol{I}_{i_1,h_1^+} \subseteq \boldsymbol{C}_{(h_1+1)^-}$ under any communication strategy since any information in $\boldsymbol{I}_{i_1,h_1^+}$ through additional sharing is in the common information. Therefore, we know $(i_1, h_1) \in \Xi_{(h_1+1)^-}$. Then, from the construction of $\widetilde{\mathcal{D}}$, it holds $\widetilde{\boldsymbol{U}}_{i_1,h_1} \in \widetilde{\boldsymbol{C}}_{(h_1+1)^-}$ from the construction of $\widetilde{\mathcal{D}}$, and $\widetilde{\boldsymbol{I}}_{i_1,h_1^+} \subseteq \widetilde{\boldsymbol{C}}_{(h_1+1)^-}$ since we only added some information into common information in the construction of $\widetilde{\mathcal{D}}$. Therefore, the input and output of $\widetilde{g}^a_{i_1,h_1}$ are both included in $\widetilde{\boldsymbol{C}}_{h^-}$, and then $\mathbb{P}_1 = \mathbb{P}_2$ holds.

Secondly, let $\widetilde{C}_{h^+} \in \widetilde{\mathcal{C}}_{h^+}, \widetilde{M}_{1:h} \in \widetilde{\mathcal{M}}_{1:h}$ and any two strategies $(\widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h-1})$ and $(\widetilde{g}^{m,'}_{1:h}, \widetilde{g}^{a,'}_{1:h-1})$ that can reach $\widetilde{C}_{h^+}$ and $\widetilde{M}_{1:h}$, we can show that

$$\widetilde{\Pi}_{h^+}(\widetilde{C}_{h^+}, \widetilde{M}_{1:h}, \widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h-1}) = \widetilde{\Pi}_{h^+}(\widetilde{C}_{h^+}, \widetilde{M}_{1:h}, \widetilde{g}^{m,'}_{1:h}, \widetilde{g}^{a,'}_{1:h-1}),$$

in the same way as above.

Part 3: For any $h \in [H]$, communication strategy $\widetilde{g}^{m,'}_{1:h} \in \widetilde{\mathcal{G}}^m_{1:h}$ and control strategy $\widetilde{g}^{a,'}_{1:h-1} \in \widetilde{\mathcal{G}}^a_{1:h-1}$ (not necessarily to be linear), and any realization $\widetilde{C}_{h^+} \in \widetilde{\mathcal{C}}_{h^+}, \widetilde{M}_{1:h} \in \widetilde{\mathcal{M}}_{1:h}$ that is reachable under $(\widetilde{g}^{m,'}_{1:h}, \widetilde{g}^{a,'}_{1:h-1})$, we aim to prove that $\widetilde{\Pi}_{h^+}(\widetilde{C}_{h^+}, \widetilde{M}_{1:h}, \widetilde{g}^{m,'}_{1:h}, \widetilde{g}^{a,'}_{1:h-1})$ is a Gaussian distribution.

Since $(\widetilde{C}_{h^+}, \widetilde{M}_{1:h})$ are reachable under $(\widetilde{g}^{m,'}_{1:h}, \widetilde{g}^{a,'}_{1:h-1})$, there exist realizations $\widetilde{Y}_{1:h}, \widetilde{U}_{1:h-1}$ such that $(\widetilde{C}_{h^+}, \widetilde{M}_{1:h}, \widetilde{Y}_{1:h}, \widetilde{U}_{1:h-1})$ are reachable under $(\widetilde{g}^{m,'}_{1:h}, \widetilde{g}^{a,'}_{1:h-1})$. Then, we construct communication strategies $\widetilde{g}^m_{1:h}$ and control strategies $\widetilde{g}^a_{1:h-1}$ as: for any $i \in [n], t \leq h$, $\widetilde{g}^m_{i,t}$ always chooses $\widetilde{M}_{i,t}$, and for any $t' < h$, $\widetilde{g}^a_{i,t'}$ always chooses $\widetilde{U}_{i,t'}$. Note that, such $\widetilde{g}^a_{1:h-1}$ are affine functions, and under $\widetilde{g}^m_{1:h}$, we know that which elements in $\widetilde{\boldsymbol{P}}_{h^+}$ and $\widetilde{\boldsymbol{C}}_{h^+}$ are fixed based on Assumption II.1. Moreover, $(\widetilde{C}_{h^+}, \widetilde{M}_{1:h}, \widetilde{Y}_{1:h}, \widetilde{U}_{1:h-1})$ are reachable under $(\widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h-1})$. Therefore, from the linearity of system dynamics, under strategy $(\widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h-1})$, it holds that $\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{C}}_{h^+}, \widetilde{\boldsymbol{P}}_{h^+}$ are jointly Gaussian. Then, the distribution of $\widetilde{\boldsymbol{X}}_h, \widetilde{\boldsymbol{P}}_{h^+}$ conditioned on $\widetilde{\boldsymbol{C}}_{h^+}$ is jointly Gaussian, and the covariance does not depend on $\widetilde{\boldsymbol{C}}_{h^+}$. Finally, from the SI-CIB condition, it holds that $\widetilde{\Pi}_{h^+}(\widetilde{\boldsymbol{C}}_{h^+}, \widetilde{\boldsymbol{M}}_{1:h}, \widetilde{g}^{m,'}_{1:h}, \widetilde{g}^{a,'}_{1:h-1}) = \widetilde{\Pi}_{h^+}(\widetilde{\boldsymbol{C}}_{h^+}, \widetilde{\boldsymbol{M}}_{1:h}, \widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h-1})$, which is a Gaussian distribution with the covariance independent to $\widetilde{C}_{h^+}$.

Therefore, $\widetilde{\boldsymbol{B}}_{h^+}$ is a Gaussian distribution with covariance depends only on $\widetilde{\boldsymbol{M}}_{1:h}$ and does not depend on the strategies $\widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h-1}$, and $\widetilde{\boldsymbol{C}}_{h^+}$. This means $\widetilde{\Sigma}_{h^+} = \widetilde{\Psi}^2_{h^+}(\widetilde{\boldsymbol{M}}_{1:h})$ for some $\widetilde{\Psi}^2_{h^+}$.

In the same way, we can prove that for any $h \in [H+1], \widetilde{\boldsymbol{B}}_{h^-}$ is a Gaussian distribution with covariance only depends on $\boldsymbol{M}_{1:h-1}$ and does not depend on the strategies $\widetilde{g}^m_{1:h-1}, \widetilde{g}^a_{1:h-1}$, and $\widetilde{\boldsymbol{C}}_{h^-}$, and $\widetilde{\Sigma}_{h^-} = \widetilde{\Psi}^2_{h^-}(\widetilde{\boldsymbol{M}}_{1:h-1})$ for some $\widetilde{\Psi}^2_{h^-}$.

Since $\widetilde{\mathcal{D}}$ satisfies Assumption II.1 and III.2, from Lemma .4, we know that for each $h \in [H]$,

$$\widetilde{\boldsymbol{B}}_{h^-} = \widetilde{\Phi}^1_h(\widetilde{\boldsymbol{B}}_{(h-1)^+}, \widetilde{\boldsymbol{Z}}^b_h, \widetilde{\gamma}^a_{h-1}), \widetilde{\boldsymbol{B}}_{h^+} = \widetilde{\Phi}^2_h(\widetilde{\boldsymbol{B}}_{h^-}, \widetilde{\boldsymbol{Z}}^a_h, \widetilde{\boldsymbol{M}}_h),$$

for some function Φ^1_h, Φ^2_h. Meanwhile, we know $\widetilde{\boldsymbol{B}}_{h^+}, \widetilde{\boldsymbol{B}}_{h^-}$ are Gaussian distributions that can be characterized by mean $\widetilde{\boldsymbol{\Theta}}_{h^-}, \widetilde{\boldsymbol{\Theta}}_{h^+}$ and covariance $\widetilde{\Sigma}_{h^-}, \widetilde{\Sigma}_{h^+}$. Then, it holds that $(\widetilde{\boldsymbol{\Theta}}_{h^-}, \widetilde{\Sigma}_{h^-}) = \widetilde{\Phi}^3_{h^-}(\widetilde{\boldsymbol{\Theta}}_{(h-1)^+}, \widetilde{\Sigma}_{(h-1)^+}, \widetilde{\boldsymbol{Z}}^b_h, \widetilde{\gamma}^a_{h-1}), (\widetilde{\boldsymbol{\Theta}}_{h^+}, \widetilde{\Sigma}_{h^+}) =$

$\widetilde{\Phi}^3_{h^+}(\widetilde{\boldsymbol{\Theta}}_{h^-}, \widetilde{\Sigma}_{h^-}, \widetilde{\boldsymbol{Z}}^a_h, \widetilde{\boldsymbol{M}}_h)$ for some function $\widetilde{\Phi}^3_{h^-}, \widetilde{\Phi}^3_{h^+}$. From $\widetilde{\Sigma}_{h^-} = \widetilde{\Psi}^2_{h^-}(\widetilde{\boldsymbol{M}}_{1:h-1}), \widetilde{\Sigma}_{h^-+} = \widetilde{\Psi}^2_{h^+}(\widetilde{\boldsymbol{M}}_{1:h})$, we have

$$\widetilde{\boldsymbol{\Theta}}_{h^-} = \widetilde{\Psi}^1_{h^-}(\widetilde{\boldsymbol{\Theta}}_{(h-1)^+}, \widetilde{\boldsymbol{M}}_{1:h-1}, \widetilde{\boldsymbol{Z}}^b_h, \widetilde{\gamma}^a_{h-1}), \widetilde{\boldsymbol{\Theta}}_{h^+} = \widetilde{\Psi}^1_{h^+}(\widetilde{\boldsymbol{\Theta}}_{h^-}, \widetilde{\boldsymbol{M}}_{1:h}, \widetilde{\boldsymbol{Z}}^a_h),$$

for some functions $\widetilde{\Psi}^1_{h^-}, \widetilde{\Psi}^1_{h^+}$. Furthermore, since $\widetilde{\mathcal{D}}$ satisfies SI-CIB condition, then $\widetilde{\boldsymbol{\Theta}}_{h^-}$ does not depend on $\widetilde{\gamma}^a_{h-1}$, and $\widetilde{\Psi}^1_{h^-}, \widetilde{\Psi}^1_{h^+}$ do not depend on strategy $(\widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h})$. Finally, since $\widetilde{\boldsymbol{C}}_{h^-} = (\cup_{t=1}^h \widetilde{\boldsymbol{Z}}^b_t) \cup (\cup_{t=1}^{h-1} \widetilde{\boldsymbol{Z}}^a_t)$, and $\widetilde{\boldsymbol{C}}_{h^+} = (\cup_{t=1}^h \widetilde{\boldsymbol{Z}}^b_t) \cup (\cup_{t=1}^h \widetilde{\boldsymbol{Z}}^a_t)$, we know that $\widetilde{\boldsymbol{\Theta}}_{h^-} = \widetilde{\Psi}^3_{h^-}(\widetilde{\boldsymbol{C}}_{h^-}, \widetilde{\boldsymbol{M}}_{1:h-1}), \widetilde{\boldsymbol{\Theta}}_{h^+} = \widetilde{\Psi}^3_{h^+}(\widetilde{\boldsymbol{C}}_{h^+}, \widetilde{\boldsymbol{M}}_{1:h})$, for some functions $\widetilde{\Psi}^3_{h^-}, \widetilde{\Psi}^3_{h^+}$ that do not depend on the strategies $\widetilde{g}^m_{1:h}, \widetilde{g}^a_{1:h}$. This completes the proof. ■

E. Algorithm to Solve PN JCCO with Closed-loop Communication Strategies

Given any JCCO problem $\mathcal{D}$ with closed-loop communication strategies and PN IS that satisfies Assumptions II.1, III.2, III.4, and V.1, we can expand it into a problem $\widetilde{\mathcal{D}}$ based on Eq. (.11). Then, from Theorem .6, we know that $\widetilde{\mathcal{D}}$ satisfies the SI-CIB condition and Assumptions II.1, III.2, III.4, and V.1. Furthermore, we can apply Algorithm 1 to solve for the team-optimal strategy $(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H})$ of the problem $\widetilde{\mathcal{D}}$, and then obtain a team-optimal strategy $(g^{m,*}_{1:H}, g^{a,*}_{1:H}) = \varphi^c(\widetilde{g}^{m,*}_{1:H}, \widetilde{g}^{a,*}_{1:H}, \mathcal{D})$ of the problem $\mathcal{D}$ according to Lemma .5.

Algorithm 1 Dynamic Programming for SI-CIB JCCO with Closed-loop Communication Strategies

Require: JCCO $\mathcal{D}$ satisfying the SI-CIB condition and Assumptions II.1, III.2, III.4, and V.1.

for each $M_{1:H} \in \mathcal{M}_{1:H}$, and each realization $\Theta_{(H+1)^-} \in \mathcal{X} \times \mathcal{P}_{(H+1)^-}(M_{1:H})$ **do**
  $V_{(H+1)^-}(\Theta_{(H+1)^-}, M_{1:H}) \leftarrow \mathbb{E}[\boldsymbol{X}^\top_{H+1} Q^1_{H+1} \boldsymbol{X}_{H+1} \,|\, \boldsymbol{\Theta}_{(H+1)^-} = \Theta_{(H+1)^-}, M_{1:H}]$
end for
for $h = H$ to 1 **do**
  for each $M_{1:h} \in \mathcal{M}_{1:h}$, and each realization $\Theta_{h^+} \in \mathcal{X} \times \mathcal{P}_{h^+}(M_{1:h})$ **do**
    $\{\gamma^*_{i,h}\}_{i\in[n]} \leftarrow \operatorname{argmin}_{\{\gamma_{i,h}\}_{i\in[n]} \in \{\Gamma_{i,h}(M_{1:h})\}_{i\in[n]}}$
      $\mathbb{E}[\boldsymbol{X}^\top_h Q^1_h \boldsymbol{X}_h + \boldsymbol{U}^\top_h Q^2_h \boldsymbol{U}_h + V_{(h+1)^-}(\boldsymbol{\Theta}_{(h+1)^-}, M_{1:h}) \,|\, \boldsymbol{\Theta}_{h^+} = \Theta_{h^+}, \boldsymbol{M}_{1:h} = M_{1:h}, \{\gamma_{i,h}\}_{i\in[n]}]$
    $V^*_{h^+}(\Theta_{h^+}, M_{1:h}) \leftarrow$
      $\mathbb{E}[\boldsymbol{X}^\top_h Q^1_h \boldsymbol{X}_h + \boldsymbol{U}^\top_h Q^2_h \boldsymbol{U}_h + V_{(h+1)^-}(\boldsymbol{\Theta}_{(h+1)^-}, M_{1:h}) \,|\, \boldsymbol{\Theta}_{h^+} = \Theta_{h^+}, \boldsymbol{M}_{1:h} = M_{1:h}, \{\gamma^*_{i,h}\}_{i\in[n]}]$
    for each C_{h^+} such that $\Psi^3_{h^+}(C_{h^+}, M_{1:h}) = \Theta_{h^+}$ **do**
      for $i \in [n]$ **do**
        $g^{a,*}_{i,h}(C_{h^+}, \cdot, M_{1:h}) \leftarrow \gamma^*_{i,h}(\cdot)$
      end for
    end for
  end for
  for each $M_{1:h-1} \in \mathcal{M}_{1:h-1}$, and each realization $\Theta_{h^-} \in \mathcal{X} \times \mathcal{P}_{h^-}(M_{1:h-1})$ **do**
    $M^*_h \leftarrow \operatorname{argmin}_{M_h \in \mathcal{M}_h} \mathbb{E}[\mathcal{K}_h(M_h) + V_{h^+}(\boldsymbol{\Theta}_{h^+}, M_{1:h}) \,|\, \boldsymbol{\Theta}_{h^-} = \Theta_{h^-}, \boldsymbol{M}_{1:h-1} = M_{1:h-1}, \boldsymbol{M}_h = M_h]$
    $V^*_{h^-}(\Theta_{h^-}, M_{1:h-1}) \leftarrow \mathbb{E}[\mathcal{K}_h(M_h) + V_{h^+}(\boldsymbol{\Theta}_{h^+}, M_{1:h-1}, M^*_h) \,|\, \boldsymbol{\Theta}_{h^-} = \Theta_{h^-}, \boldsymbol{M}_{1:h-1} = M_{1:h-1}, \boldsymbol{M}_h = M^*_h]$
    for each C_{h^-} such that $\Psi^3_{h^-}(C_{h^-}, M_{1:h-1}) = \Theta_{h^-}$ **do**
      for $i \in [n]$ **do**
        $g^{m,*}_{i,h}(C_{h^-}, M_{1:h-1}) \leftarrow M^*_{i,h}$
      end for
    end for
  end for
end for
return $(g^{m,*}_{1:H}, g^{a,*}_{1:H})$